


# GAME MANUAL.

An FRC off-season game by TideCrew.

TIDE CREW

RIDE THE TIDE.

GUADALAJARA, JALISCO · VERSION 1.5 · NOVEMBER 2026

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# 01. INTRODUCTION

## 1.1 About FIRST®

FIRST® (For Inspiration and Recognition of Science and Technology) was founded by inventor Dean Kamen to inspire young people's interest in science and technology. As a robotics community that prepares young people for the future, FIRST is the world's leading youth-serving nonprofit advancing STEM education. For 30 years, FIRST has combined the rigor of STEM learning with the fun and excitement of traditional sports and the inspiration that comes from community through programs that have a proven impact on learning, interest, and skill-building inside and outside of the classroom. FIRST provides programs that span a variety of age groups:

- FIRST® Robotics Competition for grades 9-12, suggested ages 14-18
- FIRST® Tech Challenge for grades 7-12, ages 12-18
- FIRST® LEGO® League for grades Pre-K-8, ages 4-16
  - FIRST® LEGO® League Challenge for grades 4-8 (ages 9-16, ages vary by country)
  - FIRST® LEGO® League Explore for grades 2-4 (ages 6-10)
  - FIRST® LEGO® League Discover for grades Pre-K-1 (ages 4-6)

Please visit the FIRST website for more information about FIRST and its programs.

| Purpose | Vision | Mission |
|---|---|---|
| FIRST exists to prepare the young people of today for the world of tomorrow. | To transform our culture by creating a world where science and technology are celebrated and where young people dream of becoming science and technology leaders. | The mission of FIRST is to provide life-changing robotics programs that give young people the skills, confidence, and resilience to build a better world. |

## 1.2 In Memoriam

In October 2019, Dr. Woodie Flowers, an innovator in design and engineering education and a Distinguished Advisor to FIRST and supporter of our mission, passed away. As thousands of heartfelt tributes to Woodie have poured in from around the world, it is clear his legacy will live on indefinitely through the gracious nature of our community and our ongoing commitment to empowering educators and building global citizens.

Figure 1-1 *Dr. Woodie Flowers, 1943-2019*



## 1.3 FIRST® Robotics Competition

FIRST® Robotics Competition combines the excitement of sport with the rigors of science and technology. Teams of students are challenged to design, build, and program industrial-size robots and compete for awards, while they also create a team identity, raise funds, hone teamwork skills, and advance respect and appreciation for STEM within the local community.

Volunteer professional mentors lend their time and talents to guide each team. It's as close to real-world engineering as a student can get. Plus, high school students gain access to exclusive scholarship opportunities from colleges, universities, and technical programs.

Each January at an event known as "Kickoff," a new, challenging game is introduced. These exciting competitions combine the practical application of science and technology with the fun, intense energy, and excitement of a championship-style sporting event. Teams are encouraged to display Gracious Professionalism®, help other teams, and cooperate while competing. This is known as Coopertition®.

In 2026, FIRST Robotics Competition is projected to reach approximately 90,000 high-school students representing more than 3,700 teams. Teams represent more than 30 countries and come from nearly every state in the United States.

> **FROZEN BLITZ is not an official FIRST game.** It is an off-season game designed and owned by TideCrew for the TideChallenge. It is built on the 2026 FIRST Robotics Competition rule set, so everything a team learns here carries straight into the official season.

This off-season game, and this manual, were presented at the TideChallenge Kickoff on Monday, October 26, 2026, at 4:15 pm Guadalajara time (UTC-6). At the Kickoff, all participating teams:

- saw the FROZEN BLITZ game for the first time, and
- learned about the FROZEN BLITZ rules and regulations.

## 1.4 FIRST Ethos and Core Values

### 1.4.1 Core Values

The FIRST Core Values are fundamental to FIRST and unique to its programs. They emphasize friendly sportsmanship, respect for the contributions of others, teamwork, learning, and community involvement and are part of our commitment to fostering, cultivating, and preserving a culture of unity.

Our community expresses the FIRST philosophies of Gracious Professionalism® and Coopertition® through the FIRST Core Values.

- **Discovery:** We explore new skills and ideas.
- **Innovation:** We use creativity and persistence to solve problems.
- **Impact:** We apply what we learn to improve our world.
- **Inclusion:** We respect each other and embrace our differences.
- **Teamwork:** We are stronger when we work together.
- **Fun:** We enjoy and celebrate what we do!

### 1.4.2 Gracious Professionalism®, a FIRST Credo

Gracious Professionalism is part of the ethos of FIRST. It's a way of doing things that encourages high quality work, emphasizes the value of others, and respects individuals and the community. Gracious Professionalism is not clearly defined for a reason. It is an aspirational ideal to always strive towards, not a goal to be achieved or a method of measuring someone, and for this reason, you can never say someone "is" or "is not" being Graciously Professional. We should each work to better embody Gracious Professionalism in all our actions. How we pursue this can and should mean different things to everyone.

Some possible meanings of Gracious Professionalism include:

- gracious attitudes and behaviors are win-win,
- gracious folks respect others and let that respect show in their actions,
- professionals possess special knowledge and are trusted by society to use that knowledge responsibly, and
- gracious professionals make a valued contribution in a manner pleasing to others and to themselves.

In the context of FIRST, this means that all teams and participants should:

- learn to be strong competitors, but also treat one another with respect and kindness in the process and
- avoid leaving anyone feeling as if they are excluded or unappreciated.

Knowledge, pride, and empathy should be comfortably and genuinely blended.

In the end, Gracious Professionalism is part of pursuing a meaningful life. When professionals use knowledge in a gracious manner and individuals act with integrity and sensitivity, everyone wins and society benefits.

Figure 1-2 *Dr. Woodie Flowers, Gracious Professionalism advocate and exemplar*



> *"The FIRST spirit encourages doing high-quality, well-informed work in a manner that leaves everyone feeling valued. Gracious Professionalism seems to be a good descriptor for part of the ethos of FIRST. It is part of what makes FIRST different and wonderful."*
>
> – Dr. Woodie Flowers (1943 – 2019), Distinguished Advisor to FIRST

It is a good idea to spend time going over this concept with your team and reinforcing it regularly. We recommend providing your team with real-life examples of Gracious Professionalism in practice, such as when a team loans valuable materials or expertise to another team that they will later face as an opponent in competition. Routinely highlight opportunities to display Gracious Professionalism at events and encourage team members to suggest ways in which they can demonstrate this quality themselves and through outreach activities.

### 1.4.3 Coopertition®

At FIRST, Coopertition is displaying unqualified kindness and respect in the face of fierce competition. Coopertition is founded on the concept and philosophy that teams can and should help and cooperate with one another even as they compete. Coopertition involves learning from teammates and mentors. Coopertition means competing always but assisting and enabling others when you can.

## 1.5 This Document & Its Conventions

The FROZEN BLITZ Game Manual is a resource for all FIRST Robotics Competition teams taking part in the TideChallenge. Its audience will find the following detail:

- a general overview of the FROZEN BLITZ game,
- detail about the FROZEN BLITZ playing FIELD,
- a description of how to play the FROZEN BLITZ game,
- rules (related to safety, conduct, game play, ROBOT construction, and inspection), and
- how the TideChallenge tournament runs, including the Single Top Score award (Section 9 Tournament).

FROZEN BLITZ is built on the 2026 FIRST Robotics Competition rule set (REBUILT™ presented by Haas, Game Manual version TU22). All participants should also study Section 14 Event Rules (E) of the 2026 FIRST Robotics Competition Game Manual, available on the FIRST Robotics Competition Season Materials webpage. Those rules apply at TideChallenge events and carry the same weight as this document.

The intent of this manual is that the text means exactly, and only, what it says. Please avoid interpreting the text based on assumptions about intent, implementation of past rules, or how a situation might be in "real life." There are no hidden requirements or restrictions. If you've read everything, you know everything.

Specific methods are used throughout this manual to highlight warnings, cautions, key words, and phrases. These conventions are used to alert the reader to important information and are intended help teams in constructing a ROBOT that complies with the rules in a safe manner.

Links to other section headings in this manual, external articles, and rule references appear in underlined text.

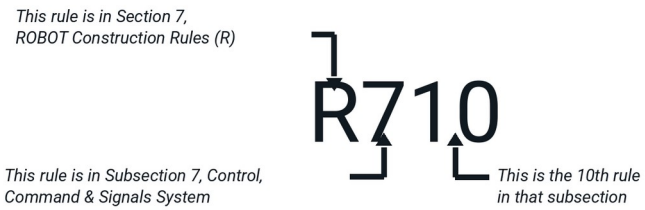
Key words that have a particular meaning within the context of the FIRST Robotics Competition and FROZEN BLITZ are defined in Section 10 Glossary and indicated in ALL CAPS throughout this document.

The rule numbering method indicates the section, subsection, and position of the rule within that subsection. The letter indicates the section in which the rule is published.

- Q for 5.7.1 Question Box (Q)
- G for 6 Game Rules (G)
- R for 7 ROBOT Construction Rules (R)
- I for 8 Inspection & Eligibility (I)

The following digit(s) represents the subsection in which the rule can be found. The final digits indicate the rule's position within that subsection.

**Figure 1-3** *Rule numbering method*




> Warnings, cautions, and notes appear in blue boxes. Pay close attention to their contents as they're intended to provide insight into the reasoning behind a rule, helpful information on understanding or interpreting a rule, and/or possible "best practices" for use when implementing systems affected by a rule.
>
> While blue boxes are part of the manual, they do not carry the weight of the actual rule (if there is an inadvertent conflict between a rule and its blue box, the rule supersedes the language in the blue box).

Imperial dimensions are followed by comparable metric dimensions in parentheses to provide metric users with the approximate size, mass, etc. Decimal values are rounded in order to limit the number of decimal places shown and the number of significant figures. Metric conversions in rules round such that the metric dimension is compliant with the rule (i.e. maximums round down, minimums round up). The metric conversions are offered for convenient reference only and do not overrule or take the place of the imperial dimensions presented in this manual and the official drawings (i.e. dimensions and rules will always defer to measurements using imperial units).

Rules include colloquial language, also called headlines, in an effort to convey an abbreviated version of the rule or rule set. There are two versions of headline formatting.

Evergreen rules, or rules which are expected to go relatively unchanged from season to season, are indicated with a leading asterisk and their rule number and headline are presented in **bold deep teal text**. "Relatively unchanged" means that the overall intent and presence of the rule from season to season is constant, but game specific terms may be updated as needed (e.g. changing FUEL to SCOOPS in a rule about what DRIVE COACHES may not contact during a MATCH). These rules also start their respective section, so their rule number is not expected to change from season to season.

Rules specific to FROZEN BLITZ use **bold Pacific Blue text**. Any disagreement between the specific language used in the rules and the colloquial language is an error, and the specific rule language is the ultimate authority. If you discover a disparity, please let us know at santi.arzate@tec.mx and we will correct it.

## 1.6 Team Updates

Team Updates are used to notify the FROZEN BLITZ community of revisions to the official documentation (e.g. the manual, drawings, etc.) or important event news. Team Updates are posted each Thursday, starting on the first Thursday after Kickoff and ending on the Thursday prior to the event.

Team Updates are posted on the official FROZEN BLITZ WhatsApp community and are published before 10:00 pm Guadalajara time (UTC-6).

Generally, Team Updates follow the following convention:

- Additions are highlighted in yellow. This is an example.
- Deletions are indicated with a strikethrough.
- Notes that are added for clarity or explanation for the change but are not retained as part of the manual appear in bold. **This is an example.**

## 1.7 Question and Answer System

The Question and Answer System (Q&A) is a resource for clarifying the FROZEN BLITZ Game Manual and the official FIELD drawings. Teams can search for previously asked questions and responses or pose new questions. Questions can include examples for clarity or reference multiple rules to understand the relationships and differences between them.

The Q&A opens on Friday, October 30, 2026, at 4:15 pm Guadalajara time (UTC-6). The link to the Q&A will be shared in a Team Update. The Q&A may result in revisions to the text in the official manual (which are communicated using the process described in 1.6 Team Updates).

The responses in the Q&A do not supersede the text in the manual, although every effort will be made to eliminate inconsistencies between the two. While responses provided in the Q&A may be used to aid discussion at each event, per section 8 Inspection & Eligibility and section 5.7 Head REFEREE and FTA Interaction, REFEREES and INSPECTORS are the ultimate authority on rules.

The Q&A is not a resource for firm predictions on how a situation will play out an event. Questions about the following will not be addressed:

- rulings on vague situations,
- challenging decisions made at past events, or

- design reviews of a ROBOT system for legality.

Weak questions are overly broad, vague, and/or include no rule references. Some examples of questions that will not be answered in the Q&A are:

- Is this part/design legal?
- How should the REFEREE have ruled when this specific game play happened?
- Duplicate questions
- Nonsense questions

Good questions ask generically about features of parts or designs, gameplay scenarios, or rules, and often reference one or more relevant rules within the question. Some examples of questions that will likely be answered in the Q&A are:

- A device we are considering using on the ROBOT comes with purple AWG 40 wire, does this comply with R?? and R??
- We're not sure how to interpret how Rule G?? applies if blue ROBOT A does X and red ROBOT B does Y, can you please clarify?
- If a ROBOT does this specific action, is it doing what this defined term is describing?

Questions from "FRC 99999" represent content asked by key volunteers (e.g., REFEREES, INSPECTORS, etc.), answered by the TideCrew Game Design Team, and are considered relevant to teams.

SECTION 02 · WHO WE ARE

# 02. TIDECHALLENGE OVERVIEW

## Every crew has a first wave. The TideChallenge is ours to share.

TideCrew was born from a family of robotics crews in Guadalajara, Jalisco. We build a place where the people who were called “too much” get to make extraordinary things, and we widen the door so the next ones get in, regardless of money or background.

For years, our mentors have designed a short challenge before January so students, rookies and veterans alike, can live what a FIRST build season really feels like before the official one begins. The first build season can be a lot. The TideChallenge exists so it doesn’t have to be: a short build season, no pressure, and one goal. Learn, and have fun doing it.

This season the door is wider. FROZEN BLITZ is open to the teams we call friends, so every crew can build, test, fail, fix, and play side by side.

THE PROMISE

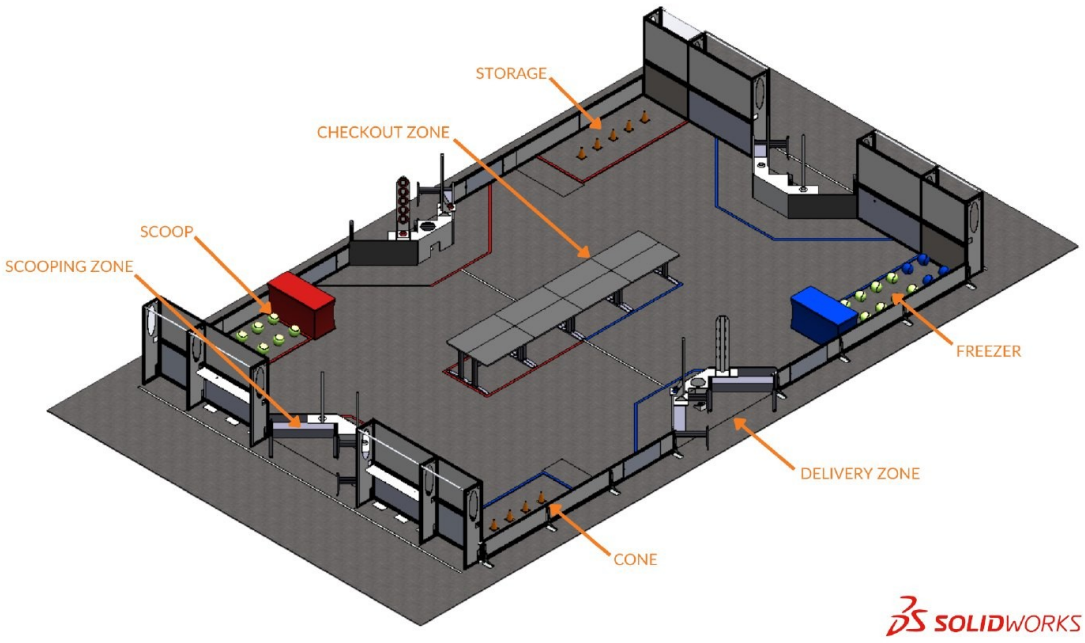
***WE WERE NEVER ABOUT THE MACHINES. THEY WERE JUST OUR PROOF.***

**What you need to know before you build:**

- **It’s a real FRC ruleset.** ROBOT construction, safety, and inspection follow the 2026 FIRST Robotics Competition rules. If it’s legal here, it’s legal in January, and the other way around.
- **The game is ours.** FROZEN BLITZ field, scoring, and game rules were designed by TideCrew. When this manual and the 2026 FIRST manual disagree on gameplay, this manual wins.
- **Questions are welcome.** Use the Q&A (section 1.7). No question about the rules is too small.

# 03. GAME OVERVIEW

Figure 3-1 *FROZEN BLITZ FIELD and key elements*




In FROZEN BLITZ, two competing ALLIANCES are invited to make ice cream and deliver it. Each ALLIANCE is assigned a SCOOP color (red or blue, based on ALLIANCE affiliation). Robots retrieve CONES and SCOOPS, build them into ICE CREAM, and deliver it to their DELIVERY ZONE. HUMAN PLAYERS assist from their SCOOPING STATION, where they place the HOOPS that make stacking possible, and from their DELIVERY ZONE, where they clear delivered ICE CREAM off the DELIVERY TRAYS, recycle its SCOOPS through the CHUTE, and throw its CONE back to their STORAGE. In the final moments of each MATCH, ALLIANCE ROBOTS race to the CHECKOUT ZONE to finish the day.

Each MATCH begins with a 15-second autonomous period, during which time ALLIANCE ROBOTS operate only on pre-programmed instructions to score points by:

- reaching their FREEZER, and
- placing CONES or stacking SCOOPS.

In the final 2 minutes and 15 seconds of the MATCH, DRIVERS take control of the ROBOTS and score points by:

- delivering ICE CREAM to their DELIVERY ZONE,
- refilling their FREEZER with SCOOPS by the end of the MATCH,
- getting CONES into their SCOOPING STATION'S SAFE ZONE, and
- checking out in the CHECKOUT ZONE as the MATCH ends.

**The ALLIANCE with the highest score at the end of the MATCH wins!**

# 04. ARENA

The ARENA includes all elements of the game infrastructure that are required to play FROZEN BLITZ: the FIELD, SCORING ELEMENTS (SCOOPS and CONES), and all equipment needed for FIELD control, ROBOT control, and scorekeeping.

## 4.1 Dimensions and Accuracy

All official models for the FROZEN BLITZ FIELD were created in SOLIDWORKS. The specification for the FROZEN BLITZ FIELD can be retrieved from a few locations:

- The 3D CAD model is the official representation of the FROZEN BLITZ FIELD and how it is constructed.
- Illustrations included in this section are for a general visual understanding of the FROZEN BLITZ ARENA, and dimensions included in the manual are nominal and no tolerances are implied. Please refer to the official drawings for exact dimensions, tolerances, and construction details.

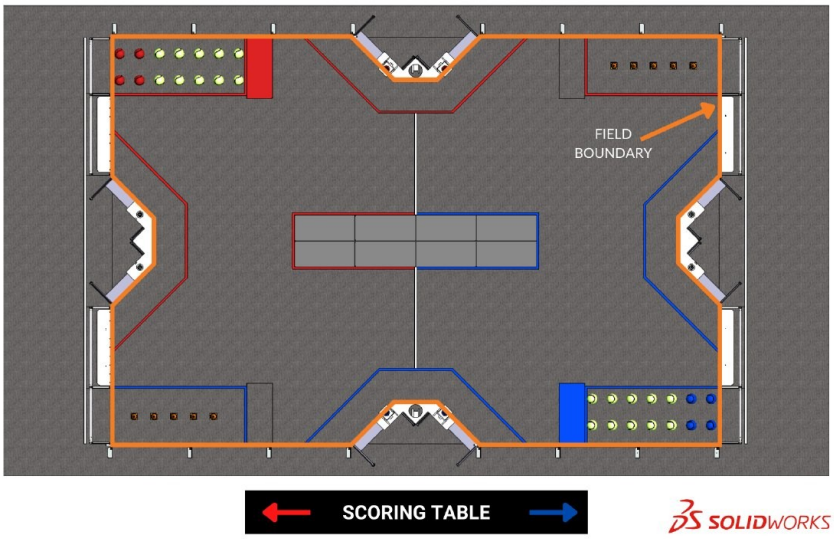
The official drawings and CAD models of important elements of the FROZEN BLITZ FIELD are posted on the official FROZEN BLITZ WhatsApp community.

> The FROZEN BLITZ ARENA is assembled in venues that differ from standard FIRST Robotics Competition venues, and some small variations occur. For details regarding assembly tolerances, please use the Question and Answer System. Successful teams will design ROBOTS that are insensitive to these variations.

## 4.2 FIELD

Each FIELD for FROZEN BLITZ is an approximately 360.0in (~9.14m) by 552.0in (~14.02m) carpeted area bounded by and including the inward- and upward-facing surfaces of the guardrails, the inward-facing surfaces of the ALLIANCE WALLS, and the front vertical faces of the ALLIANCE WALLS.

Figure 4-1 *FROZEN BLITZ FIELD boundary*

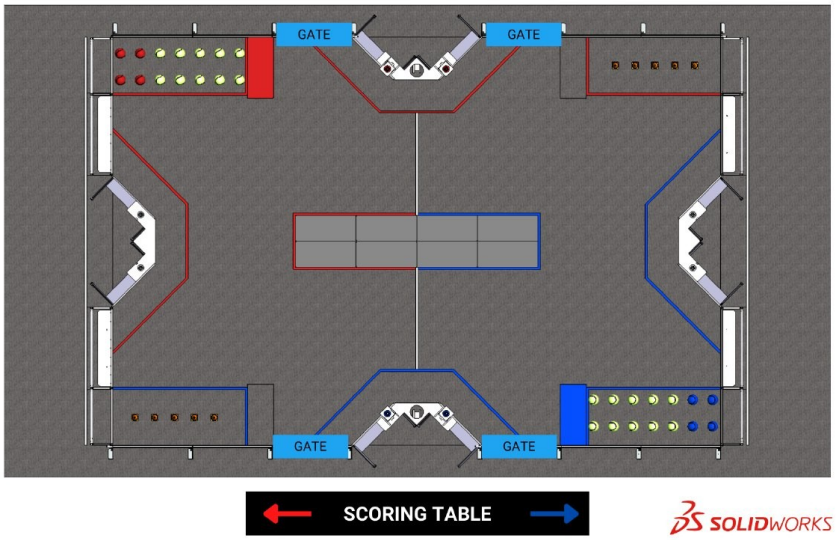



The FIELD is populated with the following elements:

- 2 FREEZERS (a red FREEZER and a blue FREEZER),
- 2 STORAGES (a red STORAGE and a blue STORAGE),
- 2 DELIVERY ZONES (a red DELIVERY ZONE and a blue DELIVERY ZONE),
- 2 SCOOPING ZONES (a red SCOOPING ZONE and a blue SCOOPING ZONE), each with a SCOOPING STATION, and
- 2 CHECKOUT ZONES (a red CHECKOUT ZONE and a blue CHECKOUT ZONE), divided by the CENTER LINE, and
- 8 AprilTags (see section 4.9 AprilTags).

The surface of the FIELD is low pile carpet. Guardrails form the long edges of the FIELD. Guardrails are a 20.0in (~50.8cm) tall system of transparent polycarbonate supported on the top and bottom by aluminum extrusion. There are 4 gates in the guardrail that allow access to the FIELD for placement and removal of ROBOTS. The gate passthrough, when open, is 38.0in (~96.5cm) wide. Gates are closed and shielded during the MATCH.

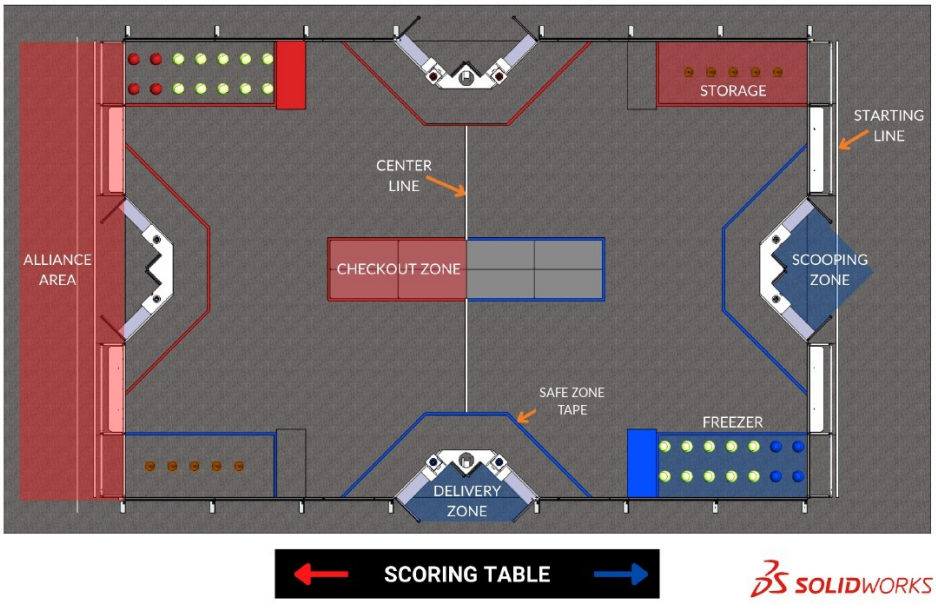
Figure 4-2 *Gate locations*




## 4.3 Areas, Zones, & Markings

FIELD areas, zones, and markings of consequence are described below. Unless otherwise specified, the tape used to mark lines and zones throughout the FIELD is 2.0in (~5.1cm) wide gaffers tape, as used on official FIRST Robotics Competition FIELDS. The tape brand and exact color may vary from event to event.

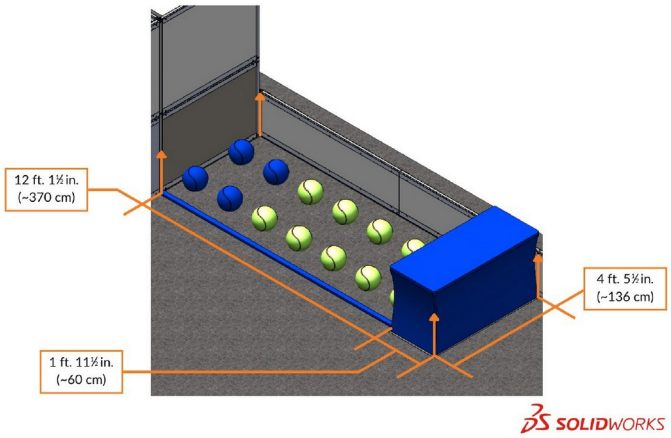
Figure 4-3 *FROZEN BLITZ FIELD areas, zones, and markings*




- **ALLIANCE AREA:** a 330.0in (~8.38m) wide by 106.0in (~2.69m) deep, infinitely tall volume formed by, and including, the ALLIANCE STATIONS and ALLIANCE colored tape. It includes neither the FREEZER nor the STORAGE.
- **CENTER LINE:** a white line at the center of the FIELD, parallel to the ALLIANCE WALL, that bisects the length of the FIELD.

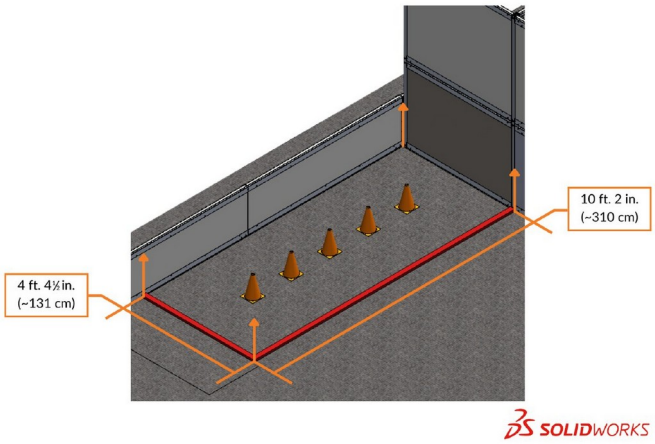
- **FREEZER:** a 122.0in (~3.10m) wide, 53.5in (~1.36m) deep, infinitely tall volume defined by the ALLIANCE STATION, guardrail, and ALLIANCE colored boundaries.

Figure 4-4 *FREEZER*



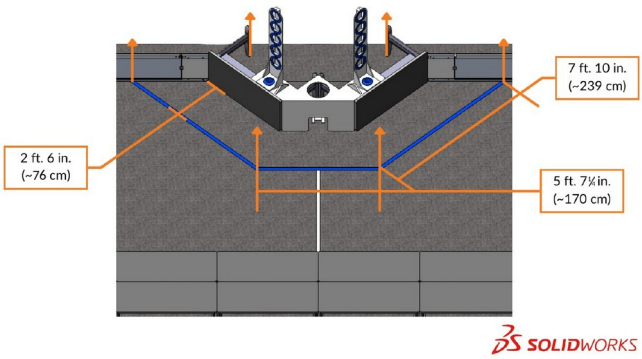

- **STORAGE:** a 122.0in (~3.10m) wide, 52.5in (~1.33m) deep, infinitely tall volume defined by the ALLIANCE STATION, guardrail, and ALLIANCE colored boundaries.

Figure 4-5 *STORAGE*



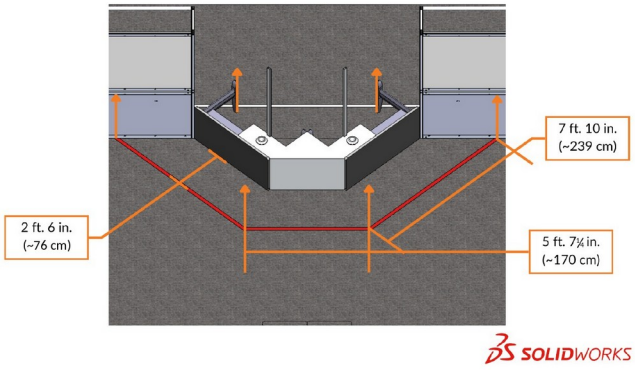

- **STARTING LINE:** a white line spanning the width of the carpet and located 28.0in (~71.1cm) from the back of the DRIVER STATION diamond plate panel to the near edge of the tape.
- **DELIVERY ZONE:** 1 of 2 (1 per ALLIANCE) 200.0in (~5.08m) wide by 67.25in (~1.71m) deep, infinitely tall volumes bounded by and including the ALLIANCE colored tape. Each DELIVERY ZONE includes a SAFE ZONE.

Figure 4-6 *DELIVERY ZONE*



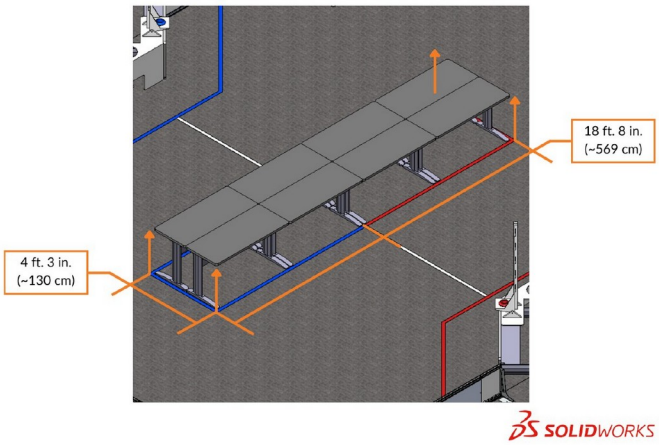

- **SCOOPING ZONE:** 1 of 2 (1 per ALLIANCE) 200.0in (~5.08m) wide by 67.25in (~1.71m) deep, infinitely tall volumes bounded by and including the ALLIANCE colored tape, containing the ALLIANCE'S SCOOPING STATION.
- **SCOOPING STATION:** the ALLIANCE structure inside each SCOOPING ZONE where the HUMAN PLAYER places HOOPS and ROBOTS build stacks. Each SCOOPING STATION includes a SAFE ZONE. SCOOPING STATION dimensions are given in the official CAD model.

Figure 4-7 *SCOOPING ZONE and SCOOPING STATION*




- **SAFE ZONE:** 1 per SCOOPING STATION and 1 per DELIVERY ZONE structure (4 in total, 2 per ALLIANCE): the infinitely tall volume between the structure and the ALLIANCE colored tape located 30.0in (~76.2cm) from it, bounded by and including that tape. A ROBOT at least half within its own SAFE ZONE is protected from opponent contact, and opponent ROBOTS may not enter it (see G425 and G426).
- **CHECKOUT ZONE:** 1 of 2 (1 per ALLIANCE) 112.0in (~2.84m) wide by 51.0in (~1.30m) deep, infinitely tall volumes bounded by and including the ALLIANCE colored tape and divided by the CENTER LINE.

Figure 4-8 *CHECKOUT ZONE*




## 4.4 DELIVERY ZONE

One DELIVERY ZONE is located at the center of each long side of the FIELD. Each DELIVERY ZONE belongs to an ALLIANCE and consists of 2 DELIVERY TRAYS, 1 CHUTE, 1 PURPLE PLANE, the supporting structure shown, and a SAFE ZONE in front of the structure.

Each DELIVERY TRAY holds 1 ICE CREAM at a time and has 2 positions: raised (ready to receive ICE CREAM) and lowered (pivoted back toward the HUMAN PLAYER so the ICE CREAM can be removed). A lever locks each tray in the raised position; pulling the lever releases the mechanism and lets the tray pivot back. Only the HUMAN PLAYER at the DELIVERY ZONE moves the trays. The CHUTE sits between the 2 DELIVERY TRAYS of each DELIVERY ZONE structure. SCOOPS dropped into it roll straight out into the DELIVERY ZONE, right where a ROBOT that has just delivered is likely to be.

Every ICE CREAM goes through the same **delivery cycle**:

A. A ROBOT places the ICE CREAM on an empty, raised DELIVERY TRAY and releases it.
B. The ICE CREAM remains stable on the tray for 3 seconds with no ROBOT CONTROL. It is now delivered, and its points (including any TOPPER) are locked in.
C. The HUMAN PLAYER pulls the tray's release lever, pivots the tray back (lowered), and removes the ICE CREAM.
D. The HUMAN PLAYER drops each SCOOP into the CHUTE, which returns it to the FIELD (see G437).
E. The HUMAN PLAYER throws or rolls the CONE back into their ALLIANCE'S STORAGE (see G438).
F. The HUMAN PLAYER sets the HOOPS aside. HOOPS removed from delivered ICE CREAM stay out of play for the rest of the MATCH.
G. The HUMAN PLAYER raises the tray until the lever locks it, ready for the next ICE CREAM.

> Lowering a tray before its ICE CREAM has been stable for 3 seconds means that ICE CREAM is not delivered and scores nothing. Count to 3. It's ice cream; it can wait (a little).

Figure 4-9 *DELIVERY ZONE elements*

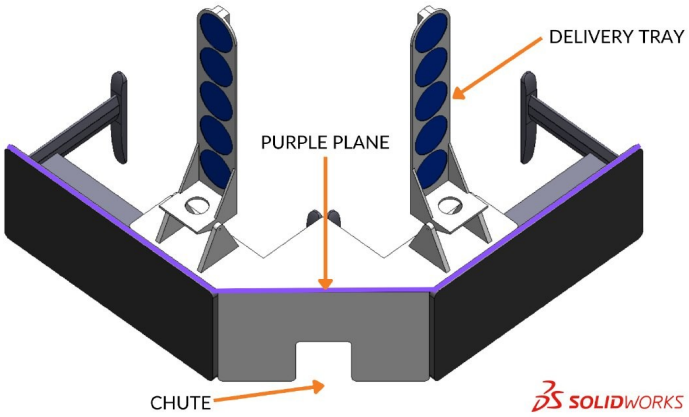



Figure 4-10 *DELIVERY ZONE structure dimensions*

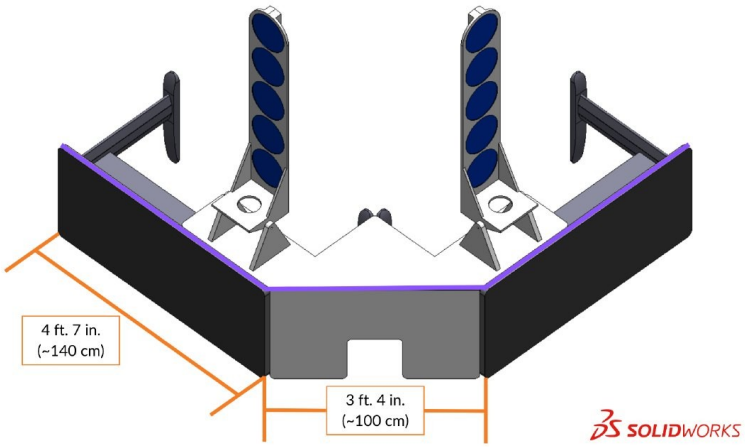
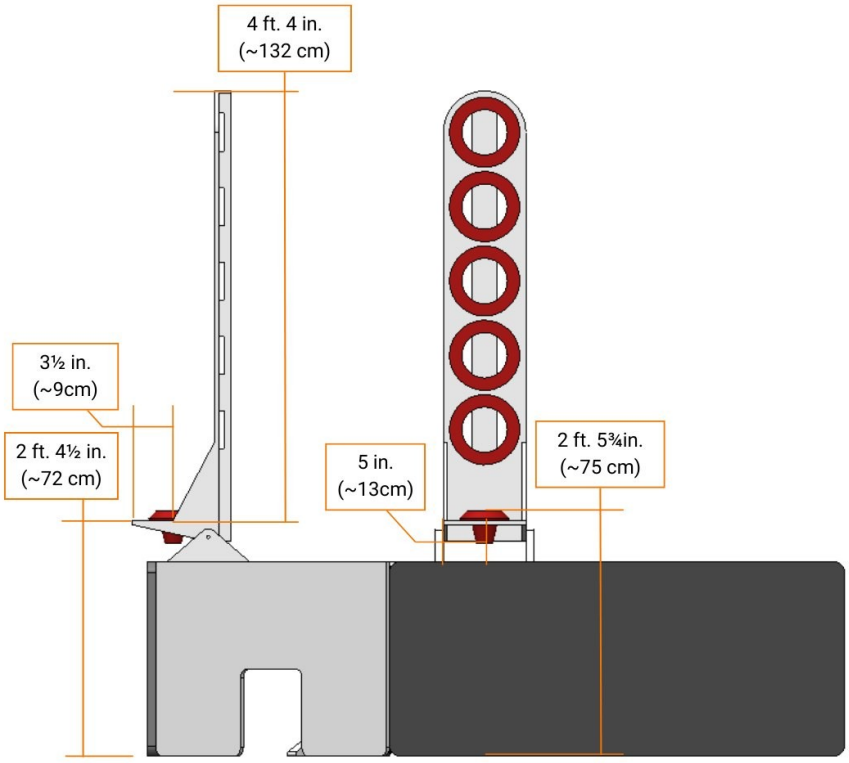


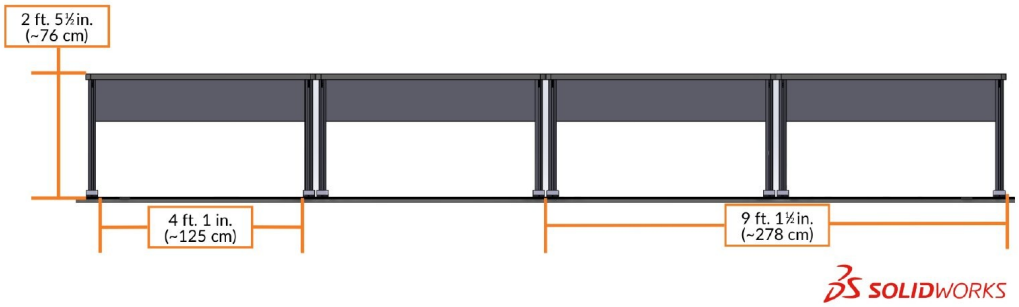
Figure 4-11 *DELIVERY TRAY and CHUTE elevation*




## 4.5 CHECKOUT ZONE

The 2 CHECKOUT ZONES are located at the center of the FIELD, 1 on each side of the CENTER LINE. Together they form an assembly of tables, delimited with ALLIANCE colored tape according to the side of the FIELD. The tables have an inner clearance of 49.0in (~1.24m) and are 29.5in (~74.9cm) tall.

Figure 4-12 *CHECKOUT ZONE tables*




## 4.6 SCORING ELEMENTS

There are 2 types of SCORING ELEMENTS: SCOOPS and CONES. SCOOPS are stacked on a CONE at the SCOOPING STATION, with a HOOP between each pair of SCOOPS. A stack becomes an ICE CREAM once it

leaves the SCOOPING STATION (see 5.5.1 SCORING ELEMENT Scoring Criteria and G408). HOOPS are not SCORING ELEMENTS.

### 4.6.1 SCOOP

Figure 4-13 *SCOOPS: neutral, red ALLIANCE, and blue ALLIANCE (colors are examples)*




FROZEN BLITZ is played with oversized, felt-covered tennis balls called SCOOPS. A SCOOP has a rubber core with a felt cover and is 8.0in (~20.3cm) in diameter when inflated. Neutral SCOOPS are green (classic tennis ball color). Red ALLIANCE SCOOPS are pink or red, and blue ALLIANCE SCOOPS are blue or light blue.

> Any brand of 8in oversized tennis ball works, as long as it meets the size above. Event organizers will use similar colors for each SCOOP type, but a particular shade cannot be guaranteed. Design vision and sensing to tell the 3 SCOOP types apart, not to match an exact color.
>
> SCOOP weight and bounce are not specified. Teams are encouraged to buy a few SCOOPS and characterize them.

SCOOPS are usually shipped deflated and are inflated with a hand pump until they measure 8.0in ±0.25in (~20.3cm ±0.6cm) in diameter, measured across the ball with a tape measure or caliper. FIELD STAFF check SCOOP size before each MATCH block and re-inflate as needed. Teams should design ROBOTS that tolerate small variations in SCOOP size and firmness.

The SCOOPS used at the TideChallenge are:

- **neutral:** Hanaive 8in inflatable dog tennis balls, green (4-pack with pump, ASIN B0BL14BYRX), available on Amazon México, and
- **ALLIANCE colored:** Fydun 8in jumbo inflatable tennis balls in pink (red ALLIANCE) and blue (blue ALLIANCE) (ASIN B0D8M9BGJN), available on Amazon México.

### 4.6.2 CONE

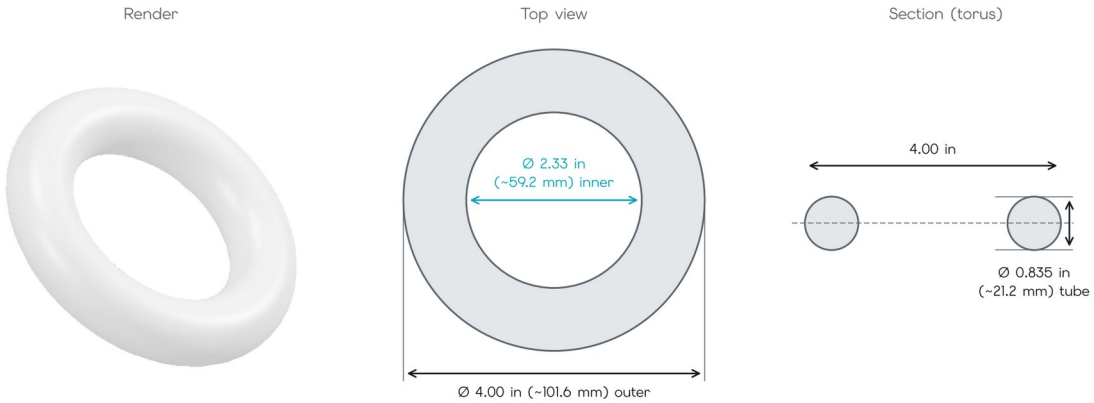
Figure 4-14 *CONES*



FROZEN BLITZ is also played with CONES. A CONE is a 9.0in (~22.9cm) tall orange plastic sport training cone with a 4.92in (~12.5cm) base diameter. CONE weight is not specified. CONES are Hirola 9in training cones (24-pack, ASIN B0BWSJHFBG), available on Amazon México.

### 4.6.3 HOOP

Figure 4-15 *HOOP dimensions*




A HOOP is a ring that sits on top of a SCOOP so that another SCOOP can rest on it. Without a HOOP, a SCOOP cannot be stacked on another SCOOP. HOOPS are introduced into the MATCH only by the HUMAN PLAYER at each SCOOPING STATION, who places them between SCOOPS as a stack is built. HOOPS are not SCORING ELEMENTS: they do not score points on their own and do not count toward the TIER of an ICE CREAM.

A HOOP is a torus (a solid ring with a round cross section) with a 4.00in (~10.2cm) outer diameter, a 2.33in (~5.9cm) inner diameter, and a 0.835in (~21.2mm) diameter cross section. HOOPS are white for both ALLIANCES and are 3D printed in PLA at 35% infill, which makes them quite light; HOOP weight is not specified.

## 4.7 The FIELD Management System

The FIELD Management System (FMS) is all the electronics responsible for sensing and controlling the FROZEN BLITZ FIELD. The FMS encompasses all FIELD electronics, including computers, REFEREE controls, the wireless access point, sensors, stack lights, E-Stops, etc.

When a DRIVE TEAM connects the Ethernet cable from their assigned DRIVER STATION to their OPERATOR CONSOLE, the Driver Station Software on the OPERATOR CONSOLE computer communicates with the FMS. Once connected, the open ports available are described in Table 7-5.

Note that ROBOT code cannot be deployed while connected to the FMS.

> The FMS software, wireless radio programming, and DRIVER STATION setup used at TideChallenge events are announced in Team Updates ahead of the event.

The FMS alerts participants to milestones in the MATCH using audio cues detailed in Table 4-1. Please note that audio cues are intended as a courtesy to participants and not intended as official MATCH markers. If there is a discrepancy between an audio cue and the FIELD timers, the FIELD timers are the authority.

Table 4-1 *Audio cues*

| Event | Timer Value | Audio Cue |
|---|---|---|
| MATCH start | 0:15 (for AUTO) | “Cavalry Charge” |
| AUTO ends | 0:00 (for AUTO) | “Buzzer” |
| TELEOP begins | 2:15 | “3 Bells” |
| END GAME begins (CHECKOUT ZONE protection engaged) | 0:30 | “Steam Whistle” |
| MATCH end | 0:00 | “Buzzer” |
| MATCH stopped | n/a | “Foghorn” |

## 4.8 FIELD STAFF

FIELD STAFF are responsible for making sure the MATCHES are cycled through efficiently, fairly, safely, and with a spirit of cooperation, Gracious Professionalism, and generosity of spirit. There are 3 FIELD-side key volunteer roles with whom teams should be familiar:

- **Head REFEREE** – trains, directs, and supervises REFEREES. They oversee all scoring processes and procedures in collaboration with the FTA. The Head REFEREE has final authority for decisions regarding MATCH scores, penalties, and YELLOW and RED CARD assignments.
- **FIRST Technical Advisor (FTA)** – ensures the event runs smoothly and safely, is the point of contact for all things related to the FIELD, ROBOTS, and network, acts as a team advocate, and is a major point of escalation and conflict resolution for the event.

- **FIELD Supervisor** – directs activity on the FIELD to ensure efficient execution and pacing of MATCHES. FIELD Supervisors make sure the FIELD is intact and lead the FIELD Reset team that resets the FIELD after each MATCH.

## 4.9 AprilTags

AprilTags help ROBOTS find their position on the FIELD. FROZEN BLITZ uses the same AprilTag family and size as the 2026 FIRST Robotics Competition FIELD, so existing vision code only needs the FROZEN BLITZ FIELD layout.

- All markers are from the 36h11 tag family, IDs 1 to 8. There are 8 unique markers on the FIELD, and no other IDs are used.
- Each AprilTag is an 8.125in (~20.64cm) square target, including its white border; the black square is 6.5in (~16.51cm).
- AprilTags are printed on a flat, matte material and mounted vertically, centered on the face they are on, facing into the FIELD.
- 4 AprilTags are on the SCOOPING STATIONS, 1 on each of the 2 outward-facing walls of each SCOOPING STATION. Their top edge is 1.0in (~2.5cm) below the top of the wall.
- 4 AprilTags are on the DELIVERY ZONE structures, 1 on each of the 2 outward-facing angled walls of each structure. Their top edge is 1.0in (~2.5cm) below the top of the wall.

Figure 4-16 *AprilTag locations*

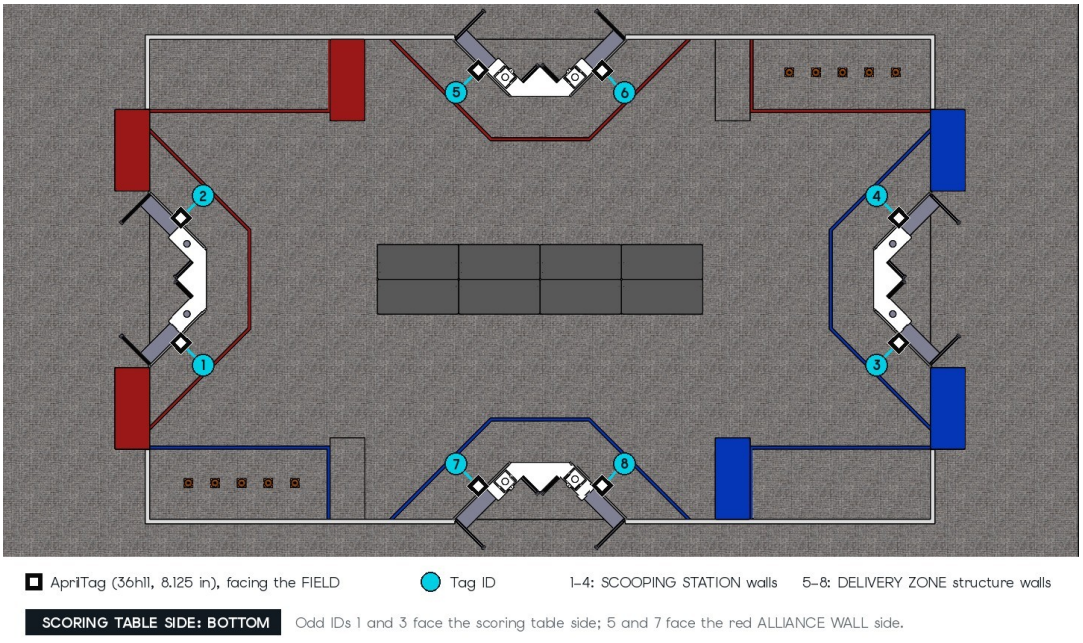



Table 4-2 *AprilTag IDs*

| ID | FIELD element | Face |
|---|---|---|
| 1 | red SCOOPING STATION | wall facing the scoring table side |
| 2 | red SCOOPING STATION | wall facing the far side |
| 3 | blue SCOOPING STATION | wall facing the scoring table side |

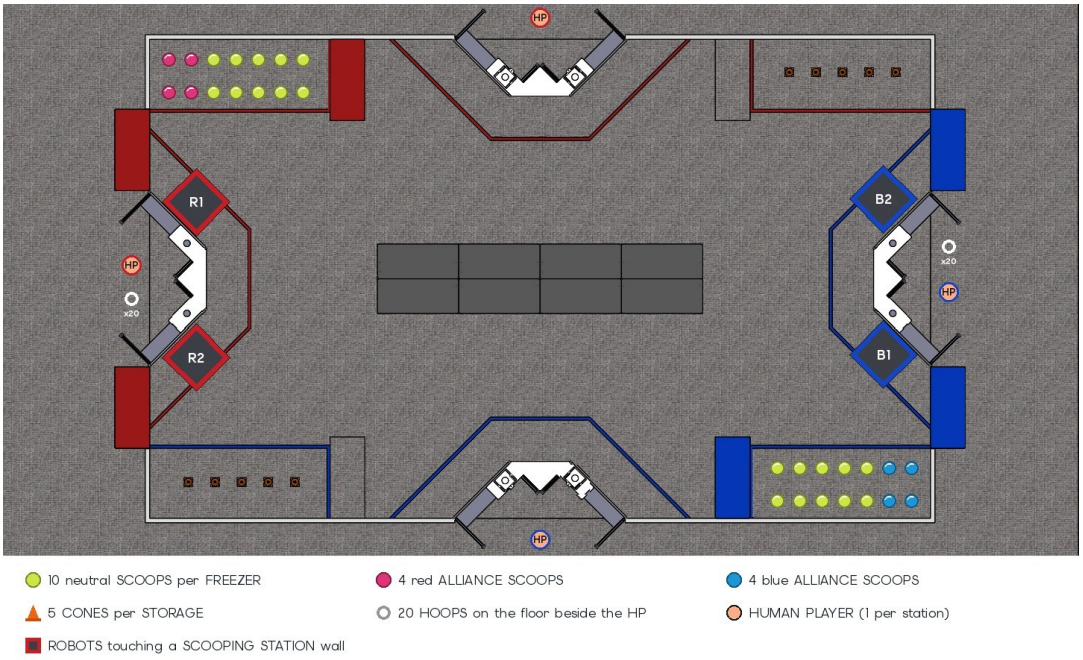
| ID | FIELD element | Face |
|---|---|---|
| 4 | blue SCOOPING STATION | wall facing the far side |
| 5 | red DELIVERY ZONE structure | wall on the red ALLIANCE WALL side |
| 6 | red DELIVERY ZONE structure | wall on the blue ALLIANCE WALL side |
| 7 | blue DELIVERY ZONE structure | wall on the red ALLIANCE WALL side |
| 8 | blue DELIVERY ZONE structure | wall on the blue ALLIANCE WALL side |

The exact position of every AprilTag is in the official CAD model and FIELD drawings. If AprilTags wear or get marked during MATCHES, they are repaired with gaffers tape.

> ROBOTS staged against a SCOOPING STATION wall (per G303) and ROBOTS working at a DELIVERY ZONE may cover AprilTags. Plan your vision for it.

# 05. GAME DETAILS

Figure 5-1 *FROZEN BLITZ at MATCH start (ROBOT positions are examples)*




In FROZEN BLITZ, 2 ALLIANCES (an ALLIANCE is a cooperative of up to 4 FIRST Robotics Competition teams) play MATCHES, set up and implemented per the details described below. Each ALLIANCE plays each MATCH with 2 ROBOTS.

## 5.1 MATCH Overview

Each MATCH cycle consists of pre-MATCH setup, the 2-minute and 30-second MATCH, and the post-MATCH reset.

During the MATCH, ROBOTS collect CONES and SCOOPS, build them into ICE CREAM at their SCOOPING STATION, deliver it to their DELIVERY ZONE, and refill their FREEZER. HUMAN PLAYERS place HOOPS at their SCOOPING STATION and, at their DELIVERY ZONE, clear each delivered ICE CREAM, recycle its SCOOPS through the CHUTE, and return its CONE to their STORAGE. ROBOTS conclude the MATCH in the CHECKOUT ZONE.

## 5.2 DRIVE TEAM

A DRIVE TEAM is a set of up to 5 people from the same FIRST Robotics Competition team responsible for team performance for a specific MATCH. There are 4 specific roles on a DRIVE TEAM which ALLIANCES can use to assist ROBOTS, and no more than 1 member of the DRIVE TEAM may be a non-STUDENT.

> The intent of the definition of DRIVE TEAM and DRIVE TEAM related rules is that, barring extenuating circumstances, the DRIVE TEAM consists of people who arrived at the event affiliated with that team and are responsible for their team's and ROBOT'S performance at the event (this means a person may be affiliated with more than 1 team).
>
> The intent is not to allow teams to "adopt" members of other teams for strategic advantage for the loaning team, borrowing team, and/or their ALLIANCE (e.g. an ALLIANCE CAPTAIN believes 1 of their DRIVERS has more experience than a DRIVER on their first pick, and the teams agree the first pick team will "adopt" that DRIVER and make them a member of their DRIVE TEAM for Playoffs).
>
> The definition isn't stricter for 2 main reasons. First, to avoid additional bureaucratic burden on teams and event volunteers. Second, to provide space for exceptional circumstances that give teams the opportunity to display Gracious Professionalism (e.g. a bus is delayed, a DRIVE COACH has no DRIVERS, and their pit neighbors agree to help by loaning DRIVERS as temporary members of the team until their bus arrives).

Table 5-1 *DRIVE TEAM roles*

| Role | Description | Max./DRIVE TEAM | Criteria |
|---|---|---|---|
| DRIVE COACH | a guide or advisor | 1 | any team member, must wear "DRIVE COACH" button |
| TECHNICIAN | a resource for ROBOT troubleshooting, setup, and removal from the FIELD | 1 | any team member, must wear "TECHNICIAN" button |
| DRIVER | an operator and controller of the ROBOT | 3 (combined) | STUDENT, must wear a "DRIVE TEAM" button |
| HUMAN PLAYER | a SCORING ELEMENT manager | | |

A STUDENT is a person who has not completed high-school, secondary school, or the comparable level as of September 1 prior to Kickoff.

The TECHNICIAN provides teams with a technical resource for pre-MATCH setup, ROBOT connectivity, OPERATOR CONSOLE troubleshooting, and post-MATCH removal of the ROBOT. Some pre-MATCH responsibilities for the TECHNICIAN may include, but are not limited to:

- location of the ROBOT radio, its power connection, and understanding of its indicator lights
- location of the roboRIO and understanding of its indicator lights
- username and password for the OPERATOR CONSOLE
- restarting the Driver Station and Dashboard software on the OPERATOR CONSOLE
- changing the bandwidth utilization (e.g. camera resolution, frame rate, etc.)

- changing a battery
- charging pneumatics

> While the TECHNICIAN may be the primary technical member of the DRIVE TEAM, all members of the DRIVE TEAM are encouraged to have knowledge of the basic functionality of the ROBOT, such as the location and operation of the main circuit breaker, connecting and resetting joysticks or gamepads from the OPERATOR CONSOLE, and removing the ROBOT from the FIELD.

Each ALLIANCE needs 2 HUMAN PLAYERS during a MATCH: 1 at its SCOOPING STATION, who places HOOPS, and 1 at its DELIVERY ZONE, who runs the delivery cycle (lowering the DELIVERY TRAYS, recycling SCOOPS through the CHUTE, and returning CONES to the STORAGE). Per Table 5-1, each team is allotted up to 3 STUDENTS who may be assigned as DRIVERS or HUMAN PLAYERS. If an ALLIANCE does not have at least 2 HUMAN PLAYERS available, 1 of the ALLIANCE'S teams may substitute a STUDENT TECHNICIAN as a HUMAN PLAYER for that MATCH only. In this case,

- the Head REFEREE must be notified,
- all HUMAN PLAYER rules now apply to this DRIVE TEAM member, and
- this DRIVE TEAM member is no longer considered a TECHNICIAN for that MATCH.

## 5.3 Setup

Before each MATCH begins, FIELD STAFF stage SCORING ELEMENTS as described in section 5.3.4 SCORING ELEMENTS. DRIVE TEAMS stage their ROBOTS (as described in section 5.3.3 ROBOTS) and OPERATOR CONSOLES (as described in section 5.3.2 OPERATOR CONSOLES). Then, DRIVE TEAMS take their places as described in section 5.3.1 DRIVE TEAMS.

### 5.3.1 DRIVE TEAMS

DRIVE TEAMS prepare for a MATCH by staging in the appropriate areas, according to their role on the DRIVE TEAM, and by identifying themselves accordingly. DRIVE TEAM starting conditions are listed below, and a DRIVE TEAM obstructing or delaying any of the conditions is at risk of violating G301.

A. only DRIVE TEAM members assigned to the upcoming MATCH are present,
B. for Qualification MATCHES only DRIVE TEAM members whose ROBOTS have passed initial, complete inspection are present,
C. exactly 1 HUMAN PLAYER per ALLIANCE is staged at each of their SCOOPING STATION and DELIVERY ZONE,
D. DRIVERS, DRIVE COACHES, and any remaining HUMAN PLAYERS are in their ALLIANCE AREA and behind the STARTING LINE,
E. TECHNICIANS are in the event-designated area near the FIELD,
F. DRIVE TEAM members clearly display their designated buttons above their waists (DRIVE COACH - "Drive Coach", DRIVERS and HUMAN PLAYERS - "Drive Team", and TECHNICIAN - "Technician"), and
G. if a Playoff MATCH, the ALLIANCE CAPTAIN clearly displays their designated ALLIANCE CAPTAIN identifier (e.g. hat or armband).

### 5.3.2 OPERATOR CONSOLES

DRIVE TEAMS set up their OPERATOR CONSOLE as soon as the DRIVE TEAM from the previous MATCH has cleared the area. OPERATOR CONSOLES must be compliant with all relevant rules, specifically those in section 7.9 OPERATOR CONSOLE. The OPERATOR CONSOLE is plugged into the team's assigned DRIVER STATION, as indicated on the team sign. Any control devices worn or held by their HUMAN PLAYERS and/or DRIVERS during the MATCH must be disconnected from or set on or beside the OPERATOR CONSOLE before the MATCH can begin. A DRIVE TEAM obstructing or delaying OPERATOR CONSOLE setup is at risk of violating G301.

> For the purposes of FIRST Robotics Competition, any device connected to the OPERATOR CONSOLE is considered a control device because REFEREES are not expected to differentiate between devices that can or cannot control the ROBOT.

### 5.3.3 ROBOTS

DRIVE TEAMS stage their ROBOT in accordance with G303. A DRIVE TEAM obstructing or delaying ROBOT setup requirements is at risk of violating G301.

If order of placement matters to either or both ALLIANCES, the ALLIANCE notifies the Head REFEREE during setup for that MATCH, and the Head REFEREE instructs ALLIANCES to alternate placement of ROBOTS. In a Qualification MATCH, ROBOTS are placed in the following order:

1. red DRIVER STATION 1 ROBOT
2. blue DRIVER STATION 1 ROBOT
3. red DRIVER STATION 2 ROBOT
4. blue DRIVER STATION 2 ROBOT

In a Playoff MATCH, the same pattern is applied, but instead of blue ALLIANCE placing last, the higher seeded ALLIANCE (regardless of color) places last.

> FIELD STAFF may ask teams to indicate their intended location and are not required to wait for a team to stage their ROBOT in its exact location before moving to the next team.

### 5.3.4 SCORING ELEMENTS

38 SCORING ELEMENTS, 28 SCOOPS (4 red, 4 blue, and 20 neutral) and 10 CONES, are staged for each MATCH as follows:

A. 10 neutral SCOOPS and the 4 SCOOPS of the corresponding ALLIANCE color are staged in each FREEZER,

B. 5 CONES are centered in each ALLIANCE'S STORAGE, and

C. each DRIVE TEAM may preload 1 SCOOP or 1 CONE, taken from their ALLIANCE'S FREEZER or STORAGE, such that it is fully supported by their ROBOT. Any SCORING ELEMENT not preloaded remains staged in its FREEZER or STORAGE.

In addition, 20 HOOPS are staged at each SCOOPING STATION, on the floor beside its HUMAN PLAYER.

## 5.4 MATCH Periods

The first period of each MATCH is 15 seconds long and called the Autonomous Period (AUTO). During AUTO, ROBOTS operate without any DRIVE TEAM control or input. ROBOTS attempt to reach their ALLIANCE'S FREEZER, place CONES, and stack SCOOPS.

The second period of each MATCH is the remaining 2 minutes and 15 seconds (2:15) and called the Teleoperated Period (TELEOP). During TELEOP, DRIVERS remotely operate ROBOTS to retrieve and place CONES in the SCOOPING ZONE, stack SCOOPS at the SCOOPING STATION, deliver ICE CREAM to the DELIVERY ZONE, refill the FREEZER with SCOOPS, and engage with the CHECKOUT ZONE. The final 30 seconds of TELEOP are called END GAME.

Table 5-2 *MATCH periods*

| MATCH Period | MATCH Timeframe | Duration | Timer Values |
|---|---|---|---|
| AUTO | AUTO | 15 seconds | 0:15 – 0:00 |
| TELEOP | TELEOP | 2 minutes 15 seconds | 2:15 – 0:00 |
| | END GAME | 30 seconds | 0:30 – 0:00 |

## 5.5 Scoring

ALLIANCES are rewarded for accomplishing various actions throughout a MATCH, including placing CONES, stacking SCOOPS, delivering ICE CREAM, refilling their FREEZER, checking out, and winning or tying MATCHES.

Rewards are granted either via MATCH points or Ranking Points (often abbreviated to RP, which increase the measure used to rank teams in the Qualification Tournament).

Scores are assessed at different moments of the MATCH:

- **REACH** is assessed throughout AUTO.
- **AUTO CONE and SCOOP** points are assessed when the ARENA timer displays 0:00 following AUTO. Only what is still placed or stacked at that moment counts.
- **ICE CREAM** and **TOPPER** points are assessed at the moment of delivery and are locked in from then on. Lowering the tray, removing the ICE CREAM, and recycling its pieces do not remove those points.
- **HANDING** points are assessed throughout TELEOP.
- **REFILL** and **CHECKOUT** points are assessed 3 seconds after the ARENA timer displays 0:00 following TELEOP, or when all ROBOTS have come to rest following the conclusion of the MATCH, whichever happens first.

> Stability and CHECKOUT criteria are evaluated by human volunteers. Teams are encouraged to make sure that it is obvious and unambiguous that the criteria are met.

### 5.5.1 SCORING ELEMENT Scoring Criteria

- A CONE is **placed** once it is in the SCOOPING ZONE and no longer CONTROLLED by a ROBOT.

- A SCOOP is **stacked** once it rests on top of a CONE, or on a HOOP placed on another SCOOP, at the SCOOPING STATION and remains stable for 2 seconds.
- A CONE with 1 or more SCOOPS stacked on it is a **stack**. While the tip of its CONE has not completely left the SCOOPING STATION, the CONE and each SCOOP in a stack are individual SCORING ELEMENTS (see G408).
- Once the tip of its CONE has completely left the SCOOPING STATION, a stack becomes an **ICE CREAM**, a single SCORING ELEMENT. The TIER of an ICE CREAM is the number of SCOOPS it holds (TIER 1 to TIER 5); HOOPS do not count.
- An ICE CREAM is **delivered** once it is fully supported by a raised DELIVERY TRAY in its ALLIANCE'S DELIVERY ZONE, is no longer CONTROLLED by a ROBOT, and remains stable for 3 seconds before the HUMAN PLAYER lowers the tray. If the tray is lowered sooner, the ICE CREAM is not delivered.
- A **TOPPER** is the top SCOOP of a delivered ICE CREAM of TIER 2 or higher when that SCOOP is the delivering ALLIANCE'S color. TOPPER points are added to the points of the ICE CREAM it tops.
- A SCOOP is a **REFILL** if, when REFILL points are assessed at the end of the MATCH, it is inside its ALLIANCE'S FREEZER, is neutral or of that ALLIANCE'S color, and is not in contact with a ROBOT. An ALLIANCE cannot earn more REFILLS than the number of SCOOPS it delivered in ICE CREAM during that MATCH. If this cap applies, ALLIANCE color SCOOPS are counted first.
- A CONE is **HANDED** once it is delivered to the SAFE ZONE of its ALLIANCE'S SCOOPING STATION after having contacted the floor at least momentarily. A CONE can be HANDED only once until a HUMAN PLAYER returns it to a STORAGE as part of the delivery cycle.

> TOPPER points stack on top of ICE CREAM points. A TIER 5 ICE CREAM is worth 10 points; the same ICE CREAM with a TOPPER is worth 30. Each ALLIANCE has only 4 SCOOPS of its color.

### 5.5.2 ROBOT Scoring Criteria

- **REACH:** a ROBOT whose BUMPERS have partially entered their ALLIANCE'S FREEZER at any point during AUTO.
- **CHECKOUT (in):** a ROBOT whose BUMPERS are completely within its ALLIANCE'S CHECKOUT ZONE.
- **CHECKOUT (on top):** a ROBOT whose BUMPERS are completely above its ALLIANCE'S CHECKOUT ZONE tables and which is fully supported by them. A ROBOT earns only 1 CHECKOUT award, the higher one.

### 5.5.3 Point Values

Point values for tasks in FROZEN BLITZ are detailed in Table 5-3.

Table 5-3 *FROZEN BLITZ point values*

| Award | Awarded for... | MATCH points | | Ranking Points |
|---|---|---|---|---|
| | | AUTO | TELEOP | |
| REACH | each ROBOT that achieves REACH | 3 | | |
| REACH BONUS | both ALLIANCE ROBOTS achieve REACH | 4 | | |
| CONE | each CONE placed in the SCOOPING ZONE at the end of AUTO | 2 | | |

| Award | Awarded for... | MATCH points | | Ranking Points |
|---|---|---|---|---|
| | | AUTO | TELEOP | |
| SCOOP | each SCOOP stacked on top of a CONE or another SCOOP at the end of AUTO | 4 | | |
| ICE CREAM | TIER 1: 1 SCOOP on a CONE, delivered | | 2 | |
| | TIER 2: 2 SCOOPS on a CONE, delivered | | 4 | |
| | TIER 3: 3 SCOOPS on a CONE, delivered | | 6 | |
| | TIER 4: 4 SCOOPS on a CONE, delivered | | 8 | |
| | TIER 5: 5 SCOOPS on a CONE, delivered | | 10 | |
| TOPPER | on a TIER 2 ICE CREAM | | 5 | |
| | on a TIER 3 ICE CREAM | | 10 | |
| | on a TIER 4 ICE CREAM | | 15 | |
| | on a TIER 5 ICE CREAM | | 20 | |
| REFILL | each neutral SCOOP in the ALLIANCE'S FREEZER at the end of the MATCH (capped, see 5.5.1) | | 1 | |
| COLOR REFILL | each ALLIANCE color SCOOP in that ALLIANCE'S FREEZER at the end of the MATCH (capped, see 5.5.1) | | 2 | |
| HANDING | each CONE HANDED (once per trip out of STORAGE) | | 2 | |
| CHECKOUT | each ROBOT completely within the CHECKOUT ZONE | | 5 | |
| | each ROBOT completely above the CHECKOUT ZONE and fully supported by it | | 15 | |
| ICE CREAM BONUS | 9 or more SCOOPS delivered in ICE CREAM. If the REACH BONUS was achieved, the threshold drops to 8. | | | 1 |
| CHECKOUT BONUS | ALLIANCE is credited with at least 20 CHECKOUT points (e.g. 1 ROBOT on top and 1 ROBOT in) | | | 1 |
| Win | completing a MATCH with more MATCH points than your opponent | | | 3 |
| Tie | completing a MATCH with the same number of MATCH points as your opponent | | | 1 |

An ALLIANCE can earn up to 5 Ranking Points per Qualification MATCH as described in Table 5-3. There are no Ranking Points or comparable points for the ICE CREAM BONUS or the CHECKOUT BONUS in Playoff MATCHES.

## 5.6 Violations

Unless otherwise noted, all violations are assigned for each instance of a rule violation. A description of the penalties is listed in Table 5-4. All rules throughout the Game Rules section are called as perceived by a REFEREE.

Table 5-4 *Rule violations*

| Penalty | Description |
|---|---|
| **MINOR FOUL** | a credit of 4 points towards the opponent's MATCH point total |
| **MAJOR FOUL** | a credit of 8 points towards the opponent's MATCH point total |
| **YELLOW CARD** | issued by the Head REFEREE for egregious ROBOT or team member behavior or rule violations. A subsequent YELLOW CARD within the same tournament phase results in a RED CARD. |
| **RED CARD** | issued by the Head REFEREE for egregious ROBOT or team member behavior or rule violations which results in a team being DISQUALIFIED for the MATCH. |
| **DISABLED** | the state in which a ROBOT is commanded to deactivate all outputs, rendering the ROBOT inoperable for the remainder of the MATCH. |
| **DISQUALIFIED** | the state of a team in which they receive 0 MATCH points and 0 Ranking Points in a Qualification MATCH or causes their ALLIANCE to receive 0 MATCH points in a Playoff MATCH |
| **VERBAL WARNING** | a warning issued to a team by event staff or the Head REFEREE. |
| **ALLIANCE is ineligible for RP** | an ALLIANCE is ineligible for the specified RP for that MATCH. This overrides any RP awarded through normal MATCH play or other rule violations. |

### 5.6.1 YELLOW and RED CARDS

In addition to rule violations explicitly listed throughout this document, YELLOW CARDS and RED CARDS are used in FIRST Robotics Competition to address team and ROBOT behavior that does not align with the mission, values, and culture of FIRST.

The Head REFEREE may assign a YELLOW CARD as a warning, or a RED CARD for egregious behavior inappropriate at a FIRST Robotics Competition event. If an action resulting in the assignment of a YELLOW or RED CARD is determined to be the result of an ARENA FAULT, the CARD will be rescinded.

YELLOW CARDS are additive, meaning that a second YELLOW CARD is automatically converted to a RED CARD. A team is issued a RED CARD for any subsequent incident in which they receive an additional YELLOW CARD, including earning a second YELLOW CARD during a single MATCH. A team that has received either a YELLOW CARD or a RED CARD carries a YELLOW CARD into subsequent MATCHES, except as noted below.

All YELLOW CARDS and G301 VERBAL WARNINGS are cleared at the conclusion of Practice, Qualification, and Playoff MATCHES. Other VERBAL WARNINGS issued by the Head REFEREE are cleared after Practice MATCHES and persist from Qualification MATCHES through subsequent tournament phases.

### 5.6.2 YELLOW and RED CARD application

YELLOW and RED CARDS are applied based on the following:

Table 5-5 *YELLOW and RED CARD application*

| Time YELLOW or RED CARDS earned | MATCH to which CARD is applied |
|---|---|
| prior to Qualification MATCHES | REFEREES may or may not be present during Practice MATCHES. With input from event staff, the Head REFEREE may opt to perpetuate a VERBAL WARNING or YELLOW CARD earned prior to Qualification MATCHES to the first Qualification MATCH for particularly egregious behavior. |
| during the Qualification MATCHES | team's current (or just-completed) MATCH. In the case where the team participated as a SURROGATE in the current (or just completed) MATCH, the card is applied to the team's previous MATCH. |
| between the end of Qualification MATCHES and the start of Playoff MATCHES | ALLIANCE'S first Playoff MATCH |
| during the Playoff MATCHES | ALLIANCE'S current (or just-completed) MATCH |

> A MATCH is no longer the current MATCH once the results of the MATCH have been posted or the Head REFEREE or their designee has indicated that teams can collect their ROBOTS, whichever is later.

### 5.6.3 YELLOW and RED CARDS during Playoff MATCHES

During Playoff MATCHES, YELLOW and RED CARDS are assigned to the violating team's entire ALLIANCE instead of to only the violating team. If an ALLIANCE receives 2 YELLOW CARDS, the entire ALLIANCE is issued a RED CARD which results in DISQUALIFICATION for the associated MATCH. If both ALLIANCES receive a RED CARD, the ALLIANCE assessed the first RED CARD, chronologically, is DISQUALIFIED and loses the MATCH.

### 5.6.4 Violation Details

There are several styles of violation wording used in this manual. Below are some example violations and a clarification of the way the violation would be assessed. The examples shown do not represent all possible violations, but rather a representative set of combinations.

Table 5-6 *Violation examples*

| Example Violation | Expanded Interpretation |
|---|---|
| **MINOR FOUL** | Upon violation, a MINOR FOUL is assessed against the violating ALLIANCE. |
| **MAJOR FOUL and YELLOW CARD** | Upon violation, a MAJOR FOUL is assessed against the violating ALLIANCE. After the MATCH, the Head REFEREE presents the violating team with a YELLOW CARD. |

| Example Violation | Expanded Interpretation |
|---|---|
| **MINOR FOUL per additional SCORING ELEMENT. If strategic, YELLOW CARD** | Upon violation, a number of MINOR FOULS are assessed against the violating ALLIANCE equal to the number of additional SCORING ELEMENTS beyond the permitted quantity. Additionally, if the REFEREES determine that the action was strategic, the Head REFEREE presents the violating team with a YELLOW CARD after the MATCH. |
| **MINOR FOUL. MAJOR FOUL if REPEATED** | Upon initial violation in a MATCH, a MINOR FOUL is assessed against the violating ALLIANCE. If the ALLIANCE ROBOT repeats the infraction in the MATCH, a MAJOR FOUL is assessed against the violating ALLIANCE. |
| **MAJOR FOUL, and for every 3 seconds in which the situation is not corrected, a MAJOR FOUL is assessed** | Upon violation, a MAJOR FOUL is assessed against the violating ALLIANCE and the REFEREE begins to count. Their count continues until the criteria to discontinue the count are met, and for each 3 seconds within that time, an additional MAJOR FOUL is assessed against the violating ALLIANCE. A ROBOT in violation of this type of rule for 15 seconds receives a total of 6 MAJOR FOULS (assuming no other rules were being simultaneously violated). |
| **RED CARD for the ALLIANCE** | After the MATCH, the Head REFEREE presents the violating ALLIANCE with a RED CARD in the following fashion:<br>– In a Playoff MATCH, a single RED CARD is assessed to the ALLIANCE.<br>– In all other scenarios, each team on the ALLIANCE is issued a RED CARD. |

## 5.7 Head REFEREE and FTA Interaction

The Head REFEREE has the ultimate authority in the ARENA during the event, but may receive input from additional sources, e.g. the TideCrew Game Design Team, FTA, and other event staff. The Head REFEREE rulings are final. No event staff, including the Head REFEREE, will review video, photos, artistic renderings, etc. of any MATCH, from any source, under any circumstances.

### 5.7.1 Question Box (Q)

Each ALLIANCE has a designated Question Box near the scoring table. If a DRIVE TEAM has a question about a MATCH, the FIELD, etc., they may send up to 2 DRIVE TEAM members to their corresponding Question Box. Depending on timing, the Head REFEREE or FTA may postpone any requested discussion until the end of the subsequent MATCH.

Technical questions regarding FIELD or ROBOT operation are addressed by the FTA, and additional team members are invited to participate in these conversations if necessary. If a DRIVE TEAM needs clarification on a ruling or score, per Q101, up to 2 DRIVE TEAM members should address the Head REFEREE after the ARENA Reset Signal (e.g. FIELD lights turn green).

> Any reasonable question is fair game in the Question Box, and Head REFEREES will make good faith efforts to provide helpful feedback (e.g. how/why certain MINOR FOULS are being called, why a particular ROBOT may be susceptible to certain MINOR FOULS based on its design or game play, how specific rules are being called or interpreted), but please know that they may not be able to supply

specific details.

**Q101** ***Head REFEREE Interactions.** A team may only address the Head REFEREE with maximum of 2 people, 1 of which must be a STUDENT.

***Violation:** The Head REFEREE will not address non-compliant team members or peripheral conversations.*

> Team members are permitted to swap buttons within their team as needed to access the Question Box.
>
> As much as possible, the STUDENT should be an active participant in the conversation.

**Q102** **Keep it cool. It's an ice cream game.** Any team interaction with the Head REFEREE must be appropriate.

***Violation:** The Head REFEREE may terminate the conversation to allow time for parties to cool down. A person with subsequent violations will not be allowed to interact with the Head REFEREE.*

> Examples of inappropriate behavior are outlined in G201.

## 5.8 Other Logistics

SCORING ELEMENTS that leave the FIELD are placed back into the FIELD approximately at the point of exit by FIELD STAFF at the earliest safe opportunity. The exception is SCORING ELEMENTS that leave the FIELD into a DELIVERY ZONE station: the HUMAN PLAYER there returns SCOOPS through the CHUTE and CONES to the STORAGE, as in the delivery cycle (section 4.4). HOOPS that leave the FIELD stay out of play.

> Note that ROBOTS and HUMAN PLAYERS may not deliberately cause SCORING ELEMENTS to leave the FIELD (see G405).

An ARENA FAULT (an error in ARENA operation) is not called for MATCHES that accidentally begin with damaged SCORING ELEMENTS. Damaged SCORING ELEMENTS are not replaced until the next ARENA reset period. DRIVE TEAMS should alert the FIELD STAFF to any missing or damaged SCORING ELEMENTS prior to the start of the MATCH.

Once the MATCH is over and the Head REFEREE determines that the FIELD is safe for FIELD STAFF and DRIVE TEAMS, they or their designee change the FIELD lights to green and DRIVE TEAMS may retrieve their ROBOT.

In addition to the 2 minutes and 30 seconds (2:30) of game play, each MATCH also has pre- and post-MATCH time to reset the ARENA. During ARENA reset, the ARENA is cleared of ROBOTS and OPERATOR CONSOLES from the MATCH that just ended, ROBOTS and OPERATOR CONSOLES for the subsequent MATCH are loaded into the ARENA by DRIVE TEAMS, and FIELD STAFF reset ARENA elements.

FIRST Robotics Competition uses 3 words in the context of how durations and actions are assessed with regards to evaluation of rules and assignment of violations. These words provide general guidance to describe benchmarks to be used across the program. It is not the intent for REFEREES to provide a count during the time periods.

- **MOMENTARY** describes durations that are fewer than approximately 3 seconds.
- **CONTINUOUS** describes durations that are more than approximately 10 seconds.
- **REPEATED** describes actions that happen more than once within a MATCH.

# 06. GAME RULES (G)

FROZEN BLITZ game rules follow the 2026 FIRST Robotics Competition structure: personal safety, conduct, pre-MATCH, in-MATCH, and post-MATCH. Evergreen rules keep their 2026 wording; rules unique to FROZEN BLITZ follow them in each subsection.

## 6.1 Personal Safety

**G101** ***Humans, remain outside the FIELD.** A team member may not reach into the FIELD with any part of their body during a MATCH.

A HUMAN PLAYER placing a HOOP at their SCOOPING STATION per G434 is an exception to this rule.

***Violation:** VERBAL WARNING. YELLOW CARD if subsequent violations during the event.*

> Violations of this rule apply to the entire team, not specifically to any 1 individual. For example, a member of Team 9999 reaches onto the FIELD during MATCH 3, and a different member reaches onto the FIELD during MATCH 25. The team receives a VERBAL WARNING for the first violation and a YELLOW CARD for the second.
>
> HUMAN PLAYERS working at their SCOOPING STATION or DELIVERY ZONE are governed by G430, G434, and G435.

**G102** ***Never step over the guardrail.** A team member may only enter or exit the FIELD through open gates and only enter if FIELD lighting (FIELD facing side of the team signs and timers) is green, unless explicitly instructed by a REFEREE or an FTA.

***Violation:** VERBAL WARNING. YELLOW CARD if subsequent violations during the event.*

> Teams are encouraged to ensure that all members of their DRIVE TEAM are aware of this rule. It's easy to violate, particularly when teams are doing their best to move on and off the FIELD quickly. The violations of this rule are intended to avoid nuisance penalties, but still enforce safety requirements around the FIELD. There is the potential for injury when stepping over the guardrail.

**G103** ***Be careful what you interact with.** A team member is prohibited from the following actions with regards to interaction with ARENA elements.

A. climbing on or inside,

B. hanging from,

C. manipulating such that it doesn't return to its original shape without human intervention, and

D. damaging.

***Violation:*** *VERBAL WARNING. YELLOW CARD if subsequent violations during the event.*

> The CHECKOUT ZONE tables are FIELD elements built for ROBOTS, not people. Sitting or standing on them is a violation of this rule.

**G104** ***Teams may not enable their ROBOTS on the FIELD.** Teams may not tether to the ROBOT while on the FIELD except in special circumstances (e.g. after Opening Ceremonies, before an immediate MATCH replay, etc.) and with the express permission from the FTA or a REFEREE.

***Violation:*** *VERBAL WARNING. YELLOW CARD if subsequent violations during the event.*

> Teams are encouraged to consider this rule when developing their ROBOTS. FMS will not enable ROBOTS after the conclusion of the MATCH.
>
> For the purposes of this rule, tethering includes any wired or wireless connection used to electrically energize and/or control elements on the ROBOT. The safety of teams and volunteers in close proximity to ROBOTS and ARENA elements on the FIELD is of the utmost importance, therefore ROBOTS or ROBOT COMPONENTS may not be enabled in any way on the FIELD before or after the MATCH.
>
> ROBOTS need to be safely transported off the FIELD and back to the pits after the MATCH, and there may be bystanders, doorways, or height restrictions along the route.

## 6.2 Conduct

**G201** ***Be a good person.** All teams must be civil toward everyone and respectful of team and event equipment while at a FIRST Robotics Competition event.

***Violation:*** *VERBAL WARNING. YELLOW CARD if subsequent violations during the event.*

> Examples of inappropriate behavior include, but are not limited to, the use of offensive language or other uncivil conduct, misuse of venue furniture, and using a CONE as a projectile in an attempt to inhibit an opponent ROBOT.
>
> Examples of particularly contemptible behavior that is likely to result in ARENA ejection include, but are not limited to, the following:
>
> A. assault, e.g. throwing something that hits another person (even if unintended),
> B. threat, e.g. saying something like “if you don’t reverse that call, I’ll make you regret it,”
> C. harassment, e.g. badgering someone with no new information after a decision’s been made or a question’s been answered,
> D. bullying, e.g. using body or verbal language to cause another person to feel inadequate,
> E. insulting, e.g. telling someone they don’t deserve to be on a DRIVE TEAM,
> F. swearing at another person (versus swearing under one’s breath or at oneself), and
> G. yelling at another person(s) in anger or frustration.

**G202** ***Don't bang on the glass.** A team member may never strike or hit the DRIVER STATION plastic windows.

***Violation:*** *VERBAL WARNING. YELLOW CARD if subsequent violations during the event.*

> We know. Your ROBOT just stacked a TIER 5 ICE CREAM and the crowd went wild. Celebrate with your voice, not your fists: the glass doesn't pay for your broken wrist, and neither does the FIELD.

**G203** ***Asking other teams to throw a MATCH – not cool.** A team may not encourage an ALLIANCE of which it is not a member to play beneath its ability.

NOTE: This rule is not intended to prevent an ALLIANCE from planning and/or executing its own strategy in a specific MATCH in which all the teams are members of the ALLIANCE.

***Violation:*** *VERBAL WARNING. RED CARD if subsequent violations during the event.*

> Example 1: A MATCH is being played by Teams A and B, in which Team B is encouraged by Team D to not engage with the CHECKOUT ZONE at the end of the MATCH, resulting in Teams A and B not earning the CHECKOUT BONUS. Team D's motivation for this behavior is to prevent Team A from rising in the Tournament rankings and negatively affecting Team D's ranking. Team D has violated this rule.
>
> Example 2: A MATCH is being played by Teams A and B, in which Team A is assigned to participate as a SURROGATE. Team D encourages Team A to not participate in the MATCH so that Team D gains ranking position over Team B. Team D has violated this rule.
>
> Example 3: Asking a team to "no show" for a MATCH.
>
> FIRST considers the action of a team influencing another team to throw a MATCH, to deliberately miss Ranking Points, etc. incompatible with FIRST values and not a strategy any team should employ.

**G204** ***Letting someone coerce you into throwing a MATCH – also not cool.** A team, as the result of encouragement by a team not on their ALLIANCE, may not play beneath its ability.

NOTE: This rule is not intended to prevent an ALLIANCE from planning and/or executing its own strategy in a specific MATCH in which all the ALLIANCE members are participants.

***Violation:*** *VERBAL WARNING. RED CARD if subsequent violations during the event.*

> Example 1: A MATCH is being played by Teams A and B. Team D requests Team B ignore the CHECKOUT ZONE at the end of the MATCH, resulting in Teams A and B not being able to earn the CHECKOUT BONUS. Team B accepts this request from Team D. Team D's motivation for this behavior is to prevent Team A from rising in the Tournament rankings negatively affecting Team D's ranking. Team B has violated this rule.
>
> Example 2: A MATCH is being played by Teams A and B, in which Team A is assigned to participate as a SURROGATE. Team A accepts Team D's request not to participate in the MATCH so that Team D gains ranking position over Team B. Team A has violated this rule.

**G205** ***Throwing your own MATCH is bad.** A team may not intentionally lose a MATCH or sacrifice Ranking Points in an effort to lower their own ranking or manipulate the rankings of other teams.

***Violation:*** *VERBAL WARNING. RED CARD if subsequent violations during the event.*

> The intent of this rule is not to punish teams who are employing alternate strategies, but rather to ensure that it is clear that throwing MATCHES to negatively affect your own rankings, or to manipulate the rankings of other teams (i.e. throw a MATCH to lower a partner's ranking, and/or increase the ranking of another team not in the MATCH) is incompatible with FIRST values and not a strategy any team should employ.

**G206** ***Don't violate rules for RPs.** A team or ALLIANCE may not collude with another team to each purposefully violate a rule in an attempt to influence Ranking Points.

***Violation:*** *YELLOW CARD and the ALLIANCE is ineligible for the ICE CREAM BONUS and CHECKOUT BONUS RPs.*

> For example, if Team A on the blue ALLIANCE agrees with Team F on the red ALLIANCE to contact each other in the CHECKOUT ZONE during END GAME so that both ALLIANCES are awarded CHECKOUT points through G424.

**G207** ***Don't abuse ARENA access.** A team member (except DRIVERS, HUMAN PLAYERS, and DRIVE COACHES) granted access to restricted areas in and around the ARENA (e.g. via TECHNICIAN button, event issued Media badges, etc.) may not assist or use signaling devices during the MATCH. Exceptions will be granted for inconsequential infractions and in cases concerning safety.

***Violation:*** *VERBAL WARNING. YELLOW CARD if subsequent violations during the event.*

> The TECHNICIAN'S role is to help the team prepare the ROBOT so it can perform at its full potential during a MATCH. The TECHNICIAN, except as described at the end of section 5.2 DRIVE TEAM, is not an additional DRIVE COACH, DRIVER, or HUMAN PLAYER.
>
> Team members in open-access spectator seating areas are not considered to be in a restricted area and are not prevented from assisting or using signaling devices.

**G208** ***Show up to your Qualification MATCHES.** If a ROBOT has passed initial, complete inspection, at least 1 member of its DRIVE TEAM must report to the ARENA and participate in each of their assigned Qualification MATCHES.

***Violation:*** *DISQUALIFIED from the current MATCH.*

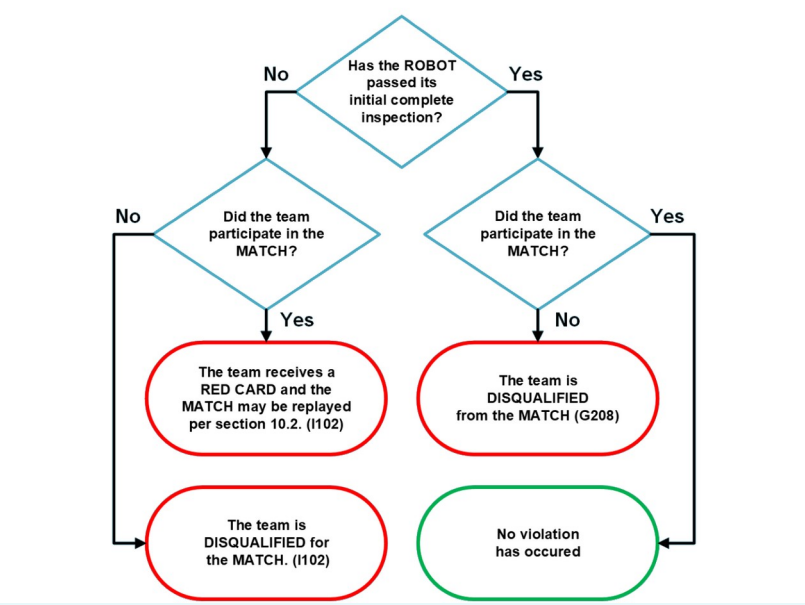



*Figure 6-1 MATCH participation flowchart*

The team should inform the Lead Queuer if the team's ROBOT is not able to participate.

**G209** ***Keep your ROBOT together.** A ROBOT may not intentionally detach or leave a part on the FIELD.

***Violation:*** *RED CARD.*

**G210** ***Don't expect to gain by doing others harm.** A strategy not consistent with standard gameplay and clearly aimed at forcing the opponent ALLIANCE to violate a rule is not in the spirit of FIRST Robotics Competition and not allowed. Rule violations forced in this manner will not result in an assignment of a penalty to the targeted ALLIANCE.

***Violation:*** *MAJOR FOUL. YELLOW CARD if REPEATED.*

This rule does not apply for strategies consistent with standard gameplay, for example:

A. a red ALLIANCE ROBOT in their CHECKOUT ZONE during END GAME contacts a blue ALLIANCE ROBOT.

This rule requires an intentional act with limited or no opportunity for the team being acted on to avoid the penalty, such as:

A. forcing an opposing ROBOT to have greater-than-MOMENTARY CONTROL of 2 SCORING ELEMENTS (see G407), or

B. forcing an opposing ROBOT against a FIELD element with the intent of breaking down an ICE CREAM.

**G211** ***Egregious or exceptional violations.** Egregious behavior beyond what is listed in the rules or subsequent violations of any rule or procedure during the event is prohibited.

In addition to rule violations explicitly listed in this manual and witnessed by a REFEREE, the Head REFEREE may assign a YELLOW or RED CARD for egregious ROBOT actions or team member behavior at any time during the event.

Intentionally violating a rule that only results in a YELLOW CARD in an ALLIANCE'S potential last MATCH (such as a team's last Qualification MATCH or a Playoff MATCH that could eliminate the ALLIANCE) will be under increased scrutiny and will likely result in a RED CARD.

Please see section 5.6.1 YELLOW and RED CARDS for additional detail.

***Violation:*** *YELLOW or RED CARD.*

> The intent of this rule is to provide the Head REFEREES the flexibility necessary to keep the event running smoothly, as well as keep the safety of all the participants as the highest priority. Behaviors that put the FIRST community or integrity of the game at risk are not allowed and are violations of this rule. Those behaviors include, but are not limited to the list below:
>
> A. inappropriate behavior as outlined in the blue box of G201,
> B. jumping over the guardrail,
> C. pushing past the FIELD reset person blocking an open gate to get on the FIELD,
> D. reaching into the FIELD and grabbing a ROBOT during a MATCH,
> E. a single PIN in excess of 15 seconds,
> F. exploiting the 3-second window after a MATCH described in section 5.5 Scoring to avoid rule violations (e.g. triggering an over-extension that enables CHECKOUT points),
> G. interfering with FMS or FIELD operation,
> H. intentionally crossing the CENTER LINE in AUTO and contacting and/or blocking an opponent ROBOT,
> I. intentionally ejecting a large quantity of SCORING ELEMENTS out of the FIELD,
> J. intentionally exceeding the expansion limits for strategic benefit, and
> K. climbing on the CHECKOUT ZONE tables or DELIVERY ZONE structures (team members).
>
> The Head REFEREE may assign a YELLOW or RED CARD for a single instance of a rule violation such as the examples given in items above, or for multiple instances of any single rule violation. Teams should be aware that any rule in this manual could escalate to a YELLOW or RED CARD. The Head REFEREE has final authority on all rules and violations at an event.

**G212** ***All teams can play.** A team may not encourage another team to exclude their ROBOT or be BYPASSED from a Qualification MATCH for any reason.

***Violation:*** *YELLOW CARD. RED CARD if the ROBOT does not participate in the MATCH.*

## 6.3 Pre-MATCH

**G301** ***Be prompt.** A DRIVE TEAM member may not cause significant delays to the start of their MATCH. Causing a significant delay requires both of the following to be true:

A. The expected MATCH start time has passed, and

B. The DRIVE TEAM has access to the FIELD and is neither MATCH ready nor making a good faith effort, to quickly become MATCH ready.

***Violation:** If a Qualification MATCH, VERBAL WARNING. MAJOR FOUL for the upcoming MATCH if a subsequent violation occurs within the tournament phase. If the DRIVE TEAM is not MATCH ready within 2 minutes of the VERBAL WARNING/MAJOR FOUL and the Head REFEREE perceives no good faith effort by the DRIVE TEAM to quickly become MATCH ready, DISABLED. If a Playoff MATCH, VERBAL WARNING issued to the ALLIANCE. MAJOR FOUL for the ALLIANCE'S upcoming MATCH if a subsequent violation occurs within the tournament phase. If the ALLIANCE is not MATCH ready within 2 minutes of the VERBAL WARNING/MAJOR FOUL and the Head REFEREE perceives no good faith effort by the DRIVE TEAM(S) to quickly become MATCH ready, the offending team's ROBOT is DISABLED.*

> Event volunteers communicate schedule delays with teams to the best of their ability. Announcements on the FIELD and in the pits also provide information on delays, and any team uncertain of when to queue for a MATCH should communicate with queuing volunteers.
>
> Teams that have violated G208 or have 1 DRIVE TEAM member present and have informed event staff that their ROBOT will not be participating in the MATCH are considered MATCH ready and not in violation of this rule.
>
> Being "MATCH ready" requires that the ROBOT is on the FIELD, in its STARTING CONFIGURATION, and turned on. Additionally, the DRIVE TEAM members must be in their starting positions.
>
> Examples of good faith efforts to quickly become MATCH ready include but are not limited to:
>
> A. walking safely towards the FIELD with a ROBOT that a team is not actively modifying,
>
> B. applying quick fixes such as tape or cable ties to make the ROBOT compliant with STARTING CONFIGURATION requirements,
>
> C. waiting for an OPERATOR CONSOLE computer to boot, and
>
> D. working with FIELD STAFF to get the ROBOT connected to the FIELD.
>
> Examples that are not considered good faith efforts include but are not limited to: a ROBOT not moving to the FIELD, a ROBOT being actively modified on its way to the FIELD, a DRIVE TEAM member remaining on the FIELD once a MATCH is ready to begin, installing BUMPERS or charging pneumatic systems once on the FIELD, time-consuming use of alignment devices external to the ROBOT, and repairs more substantial than quick fixes.

**G302** ***Limit what you use during a MATCH.** Items used during a MATCH must fit on your team's DRIVER STATION shelf, be worn or held by members from your DRIVE TEAM, or be an item used as an accommodation (e.g. stools, crutches, etc.). Regardless of if the equipment fits the criteria above, it may not:

A. be employed in a way that introduces a safety hazard,

B. extend more than 78.0in (1.981m) above the floor,

C. communicate with anything or anyone outside of the ARENA with the exception of medically required equipment,

D. block visibility for FIELD STAFF or audience members, or

E. jam or interfere with the remote sensing capabilities of another team or the FIELD.

Exceptions to part B are granted for MOMENTARY extensions above 78.0in (1.981m) and for individuals above 78.0in (1.981m) wearing PPE or reasonable decorative apparel such as hats, headbands, etc.

***Violation:*** *MATCH will not start until the situation is remedied. If discovered or used inappropriately during a MATCH, YELLOW CARD.*

> Examples of equipment that may be considered a safety hazard in the confined space of the ALLIANCE AREA include, but are not limited to, a folding step stool, ladder, or a large signaling device.
>
> Using an item that has wireless communications disabled complies with C above.
>
> Examples of jamming or interfering with remote sensing capabilities include, but are not limited to, mimicking FIELD vision targets and shining bright lighting or laser pointers onto the FIELD.

**G303** ***Start your ROBOTS.** A ROBOT must meet all following MATCH-start requirements:

A. it does not pose a hazard to humans, FIELD elements, or other ROBOTS,

B. has passed initial, complete inspection, i.e. it's compliant with all ROBOT rules (for exceptions regarding Practice MATCHES, see section 8 Inspection & Eligibility (I)),

C. if modified after initial inspection, it's compliant with I104,

D. it is positioned such that one of its sides is parallel to, and its BUMPERS fully contact, 1 of its ALLIANCE'S SCOOPING STATION walls,

E. it's the only team-provided item left on the FIELD,

F. it's not attached to, entangled with, or suspended from any FIELD element,

G. it's confined to its STARTING CONFIGURATION (reference R102 and R104), and

H. it fully and solely supports not more than 1 SCORING ELEMENT (as described in section 5.3.4 SCORING ELEMENTS).

***Violation:*** *If fix is a quick remedy, the MATCH won't start until all requirements are met. If it is not a quick remedy, DISABLED and, at the discretion of the Head REFEREE, must be re-inspected. If a ROBOT not compliant with part B or C participates, its team receives a RED CARD.*

> If a ROBOT is BYPASSED prior to the start of the MATCH, the DRIVE TEAM may not remove the ROBOT from the FIELD without permission from the Head REFEREE or the FTA.
>
> For assessment of many of the items listed above, the Head REFEREE is likely to consult with the LRI.

## 6.4 In-MATCH

Rules in this section pertain to game play once a MATCH begins.

### 6.4.1 AUTO

The AUTO period is the first 15 seconds of the MATCH, and the FMS blocks any DRIVER control, so ROBOTS operate with only their pre-programmed instructions. Rules in this section apply only during the AUTO period.

**G401** ***Behind the lines.** In AUTO, each DRIVE TEAM member must remain in their staged areas. A DRIVE TEAM member staged behind the STARTING LINE may not contact anything in front of the STARTING LINE, unless for personal or equipment safety, to press the E-Stop, or granted permission by a Head REFEREE or FTA.

***Violation:** MINOR FOUL regardless of the number of items contacted.*

> Pointing, gesturing, or otherwise extending across the STARTING LINE such that contact is not made with carpet or other ARENA elements is not a violation of this rule.
>
> An example of an exception for equipment safety is if an OPERATOR CONSOLE starts to fall from, or has already fallen off of, the DRIVER STATION shelf. In that circumstance, DRIVE TEAM members may step forward to catch it or pick it up off the ground and return it to the shelf.

**G402** ***Let the ROBOT do its thing.** In AUTO, a DRIVE TEAM member may not directly or indirectly interact with a ROBOT or an OPERATOR CONSOLE unless for personal safety, OPERATOR CONSOLE safety, or pressing an E-Stop. A HUMAN PLAYER placing HOOPS at their SCOOPING STATION or performing their role at their DELIVERY ZONE is an exception to this rule.

***Violation:** MINOR FOUL and YELLOW CARD.*

**G403** **The shop isn't open yet.** In AUTO, a ROBOT may not cross the FIELD such that its BUMPERS break the plane defined by the CENTER LINE.

***Violation:** MAJOR FOUL.*

> FROZEN BLITZ is stricter than the 2026 FIRST rule: in AUTO, ROBOTS may not even partially cross the CENTER LINE. The whole first 15 seconds belong to your own side of the shop.

### 6.4.2 SCORING ELEMENTS

**G404** ***ROBOTS: use SCORING ELEMENTS as directed.** A ROBOT may not deliberately use a SCORING ELEMENT in an attempt to ease or amplify a challenge associated with a FIELD element.

***Violation:** MAJOR FOUL per SCORING ELEMENT.*

> Examples include, but are not limited to:
>
> A. placing SCORING ELEMENTS onto another ROBOT, and
>
> B. placing SCORING ELEMENTS on the CHECKOUT ZONE tables to prevent other ROBOTS

> from getting on top.

**G405** ***Keep SCORING ELEMENTS in bounds.** A ROBOT may not intentionally eject SCORING ELEMENTS from the FIELD (either directly or by bouncing off a FIELD element or other ROBOT).

***Violation:** MINOR FOUL. If REPEATED, MAJOR FOUL.*

**G406** ***Don't abuse SCORING ELEMENTS.** Neither a ROBOT nor a HUMAN PLAYER may damage a SCORING ELEMENT.

***Violation:** VERBAL WARNING. MAJOR FOUL if REPEATED in any subsequent MATCHES during the event. If via a ROBOT and the Head REFEREE determines that further damage is likely to occur, DISABLED. Corrective action (such as eliminating sharp edges, removing the damaging MECHANISM, and/or re-inspection) may be required before the ROBOT may compete in subsequent MATCHES.*

> SCORING ELEMENTS are expected to undergo a reasonable amount of wear and tear as they are handled by ROBOTS, such as scratching or marking. Gouging, tearing off pieces, deforming a CONE, or routinely marking SCORING ELEMENTS are violations of this rule.

**G407** **One SCORING ELEMENT at a time.** A ROBOT may not have greater-than-MOMENTARY CONTROL of more than 1 SCORING ELEMENT (CONE, SCOOP, or ICE CREAM) at a time, either directly or transitively through other objects.

A ROBOT is in CONTROL of a SCORING ELEMENT if:

A. the SCORING ELEMENT is fully supported by the ROBOT,
B. the SCORING ELEMENT travels across the FIELD such that when the ROBOT changes direction, the SCORING ELEMENT travels with the ROBOT,
C. the ROBOT is holding a SCORING ELEMENT against a FIELD element in an attempt to guard or shield it, or
D. the ROBOT is preventing a SCORING ELEMENT from leaving the DELIVERY ZONE.

***Violation:** MINOR FOUL per additional SCORING ELEMENT. If strategic, YELLOW CARD.*

> An ICE CREAM counts as 1 SCORING ELEMENT: having CONTROL of an ICE CREAM does not count as having CONTROL of more than 1 SCORING ELEMENT, regardless of its TIER. A stack that has not yet become an ICE CREAM does not get this treatment (see G408).

**G408** **It's not ice cream until it leaves the shop.** A stack is not an ICE CREAM until the tip of its CONE has completely left the SCOOPING STATION. Until then, the CONE and each SCOOP in the stack are individual SCORING ELEMENTS for all rules in this manual.

**G409** **Don't hoard opposing ALLIANCE SCOOPS.** A ROBOT may not restrict access to more than 1 opposing ALLIANCE color SCOOP except during END GAME.

***Violation:** MINOR FOUL, and for every 3 seconds in which the situation is not corrected, a MAJOR FOUL is assessed.*

Examples of restricting access to opposing ALLIANCE color SCOOPS include, but are not limited to:

A. holding 1 opposing ALLIANCE color SCOOP within your ROBOT and blocking an opponent from getting to more opposing ALLIANCE color SCOOPS cornered behind your ROBOT, and

B. herding several opposing ALLIANCE color SCOOPS into a corner and parking in front of them.

In a scenario where a ROBOT is PINNING (per G422), REFEREES are instructed to disregard this rule so they can focus on the PINNING.

**G410** **Stay out of the other shop's pantry.** A ROBOT may not enter the opponent's FREEZER or STORAGE (i.e. its BUMPERS may not be partially or completely within them), and may not gain CONTROL of a SCORING ELEMENT located in the opponent's FREEZER or STORAGE.

Exceptions to this rule:

A. a ROBOT retrieving a SCOOP of its own ALLIANCE color that ended up in the opponent's FREEZER or STORAGE, which may enter as needed to retrieve that SCOOP, and only that SCOOP, without penalty, and

B. a ROBOT forced into the opponent's FREEZER or STORAGE by contact from an opponent ROBOT.

***Violation:*** *MINOR FOUL per entry. MAJOR FOUL per SCORING ELEMENT if the ROBOT gains CONTROL of a SCORING ELEMENT from there (other than as permitted above).*

Your pantry is yours, theirs is theirs. Stocking up on their ingredients is not Gracious Professionalism. If one of your own color SCOOPS wanders into their pantry, go get it, and come straight back out.

**G411** **No throwing SCORING ELEMENTS.** A ROBOT may not use an active MECHANISM to shoot SCORING ELEMENTS into the air in a way that's prohibited in R207, kick them across the floor, or eject them in a forceful way (i.e. the SCORING ELEMENT is caused to move a significant distance).

***Violation:*** *MAJOR FOUL per SCORING ELEMENT. YELLOW CARD if REPEATED, or if the REFEREE perceives a strategic benefit (e.g. the SCORING ELEMENT is launched onto a stack, a DELIVERY TRAY, or into a FREEZER).*

FROZEN BLITZ is a stacking game. SCOOPS are placed, not launched.

The following are not violations of this rule:

A. pushing SCORING ELEMENTS across the carpet with BUMPERS or the ROBOT frame while driving (bulldozing),

B. releasing a SCOOP onto a stack, or a CONE onto the carpet, from a short height, even if it bounces, and

C. a SCORING ELEMENT that pops loose during contact between ROBOTS.

**G412** **They don't pay us enough.** An ICE CREAM may not be taller than 5 SCOOPS. It gets too complicated, and they don't pay us enough.

***Violation:*** *MAJOR FOUL.*

> Accidentally placing a 6th SCOOP on an ICE CREAM and immediately removing it is an exception to this rule.

**G413** **Stack in the designated zone.** A ROBOT may only stack SCOOPS at their ALLIANCE'S SCOOPING STATION.

***Violation:*** *MAJOR FOUL per SCORING ELEMENT.*

### 6.4.3 ROBOT

**G414** ***ROBOTS must be safe.** A ROBOT may not pose an undue hazard to a human, an ARENA element, or another ROBOT in the following ways:

A. the ROBOT or anything it CONTROLS, e.g. a SCORING ELEMENT, contacts anything outside the FIELD except for MOMENTARY incursions beyond the FIELD boundary at the SCOOPING STATION and DELIVERY ZONE structures, including the CHUTE,
B. its BUMPERS fail such that a segment completely detaches,
C. a corner of its ROBOT PERIMETER is exposed,
D. its team number or ALLIANCE color are indeterminate,
E. its BUMPERS leave the BUMPER ZONE (see R405) REPEATEDLY or for more than a MOMENTARY amount of time, unless the ROBOT is fully supported by the CHECKOUT ZONE tables or attempting to get on top of them, or
F. its operation or design is dangerous or unsafe.

***Violation:*** *DISABLED. Corrective action (such as repair of BUMPERS, removing the unsafe MECHANISM, and/or re-inspection) may be required before the ROBOT will be allowed to compete in subsequent MATCHES.*

> Examples of dangerous operation or designs that likely pose undue hazards include, but are not limited to:
>
> A. uncontrolled motion that cannot be stopped by the DRIVE TEAM,
> B. ROBOT parts "flailing" outside of the FIELD,
> C. ROBOTS dragging their battery, and
> D. ROBOTS that consistently extend outside the FIELD.
>
> Please be conscious of REFEREES and FIELD STAFF working around the ARENA who may be in close proximity to your ROBOT.

**G415** ***Keep your BUMPERS low.** ROBOT extensions may not interact with the carpet such that the BUMPERS are lifted out of the BUMPER ZONE (see R405).

***Violation:*** *MINOR FOUL. Corrective action (such as removing the offending MECHANISM, and/or re-inspection) may be required before the ROBOT will be allowed to compete in subsequent MATCHES.*

**G416** **Don't damage the FIELD.** A ROBOT may not damage FIELD elements.

***Violation:*** *VERBAL WARNING. If the Head REFEREE infers that additional damage is likely, DISABLED. YELLOW CARD for any subsequent damage during the event. Corrective action (such as eliminating sharp edges, removing the damaging MECHANISM, and/or re-inspection) may be required before the ROBOT will be allowed to compete in subsequent MATCHES.*

**G417** **Watch your FIELD interaction.** A ROBOT is prohibited from the following interactions with FIELD elements (with the exception of the CHECKOUT ZONE tables):

A. grabbing,
B. grasping,
C. attaching to (including the use of a vacuum or hook fastener to anchor to the FIELD carpet),
D. becoming entangled with, and
E. suspending from.

***Violation:*** *MAJOR FOUL, plus YELLOW CARD if REPEATED, or longer than MOMENTARY. If the Head REFEREE infers that damage is likely, DISABLED. Corrective action (such as removing the offending MECHANISM, and/or re-inspection) may be required before the ROBOT will be allowed to compete in subsequent MATCHES.*

**G418** **Expansion limits.** A ROBOT may not extend beyond the expansion limit described in R105. This rule does not apply during END GAME to a ROBOT that is partially or completely within its ALLIANCE'S CHECKOUT ZONE, or in contact with its ALLIANCE'S CHECKOUT ZONE tables.

Exceptions to this rule:

A. If the over-expansion is due to visible damage and not used for strategic benefit, no penalty is imposed.
B. If the over-expansion is MOMENTARY and is not used for strategic benefit, no penalty is imposed.

***Violation:*** *MINOR FOUL. If strategic (e.g. the expansion results in scoring a SCORING ELEMENT), RED CARD.*

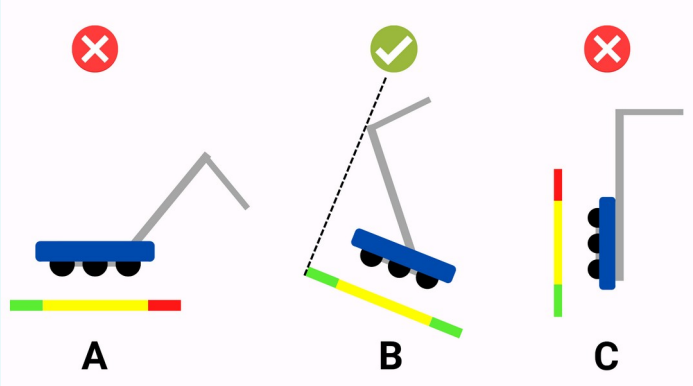



*Figure 6-2 Examples of G418 compliance and non-compliance*

Yellow bars represent the limits of the ROBOT PERIMETER and are drawn in the same orientation of the ROBOT PERIMETER. Green bars represent a measured extension from the ROBOT PERIMETER that has not been exceeded. Red bars represent a measured extension from the ROBOT PERIMETER that has exceeded the limit in R105. ROBOTS A and C violate this

rule, whereas ROBOT B does not.

Exceptions are only given for visible damage, as perceived by a REFEREE. Teams should not assume that REFEREES will give an exception for unobservable damage even if ROBOT function is affected.

### 6.4.4 Opponent Interaction

Note, G419, G420, and G421 are mutually exclusive. A single ROBOT to ROBOT interaction which violates more than 1 of these rules results in the most punitive penalty, and only the most punitive penalty, being assessed.

**G419** ***Stay out of other ROBOTS.** A ROBOT may not damage or functionally impair an opponent ROBOT by initiating contact, either directly or transitively via a SCORING ELEMENT CONTROLLED by the ROBOT:

A. inside the vertical projection of an opponent's ROBOT PERIMETER, or

B. with the opponent's BUMPER backing or mounting.

Exceptions to this rule:

A. Contact between the ROBOT'S BUMPERS or COMPONENTS inside the ROBOT PERIMETER and COMPONENTS inside an opening of an opponent's BUMPERS or in the space above the BUMPER opening

B. Damage or functional impairment because of contact with a tipped-over opponent ROBOT

C. Damage that appears to the REFEREE as cosmetic only

***Violation:** MAJOR FOUL and YELLOW CARD, or if opponent ROBOT is unable to drive, then MAJOR FOUL and RED CARD.*

FIRST Robotics Competition can be a full-contact competition and may include rigorous game play. While this rule aims to limit severe damage to ROBOTS, teams should design their ROBOTS to be robust.

For the purposes of this rule, "initiating contact" is a judgement call as to which ROBOT(S) in any interaction were responsible for the contact occurring based on direction of travel and relative speeds of both ROBOTS. Generally, for a ROBOT to have initiated contact, at minimum they have to be both moving towards the opponent ROBOT and have reasonable possibility to have avoided the contact. In a collision, it's possible for both ROBOTS to initiate contact.

"Unable to drive" means that because of the incident, for approximately 20 seconds or more, the DRIVER can no longer drive to a desired location in a reasonable time (generally). For example, if a ROBOT can only move in circles, or can only move extremely slowly, the ROBOT is considered unable to drive.

The exception in part C effectively means that ROBOTS with BUMPER gaps are at their own risk regarding damaging contact in these areas.

**G420** ***This isn't combat robotics.** A ROBOT may not intentionally and/or recklessly damage or functionally impair an opponent ROBOT.

***Violation:** MAJOR FOUL and YELLOW CARD, or if opponent ROBOT is unable to drive, then MAJOR FOUL and RED CARD.*

> The intent of this rule is to cover situations where a ROBOT intentionally and/or recklessly, as perceived by the REFEREE, damages another ROBOT anywhere on the FIELD.
>
> Situations where a REFEREE is likely to infer intent or recklessness include, but are not limited to: a ROBOT REPEATEDLY smashing into an opponent ROBOT intake, a ROBOT REPEATEDLY hitting another ROBOT with its extension, a ROBOT driving at high speed across the FIELD into an opponent with its extension, and a ROBOT violently driving into an opponent that is already firmly against a FIELD element.
>
> Examples that are not reckless include, but are not limited to: REPEATED BUMPER to BUMPER contact, and a high-speed BUMPER to BUMPER hit.
>
> Examples of functionally impairing another ROBOT include, but are not limited to, opening an opponent's relief valve such that the opponent's air pressure drops, and powering off an opponent's ROBOT (this example also clearly results in a RED CARD because the ROBOT is no longer able to drive).
>
> At the conclusion of the MATCH, the Head REFEREE may elect to visually inspect a ROBOT to confirm violations of this rule made during a MATCH and remove the violation if the damage cannot be verified.

**G421** ***Don't tip or entangle.** A ROBOT may not deliberately attach to, tip over, or entangle with an opponent ROBOT.

***Violation:** MAJOR FOUL and YELLOW CARD, or if CONTINUOUS or opponent ROBOT is unable to drive, then MAJOR FOUL and RED CARD.*

> Examples of violations of this rule include, but are not limited to:
>
> A. using a wedge-like MECHANISM to tip over opponent ROBOTS,
> B. making BUMPER-to-BUMPER contact with an opponent ROBOT that is attempting to right itself after previously falling over and causing them to fall over again, and
> C. causing an opponent ROBOT to tip over by contacting the ROBOT after it starts to tip if, in the judgement of the REFEREE, that contact could have been avoided.
>
> Tipping as an unintended consequence of normal ROBOT to ROBOT interaction, including single BUMPER to BUMPER hits that result in a ROBOT tipping, is not a violation of this rule.

**G422** ***There's a 3-count on PINS.** A ROBOT may not PIN an opponent's ROBOT for more than 3 seconds. A ROBOT is PINNING if it is preventing the movement of an opponent ROBOT by contact, either direct or transitive (such as against a FIELD element). A PIN count ends once any of the following criteria below are met:

A. the ROBOTS have separated by at least 72.0in (1.83m) from each other for more than 3 seconds,

B. either ROBOT has moved 72.0in (1.83m) from where the PIN initiated for more than 3 seconds, or

C. the PINNING ROBOT gets PINNED.

For criteria A, the PIN count pauses once ROBOTS are separated by 72.0in (1.83m) until either the PIN ends or the PINNING ROBOT moves back within 72.0in (1.83m), at which point the PIN count is resumed.

For criteria B, the PIN count pauses once either ROBOT has moved 72.0in (1.83m) from where the PIN initiated until the PIN ends or until both ROBOTS move back within 72.0in (1.83m), at which point the PIN count is resumed.

***Violation:*** *MINOR FOUL, and for every 3 seconds in which the situation is not corrected, a MAJOR FOUL is assessed.*

> A team's desired direction of travel is not a consideration when determining if a ROBOT is PINNED.

**G423** ***Don't collude with your partners to shut down major parts of game play.** 2 or more ROBOTS that appear to a REFEREE to be working together may not isolate or close off any major element of MATCH play.

***Violation:*** *MAJOR FOUL, and for every 3 seconds in which the situation is not corrected, a MAJOR FOUL is assessed.*

> Examples of violations of this rule include, but are not limited to:
>
> A. fully blocking all access to an opponent's SCOOPING STATION or DELIVERY ZONE,
>
> B. shutting down access to all opponent SCOOPS or CONES,
>
> C. quarantining all opponents to a small area of the FIELD, and
>
> D. blocking all access to the opponent's FREEZER or STORAGE.
>
> A single ROBOT blocking access to a particular area of the FIELD is not a violation of this rule. 2 ROBOTS independently playing defense on 2 opponent ROBOTS is not a violation of this rule.

**G424** **CHECKOUT ZONE protection.** During END GAME, a ROBOT may not contact, directly or transitively through a SCORING ELEMENT, an opponent ROBOT that is in contact with the opponent's CHECKOUT ZONE, regardless of who initiates contact. This rule does not apply to contact between 2 ROBOTS that are each in contact with their own ALLIANCE'S CHECKOUT ZONE.

***Violation:*** *MAJOR FOUL. If the contacted ROBOT is not supported by the carpet at the time of contact, the contacted ROBOT is awarded CHECKOUT points as if it were completely above the CHECKOUT ZONE and fully supported by it.*

> The END GAME audio cue (Table 4-1) signals that this protection is active. Transitive contact requires that each ROBOT is contacting the same SCORING ELEMENT(S) simultaneously.

**G425** **Nobody likes a dropped cone.** A ROBOT may not contact, directly or transitively through a SCORING ELEMENT, an opponent ROBOT that is at least half within that opponent's SAFE ZONE, regardless of who initiates contact. Contact initiated by a part of the protected ROBOT that is outside its SAFE ZONE is an exception to this rule. Whether a ROBOT is at least half within its SAFE ZONE is the REFEREE'S call, based on its BUMPERS, and that call is final and not subject to review.

***Violation:*** *MAJOR FOUL. If the contact causes any SCORING ELEMENT in the protected ROBOT'S stack, ICE CREAM, or CONTROL to fall, MAJOR FOUL and YELLOW CARD.*

> "At least half" is a judgment call on purpose, not a measurement. If the REFEREE says the cup was half empty, it was empty. Teams that want protection should make it obvious: get in deep, and don't argue about the tape.
>
> Transitive contact requires that each ROBOT is contacting the same SCORING ELEMENT(S) simultaneously.

**G426** **Employees only.** A ROBOT may not enter an opponent's SAFE ZONE (i.e. its BUMPERS may not be partially or completely within it). A ROBOT forced into an opponent's SAFE ZONE by contact from an opponent ROBOT is an exception to this rule.

***Violation:*** *MINOR FOUL if MOMENTARY. MAJOR FOUL if intentional or greater-than-MOMENTARY.*

> SAFE ZONES are where ice cream gets made and served, so they are off limits to the other shop whether or not a ROBOT is working in them. Contact with a protected ROBOT is penalized separately under G425.

**G427** **If an opponent's down, back off.** Fallen (i.e. tipped over) ROBOTS attempting to right themselves (either by themselves or with assistance from a partner ROBOT) have one 10-second grace period in which they may not be contacted by an opponent ROBOT. This protection lasts for either 10 seconds or until the protected ROBOT has completed the righting operation, whichever comes first.

***Violation:*** *MINOR FOUL. If intentional, YELLOW CARD.*

### 6.4.5 Human

**G428** ***No wandering.** A DRIVE TEAM member must remain in their designated area as follows:

A. DRIVERS and DRIVE COACHES may not contact anything outside their ALLIANCE AREA,

B. a DRIVER must use the OPERATOR CONSOLE in the DRIVER STATION to which they are assigned, as indicated on the team sign,

C. a HUMAN PLAYER may not contact anything outside the ALLIANCE AREA, SCOOPING STATION, or DELIVERY ZONE station in which they started the MATCH, and

D. a TECHNICIAN may not contact anything outside their designated area.

Exceptions are granted in cases concerning safety and for actions that are inadvertent, MOMENTARY, and inconsequential.

***Violation:*** *MINOR FOUL.*

> An intent of item B is to prevent unsafe situations where long tethers to OPERATOR CONSOLE devices increase tripping hazards as the operator moves about the ALLIANCE AREA. Provided the DRIVE TEAM member is within close proximity of their DRIVER STATION, there will be no repercussions. However, a DRIVE TEAM member located more than approximately half a DRIVER STATION width away from their own DRIVER STATION while using their OPERATOR CONSOLE is likely violating this rule.

**G429** ***DRIVE COACHES and other teams: hands off the controls.** A ROBOT shall be operated only by the DRIVERS and/or HUMAN PLAYERS of that team. A DRIVE COACH activating their E-Stop is the exception to this rule.

***Violation:*** *MAJOR FOUL. RED CARD if greater-than-MOMENTARY.*

> Exceptions may be made before a MATCH for major conflicts, e.g. religious holidays, major testing, transportation issues, etc.

**G430** ***DRIVE TEAMS, watch your reach.** A DRIVE TEAM member may not extend beyond the PURPLE PLANE.

***Violation:*** *MINOR FOUL.*

> Teams should be careful when placing or collecting SCORING ELEMENTS and be aware of G101.

**G431** ***Humans: use SCORING ELEMENTS as directed.** A DRIVE TEAM member may not deliberately use a SCORING ELEMENT in an attempt to ease or amplify a challenge associated with a FIELD element.

***Violation:*** *MAJOR FOUL.*

> An example of a violation of this rule is a HUMAN PLAYER using a CONE to disrupt an opponent ROBOT attempting to get on top of the CHECKOUT ZONE.

**G432** ***DRIVE COACHES, SCORING ELEMENTS are off limits.** DRIVE COACHES may not touch SCORING ELEMENTS, unless for safety purposes.

***Violation:*** *MINOR FOUL per instance.*

**G433** **Only feed your ALLIANCE ROBOTS.** HUMAN PLAYERS may not deliver SCORING ELEMENTS to opponent ROBOTS.

***Violation:*** *MINOR FOUL per SCORING ELEMENT.*

**G434** **One hand on the hoop.** HOOPS may only be introduced into the MATCH by the HUMAN PLAYER at their ALLIANCE'S SCOOPING STATION. The HUMAN PLAYER must place each HOOP using only 1 hand, on top of a SCOOP already stacked at the SCOOPING STATION.

***Violation:*** *MINOR FOUL per HOOP.*

**G435** **Hands off the merchandise.** A HUMAN PLAYER at the SCOOPING STATION may not contact a SCOOP or CONE in any way, for any purpose, including to place, arrange, or stabilize a stack.

***Violation:*** *MINOR FOUL. If the REFEREE perceives the HUMAN PLAYER is strategically arranging or stabilizing the SCOOPS below, MAJOR FOUL.*

**G436** **ROBOTS don't do HOOPS.** A ROBOT may not place a HOOP, or deliberately CONTROL a HOOP that is not part of a stack or an ICE CREAM.

***Violation:*** *MINOR FOUL per HOOP.*

**G437** **Recycle through the CHUTE.** The HUMAN PLAYER at the DELIVERY ZONE may return SCOOPS to the FIELD only by dropping them into their ALLIANCE'S CHUTE.

***Violation:*** *MINOR FOUL per SCOOP.*

**G438** **Cones go home.** The HUMAN PLAYER at the DELIVERY ZONE must return each CONE by throwing or rolling it into their ALLIANCE'S STORAGE. A CONE whose first contact after leaving the HUMAN PLAYER'S hand is with anything other than the carpet inside their ALLIANCE'S STORAGE (e.g. other carpet, a FIELD element, or a ROBOT) is a violation.

***Violation:*** *MAJOR FOUL per CONE.*

> CONES go home to STORAGE and come back out on a ROBOT. Handing or tossing a CONE to a ROBOT is not allowed. A CONE that lands in its STORAGE and then bounces or rolls out is not a violation.

**G439** **No catching the ice cream.** The HUMAN PLAYER at the DELIVERY ZONE may contact only SCORING ELEMENTS on a lowered DELIVERY TRAY, and SCORING ELEMENTS that leave the FIELD into their station (which are then returned per G437 and G438). They may not contact an ICE CREAM that is on a raised tray or still CONTROLLED by a ROBOT.

***Violation:*** *MINOR FOUL. If the REFEREE perceives the HUMAN PLAYER is strategically catching, stabilizing, or guiding an ICE CREAM, MAJOR FOUL.*

## 6.5 Post-MATCH

**G501** ***Leave promptly.** A DRIVE TEAM member may not cause significant or multiple delays to the start of a subsequent MATCH, scheduled break content, or other FIELD activities.

***Violation:*** *VERBAL WARNING. YELLOW CARD if subsequent violations at any point during the event.*

> DRIVE TEAMS are expected to remove their ROBOT and OPERATOR CONSOLE from the FIELD and DRIVER STATION safely and swiftly once the FIELD lights turn green.

# 07. ROBOT CONSTRUCTION RULES (R)

The rules listed below explicitly address legal parts and materials and how those parts and materials may be used on a FROZEN BLITZ ROBOT. A ROBOT is an electromechanical assembly built by the FIRST Robotics Competition team to play the current season's game and includes all the basic systems required to be an active participant in the game – power, communications, control, BUMPERS, and movement about the FIELD. A BUMPER is a protective assembly designed to attach to the exterior of the ROBOT and constructed as specified in section 7.4 BUMPER Rules.

> FROZEN BLITZ ROBOT Construction Rules follow the 2026 FIRST Robotics Competition rules (REBUILT, TU22). ROBOT size, weight, and extension limits are set for FROZEN BLITZ in section 7.1. For the purposes of this manual, "Kickoff" means the TideChallenge Kickoff.

There are many reasons for the structure of the rules, including safety, reliability, parity, creation of a reasonable design challenge, adherence to professional standards, impact on the competition, and compatibility with the Kit of Parts (KOP). The KOP is the collection of items listed on the current season's Kickoff Kit Checklists, distributed to the team via FIRST Choice in the current season, or paid for completely (except shipping) with a Product Donation Voucher (PDV) from the current season.

Another intent of these rules is to have all energy sources and active actuation systems on the ROBOT (e.g. batteries, compressors, motors, servos, cylinders, and their controllers) drawn from a well-defined set of options. This is to ensure that all teams have access to the same actuation resources and that the INSPECTORS are able to accurately and efficiently assess the legality of a given part.

ROBOTS are made up of COMPONENTS and MECHANISMS. A COMPONENT is any part in its most basic configuration, which cannot be disassembled without damaging or destroying the part or altering its fundamental function. A MECHANISM is an assembly of COMPONENTS that provide specific functionality on the ROBOT. A MECHANISM can be disassembled (and then reassembled) into individual COMPONENTS without damage to the parts.

Many rules in this section reference Commercial-Off-The-Shelf (COTS) items. A COTS item must be a standard (i.e. not custom order) part commonly available from a VENDOR for all teams for purchase. To be a COTS item, the COMPONENT or MECHANISM must be in an unaltered, unmodified state (with the exception of installation or modification of any software). Items that are no longer commercially available but are functionally equivalent to the original condition as delivered from the VENDOR are considered COTS and may be used.

> Example 1: A team orders 2 ROBOT grippers from RoboHands Corp. and receives both items. They put 1 in their storeroom and plan to use it later. Into the other, they drill "lightening holes" to reduce weight. The first gripper is still classified as a COTS item, but the second gripper is now a FABRICATED

ITEM, as it has been modified.

Example 2: A team obtains openly available blueprints of a drive module commonly available from Wheels-R-Us Inc. and has local machine shop "We-Make-It, Inc." manufacture a copy of the part for them. The produced part is not a COTS item, because it is not commonly carried as part of the standard stock of We-Make-It, Inc.

Example 3: A team obtains openly available design drawings from a professional publication during the pre-season and uses them to fabricate a gearbox for their ROBOT during the build period following Kickoff. The design drawings are considered a COTS item and may be used as "raw material" to fabricate the gearbox. The finished gearbox itself would be a FABRICATED ITEM, and not a COTS item.

Example 4: A COTS part that has non-functional label markings added would still be considered a COTS part, but a COTS part that has device-specific mounting holes added is a FABRICATED ITEM.

Example 5: A team has a COTS single-board processor version 1.0, which can no longer be purchased. Only the COTS single-board processor version 2.0 may be purchased. If the COTS single-board processor version 1.0 is functionally equivalent to its original condition, it may be used.

Example 6: A team has a COTS gearbox which has been discontinued. If the COTS gearbox is functionally equivalent to its original condition, it may be used.

A VENDOR is a legitimate business source for COTS items that satisfies all the following criteria:

A. has a Federal Tax Identification number. In cases where the VENDOR is outside of the United States, they must possess an equivalent form of registration or license with the government of their home nation that establishes and validates their status as a legitimate business licensed to operate within that country (in Mexico, an RFC).

B. is not a "wholly owned subsidiary" of a FIRST Robotics Competition team or collection of teams. While there may be some individuals affiliated with both a team and the VENDOR, the business and activities of the team and VENDOR must be completely separable.

C. should maintain sufficient stock or production capability so they are able to ship any general (i.e., non-FIRST unique) product within 5 business days of receiving a valid purchase request. It is recognized that certain unusual circumstances (such as a global supply chain disruption and/or 1,000 FIRST teams all ordering the same part at once from the same VENDOR) may cause atypical delays in shipping due to backorders for even the largest VENDORS. Such delays due to higher-than-normal order rates are excused. This criterion may not apply to custom-built items from a source that is both a VENDOR and a fabricator.

D. makes their products available to all FIRST Robotics Competition teams. A VENDOR must not limit supply or make a product available to just a limited number of FIRST Robotics Competition teams.

The intent of this definition is to be as inclusive as possible to permit access to all legitimate sources, while preventing ad hoc organizations from providing special-purpose products to a limited subset of teams in an attempt to circumvent the cost accounting rules.

FIRST desires to permit teams to have the broadest choice of legitimate sources possible, and to obtain COTS items from the sources that provide them with the best prices and level of service available. Teams also need to protect against long delays in availability of parts that will impact their ability to complete their ROBOT. Remember, events are not always near home – when parts fail, local access to replacement materials is often critical.

A FABRICATED ITEM is any COMPONENT or MECHANISM that has been altered, built, cast, constructed, concocted, created, cut, heat treated, machined, manufactured, modified, painted, produced, surface coated, or conjured partially or completely into the final form in which it will be used on the ROBOT.

> Note that it is possible for an item (typically raw materials) to be neither COTS nor a FABRICATED ITEM. For example, a 120.0in (3.048m) length of aluminum which has been cut into 60.0in (1.52m) pieces by the team for storage or transport is neither COTS (it's not in the state received from the VENDOR), nor a FABRICATED ITEM (the cuts were not made to advance the part towards its final form on the ROBOT).

Teams may be asked to provide documentation proving the legality of items during inspection where a rule specifies limits for a legal part (e.g. pneumatic items, current limits, COTS electronics, etc.).

Some of these rules make use of English unit requirements for parts. If your team has a question about a metric-equivalent part's legality, please submit it through the Q&A (section 1.7) for an official ruling.

Teams should acknowledge the support provided by the corporate sponsors and mentors with an appropriate display of their school and sponsors names and/or logos (or the name of the supporting youth organization, if appropriate).

FIRST Robotics Competition can be a full-contact competition and may include rigorous game play. While the rules aim to limit severe damage to ROBOTS, teams should design their ROBOTS to be robust.

## 7.1 General ROBOT Design

**R101** ***ROBOT PERIMETER must be fixed.** The ROBOT (excluding BUMPERS) must have a ROBOT PERIMETER, contained within the BUMPER ZONE and established while in the ROBOT'S STARTING CONFIGURATION, that is comprised of fixed, non-articulated structural elements of the ROBOT. Minor protrusions no greater than 0.25in (0.64cm) such as bolt heads, fastener ends, weld beads, and rivets are not considered part of the ROBOT PERIMETER.

> To determine the ROBOT PERIMETER, wrap a piece of string around the outer most parts of the ROBOT (excluding BUMPERS) at the BUMPER ZONE described in R405 and pull it taut. The string outlines the ROBOT PERIMETER.
>
> Example: A ROBOT'S chassis is shaped like the letter 'U', with a large gap between chassis elements on the front of the ROBOT. When wrapping a taut string around this chassis, the string extends across the gap and the resulting ROBOT PERIMETER is a rectangle with 4 sides.

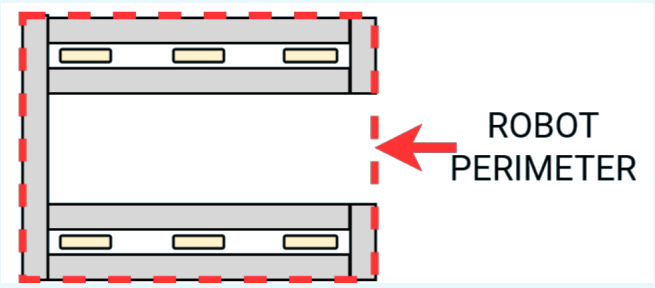



*Figure 7-1 ROBOT PERIMETER example*

**R102** ***STARTING CONFIGURATION – no overhang.** In the STARTING CONFIGURATION (the physical configuration in which a ROBOT starts a MATCH), no part of the ROBOT shall extend outside the vertical projection of the ROBOT PERIMETER, with the exception of its BUMPERS and minor protrusions such as bolt heads, fastener ends, rivets, cable ties, etc.

> If a ROBOT is designed as intended and each side is pushed up against a vertical wall (in STARTING CONFIGURATION and with BUMPERS removed), only the ROBOT PERIMETER (or minor protrusions) will be in contact with the wall.
>
> The allowance for minor protrusions in this rule is intended to allow protrusions that are both minor in extension from the ROBOT PERIMETER and cross-sectional area.
>
> If a ROBOT uses interchangeable MECHANISMS per I103, teams should be prepared to show compliance with this rule and R105 in all configurations.

**R103** ***ROBOT weight limit.** The ROBOT weight must not exceed 115.0lb (52.16kg). When determining weight, the basic ROBOT structure and all elements of all additional MECHANISMS that might be used in a single configuration of the ROBOT shall be weighed together (see I103). For the purposes of determining compliance with the weight limitations, the following items are excluded:

A. ROBOT BUMPERS,
B. ROBOT battery and its associated half of the Anderson cable quick connect/disconnect pair (including wire, the associated cable lugs, connecting bolts, and insulation), and
C. tags used for location detection systems if provided by the event.

**R104** **STARTING CONFIGURATION – max size.** A ROBOT'S STARTING CONFIGURATION may not have a ROBOT PERIMETER greater than 110.0in (2.794m) and may not be more than 52.0in (1.320m) tall.

> Be sure to consider the size of the ROBOT on its cart to make sure it will fit through doors. Also consider the size of the ROBOT to ensure that it will fit into a shipping crate, vehicle, etc.
>
> Note that rules contained in section 7.4 BUMPER Rules may impose additional restrictions on ROBOT design.

**R105** **ROBOT extension limit.** ROBOTS may not extend more than 16.0in (40.64cm) beyond the vertical projection of their ROBOT PERIMETER, except during END GAME while partially or completely within their ALLIANCE'S CHECKOUT ZONE, or in contact with their ALLIANCE'S CHECKOUT ZONE tables (see G418).

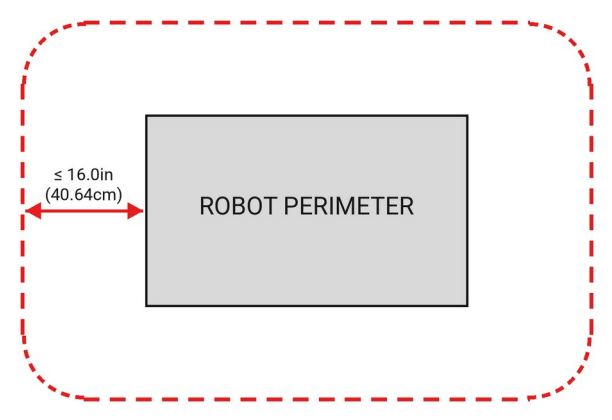



*Figure 7-2 ROBOT PERIMETER extension*

Teams should expect to have to demonstrate a ROBOT'S ability to constrain itself per above during inspection. Constraints may be implemented with either hardware or software.

See section 6 Game Rules (G) for additional restrictions for various areas of the FIELD.

**R106** **ROBOT extension floor interaction.** ROBOT extensions may not interact with the carpet such that the BUMPERS are lifted out of the BUMPER ZONE (see R405).

## 7.2 ROBOT Safety & Damage Prevention

**R201** ***No digging into carpet.** Traction devices must not have surface features that could damage the ARENA (e.g. metal, sandpaper, inflexible studs, cleats, hook-loop fasteners or similar attachments). Traction devices include all parts of the ROBOT that are designed to transmit any propulsive and/or braking forces between the ROBOT and FIELD carpet.

**R202** ***No exposed sharp edges.** Protrusions from the ROBOT and exposed surfaces on the ROBOT shall not pose hazards to the ARENA elements (including SCORING ELEMENTS) or people.

Note that the use of acrylic or other materials which may break into jagged pieces is not explicitly prohibited, but any such breakage must be corrected immediately to comply with this rule.

**R203** ***General safety.** ROBOT parts shall not be made from hazardous materials, be unsafe, cause an unsafe condition, or interfere with the operation of other ROBOTS.

Examples of items that will violate this rule include (but are not limited to):

A. shields, curtains, or any other devices or materials designed or used to obstruct or limit the vision of any DRIVE TEAM members and/or interfere with their ability to safely control

their ROBOT,

B. speakers, sirens, air horns, or other audio devices that generate sound at a level sufficient to be a distraction,

C. any devices or decorations specifically intended to jam or interfere with the remote sensing capabilities of another ROBOT, including vision systems, acoustic range finders, sonars, infrared proximity detectors, etc. (e.g. including imagery on your ROBOT that utilizes or closely mimics 36h11 AprilTags),

D. lasers other than those listed as IEC/EN 60825-1 "Class 1" or IEC/EN 62471 "Exempt,"

E. flammable gasses,

F. any device intended to produce flames or pyrotechnics,

G. hydraulic fluids or hydraulic items,

H. switches or contacts containing liquid mercury,

I. circuitry used to create voltages in excess of 24 Volts,

J. any ballast not secured sufficiently, including loose ballast e.g. sand, ball bearings, etc., such that it may become loose during a MATCH,

K. hazardous materials (e.g. lead, whether encapsulated or not) used on the ROBOT,

L. high intensity light sources used on the ROBOT (e.g. super bright LED sources marketed as 'military grade' or 'self-defense') may only be illuminated for a brief time while targeting and may need to be shrouded to prevent any exposure to participants. Complaints about the use of such light sources will be followed by re-inspection and possible disablement of the device, and

M. bright lights which flash more than approximately 5 times per second.

**R204** ***Leave SCORING ELEMENTS at the FIELD.** ROBOTS must allow removal of SCORING ELEMENTS from the ROBOT and the ROBOT from FIELD elements while DISABLED and powered off.

ROBOTS will not be re-enabled after the MATCH, so teams must be sure that SCORING ELEMENTS and ROBOTS can be quickly, simply, and safely removed. Teams are encouraged to consider G501 when developing their ROBOTS.

**R205** ***Don't contaminate the FIELD.** Liquids, gels, greases and fine particles must not contaminate the FIELD or other ROBOTS.

**R206** ***Don't damage SCORING ELEMENTS.** ROBOT elements likely to come in contact with a SCORING ELEMENT shall not pose a significant hazard to the SCORING ELEMENT.

SCORING ELEMENTS are expected to undergo a reasonable amount of wear and tear as they are handled by ROBOTS, such as scratching or marking. Gouging, tearing off pieces, or routinely marking SCORING ELEMENTS are violations of this rule.

**R207** **This is an ice cream shop, not a cannon.** A ROBOT may not be designed to shoot a SCORING ELEMENT such that it travels more than 36.0in (91.44cm) horizontally beyond its ROBOT PERIMETER (reference G411).

## 7.3 Budget Constraints & Fabrication Schedule

**R301** ***Individual item cost limit.** No individual, non-KOP item or software shall have a Fair Market Value (FMV) that exceeds $600 USD. The total cost of COMPONENTS purchased in bulk may exceed $600 USD as long as the cost of an individual COMPONENT does not exceed $600 USD.

Teams should be ready to show INSPECTORS documentation of FMV for any COMPONENTS that appear to be in the range of the $600 USD limit.

The Analog Devices IMU MXP Breakout Board, P/N ADIS16448, does not have a published FMV. This device is considered to comply with this rule regardless of its true FMV.

The FMV of a COTS item is its price defined by a VENDOR for the part or an identical functional replacement. This price must be generally available to all FIRST Robotics Competition teams throughout the build and competition season (i.e. short-term sale prices or coupons do not reflect FMV), however teams are only expected to make a good faith effort at determining the item price and are not expected to monitor prices of ROBOT items throughout the season. The FMV is the cost of the item itself and does not include any duties, taxes, tariffs, shipping, or other costs that may vary by locality.

The FMV of COTS software is the price, set by the VENDOR, to license the software (or piece of the software) that runs on the ROBOT for the period from Kickoff to the end of the FIRST Championship. The FMV of software licensed free-of-cost, including through the Virtual KOP, for use on the ROBOT is $0.

The FMV of FABRICATED parts is the value of the material and/or labor, except for labor provided by team members (including sponsor employees who are members of the team), members of other teams, and/or event provided machine shops. Material costs are accounted for as the cost of any purchasable quantity that can be used to make the individual part (i.e. the purchasable raw material is larger than the FABRICATED part).

Example 1: A team orders a custom bracket made by a company to the team's specification. The company's material cost and normally charged labor rate apply.

Example 2: A team receives a donated sensor. The company would normally sell this item for $450 USD, which is therefore its FMV.

Example 3: A team purchases titanium tube stock for $400 USD and has it machined by a local machine shop. The machine shop is not considered a team sponsor but donates 2 hours of expended labor anyway. The team must include the estimated normal cost of the labor as if it were paid to the machine shop and add it to the $400 USD.

Example 4: A team purchases titanium tube stock for $400 USD and has it machined by a local machine shop that is a recognized sponsor of the team. If the machinists are considered members of the team, their labor costs do not apply. The total applicable cost for the part would be $400 USD.

Example 5: A team purchases titanium tube stock for $400 USD and has it machined by another team. The total applicable cost for the part would be $400 USD.

Example 6: A team purchases a widget at a garage sale or online auction for $300, but it's

available for sale from a VENDOR for $700. The FMV is $700.

If a COTS item is part of a modular system that can be assembled in several possible configurations, then each individual module must fit within the price constraints defined in this rule. If the modules are designed to assemble into a single configuration, and the assembly is functional in only that configuration, then the total cost of the complete assembly including all modules must fit within the price constraints defined in this rule. In summary, if a VENDOR sells a system or a kit, a team must use the entire system/kit FMV and not the value of its COMPONENT pieces.

Example 7: VENDOR A sells a gearbox that can be used with a number of different gear sets, and can mate with 2 different motors they sell. A team purchases the gearbox, a gear set, and a motor, then assembles them together. Each part is treated separately for the purpose of determining FMV since the purchased pieces can each be used in various configurations.

Example 8: VENDOR B sells a robotic arm assembly that a team wants to use. However, it costs $630 USD, so they cannot use it. The VENDOR sells the "hand", "wrist", and "arm" as separate assemblies, for $210 USD each. A team wishes to purchase the 3 items separately, then reassemble them. This would not be legal, as they are really buying and using the entire assembly, which has a Fair Market Value of $630 USD.

Example 9: VENDOR C sells a set of wheels or wheel modules that are often used in groups of 4. The wheels or modules can be used in other quantities or configurations. A team purchases 4 and uses them in the most common configuration. Each part is treated separately for the purpose of determining FMV, since the purchased pieces can be used in various configurations.

**R302** ***MAJOR MECHANISMS, from this year only.** MAJOR MECHANISMS, as defined in I101, created before Kickoff are not permitted.

Neither this rule nor the language in this blue box define specific thresholds for how much of a MAJOR MECHANISM must be constructed after Kickoff. This rule expects and requires the team's honest assessment of whether they built the MAJOR MECHANISMS of their ROBOT after Kickoff.

Attempts to exploit loopholes in the definition of MAJOR MECHANISM in order to bypass this requirement are in the spirit of neither this rule nor FIRST Robotics Competition. Examples of exploitation include:

A. pre-assembling significant portions of a MAJOR MECHANISM prior to Kickoff and attaching those assemblies together post-Kickoff, and

B. removing a small COMPONENT of a MAJOR MECHANISM prior to Kickoff such that it is no longer a MAJOR MECHANISM and replacing it after Kickoff.

The TideChallenge is a build experience. Bringing last season's ROBOT, or its MAJOR MECHANISMS, defeats the point of it.

**R303** ***Create new designs and software, unless they're public.** ROBOT software and designs created before Kickoff are only permitted if the source files (complete information sufficient to produce the design) are available publicly prior to Kickoff.

> Example 1: A team realizes that the transmission designed and built in the fall perfectly fits their need for a transmission to drive the ROBOT arm. They build an exact copy of the transmission from the original design plans and bolt it to the ROBOT. This would be prohibited, as the transmission – although made during the competition season – was built from detailed designs developed prior to Kickoff.
>
> Example 2: A team developed an omni-directional drive system for a previous competition. They refined and improved the control software, written in C++, to add more precision and capabilities. They decided to use a similar system for FROZEN BLITZ. They copied large sections of unmodified code over into the control software of the new ROBOT, also written in C++. This would be a violation of the schedule constraint and is not allowed.
>
> Example 3: The same team decides to use Java as their software environment for FROZEN BLITZ. Following Kickoff, they use the previously developed C++ code as a reference for the algorithms and calculations required to implement their omni-directional control solution. Because they developed new Java code as they ported over their algorithms, this is permitted.
>
> Example 4: A different team develops a similar solution during the fall and plans to use the developed software on their competition ROBOT. After completing the software, they post it in a generally accessible public forum and make the code available to all teams. Because they have made their software publicly available before Kickoff, they can use it on their ROBOT.
>
> Example 5: A team develops a transmission prior to Kickoff. After completing the project, they publish the CAD files on a generally accessible public forum and make them available to all teams. Because they have made the design publicly available before Kickoff, they can use the design to create an identical transmission, fabricated after Kickoff, for use on their FROZEN BLITZ ROBOT.

## 7.4 BUMPER Rules

A BUMPER is a required assembly which attaches to the ROBOT frame. BUMPERS protect ROBOTS from damaging/being damaged by other ROBOTS and FIELD elements.

A BUMPER guide providing additional information on materials and design based on historical best practices may be found under the Mechanical Resources section of the FIRST Robotics Competition Technical Resources page.

All dimensions specified in this section are nominal and will be measured during inspection with a tolerance of 0.25in (0.63cm) unless otherwise specified. This means that maximums specified have a tolerance of +0.25in (0.63cm) and minimums specified have a tolerance of –0.25in (0.63cm). Teams are encouraged to design to the nominal dimension and reserve the tolerance for unexpected deviation such as manufacturing error or tolerance stack-up.

> **Heads up, veterans.** BUMPER rules changed a lot since 2022. Plywood is no longer required, pool noodles are one option among several foams, BUMPERS now protect the entire ROBOT PERIMETER,

and the BUMPER ZONE is 2.5in to 5.75in from the floor. Read this section with fresh eyes.

**R401** ***BUMPERS almost all around.** ROBOTS are required to use BUMPERS to protect the entire ROBOT PERIMETER. Gaps of less than 1.25in (3.17cm) between adjacent segments are permitted as long as all corners are filled per R406. A single gap larger than 1.25in (3.17cm) is permitted, as long as at least 5.0in (12.7cm) of ROBOT PERIMETER on each side of each corner is protected by BUMPER.

An arc is considered to have infinite corners and therefore may not have a gap larger than 1.25in (3.17cm).

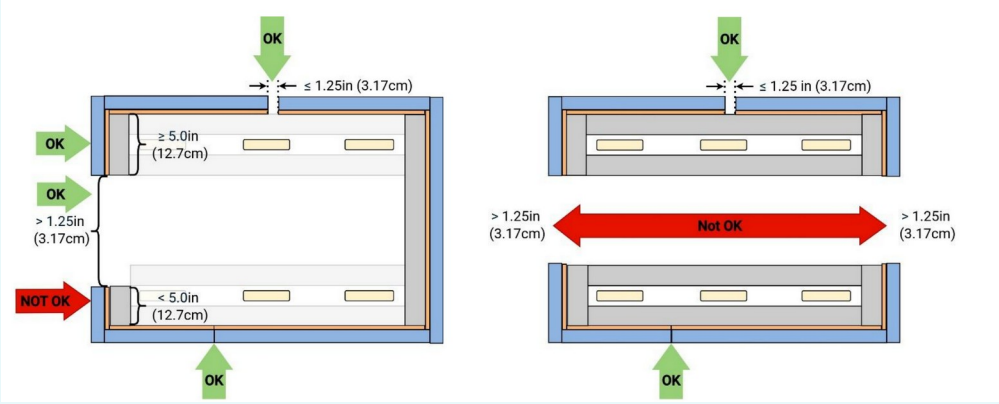



*Figure 7-3 BUMPER coverage requirements*

**R402** ***BUMPER construction.** BUMPERS must consist of the following elements. With the exception of padding and cover extending into corners and cover wrapping around the end of segments, any vertical cross section of the BUMPER must include parts A, B, and C:

A. Padding – A minimum of 2.25in (5.72cm) depth of foam padding, at least 4.5in (11.43cm) tall consisting of solid blocks, sheets, or stacked rods of one or more of the following materials: solid pool noodles or backer rod; solid polyethylene closed cell foam (including crosslinked) with density between 1.5 and 3.0lb/ft³ (24.03 to 48.05kg/m³); solid EVA closed cell foam with density between 2.0 and 6.0lb/ft³ (32.04 to 96.11kg/m³); or foam floor tiles. Multiple types, shapes, and/or layers of foam may be used within a single BUMPER.

B. Backing – A backer at least 4.5in (11.43cm) tall which supports the padding (i.e. padding is not cantilevered other than in corners) and facilitates installation and removal of the BUMPER from the ROBOT (as noted in R410). Additional elements may be added to strengthen the backing, fill space between the BUMPER and ROBOT frame, serve as part of the attachment system, or for any other purpose.

C. Cover – Cloth (as noted in R411) which covers all outward, upward and downward facing surfaces of padding such that no padding is exposed to interaction with the FIELD or other ROBOTS.

D. Fastening System – BUMPERS must attach to the ROBOT PERIMETER with a rigid fastening system to form a tight, robust connection to the main structure/frame (e.g. not attached with hook-and-loop tape, tape, or cable ties). The fastening system must be designed to withstand vigorous game play. All removable fasteners (e.g. bolts, locking pins, quick release pins, etc.) may be considered part of either the BUMPERS or ROBOT for determination of weight per R103 and R408.

Teams should be prepared to provide information about the padding material used in their BUMPERS. Teams do not need to have a separate material sample or expose padding for direct inspection to show compliance with this rule.

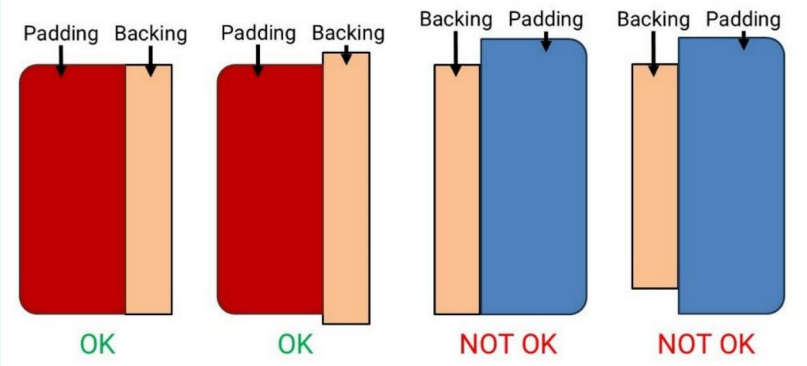



*Figure 7-4 BUMPER backing supporting padding (cross section)*

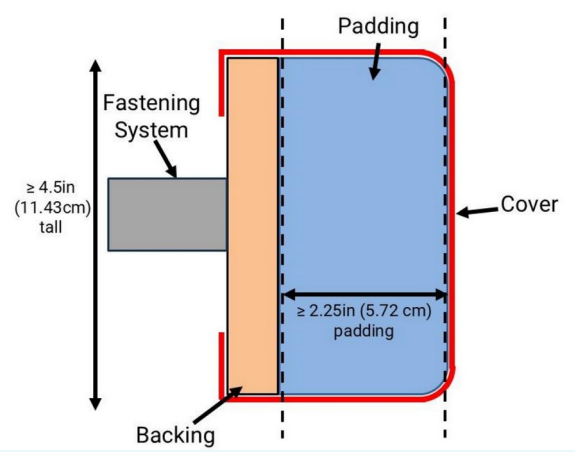



*Figure 7-5 BUMPER cross section example*

**R403** ***BUMPER extension limit.** BUMPERS must not extend more than 4.0in (10.16cm) from the ROBOT PERIMETER.

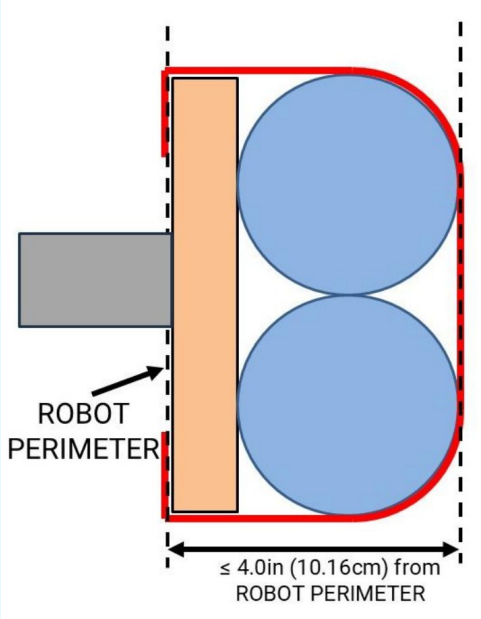



*Figure 7-6 BUMPER extension limit*

**R404** ***BUMPERS must be soft.** Hard parts of BUMPERS must not extend more than 1.25in (3.17cm) out from the ROBOT PERIMETER. Only padding (per R402-A), cover (per R402-C, including any coatings or markings), and soft fasteners used to secure padding or cover are permitted beyond this limit. Padding must extend at least 2.0in (5.08cm) beyond any hard parts of the BUMPER. There is no limit on how far BUMPER hard parts may extend inside the ROBOT PERIMETER.

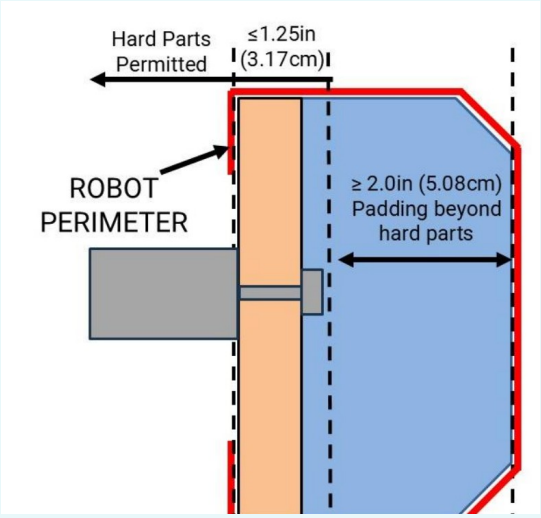



*Figure 7-7 BUMPER hard parts limit*

Hard parts include any items which are likely to cause damage to the cloth or padding of other BUMPERS such as nuts and bolts, cable ties, hard plastics, etc.

**R405** ***BUMPERS interact with BUMPERS.** While navigating the FIELD normally, all BUMPERS must have padding (per R402-A) supported by backing (per R402-B) entirely filling the BUMPER ZONE, a space between 2.5in (6.35cm) and 5.75in (14.61cm) from the floor.

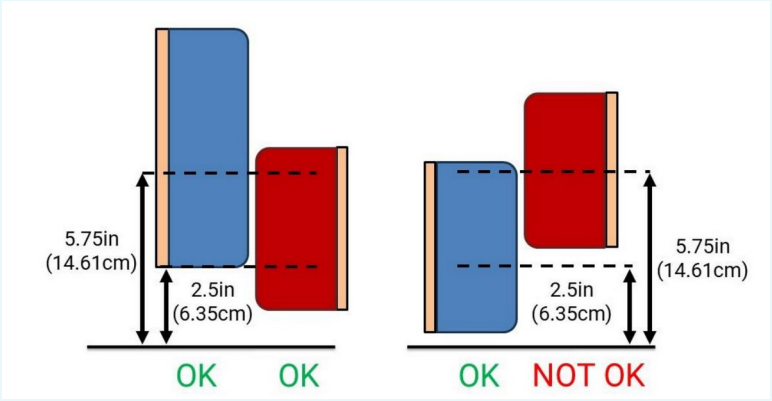



*Figure 7-8 BUMPER ZONE examples*

This measurement is intended to be made in a ROBOT'S expected configuration(s) while navigating the FIELD and is relative to whatever surface the ROBOT is currently driving on. BUMPERS leaving the BUMPER ZONE due to ROBOTS inadvertently tipping is not a violation of this rule. A ROBOT climbing onto or fully supported by the CHECKOUT ZONE tables is covered by the exception in G414-E.

**R406** ***Fill BUMPER corners.** Corner joints between BUMPERS must be filled with uncompressed padding material extending at least 2.25in (5.72cm) from the corner with no gaps or voids. Examples of implementation are shown in Figure 7-9.

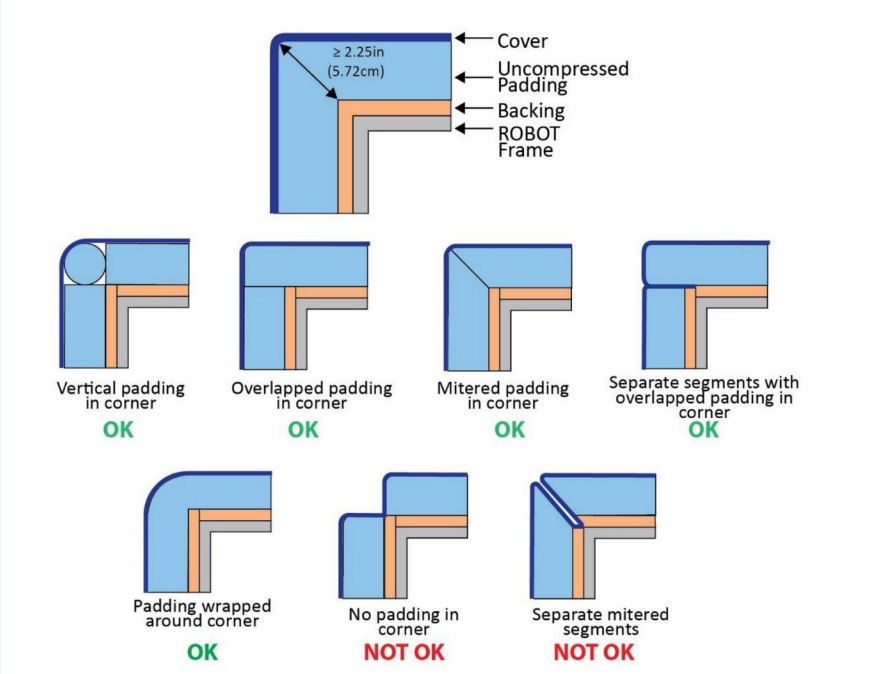



*Figure 7-9 Uncompressed corner padding*

> Separate BUMPER segments meeting at a miter in the corner are not considered to "fill" the corner due to the cloth covering and do not meet the requirements of this rule.

**R407** ***BUMPERS shouldn't be wedges.** BUMPERS must not act as wedges when interacting with other BUMPERS.

> Examples of BUMPERS which might act as a wedge, and may invite additional scrutiny, include:
>
> A. BUMPERS utilizing softer foam towards the top of a BUMPER profile,
> B. BUMPERS utilizing excessively rounded profiles such as a single semi-circle, and
> C. BUMPERS with tapered or stepped padding such that the padding is thinner or thicker towards the top or bottom.

**R408** ***Weight limit with BUMPERS.** The total weight of the ROBOT (as described in R103) with BUMPERS must not exceed 135.0lb (61.23kg).

**R409** ***BUMPERS should be passive.** BUMPERS must be fixed relative to the ROBOT PERIMETER. BUMPERS may not contain any elements which move during the MATCH or electrical elements.

> Compression and flex of BUMPER covers and/or padding materials, and incidental, inconsequential compression and flex in backing and/or fastening systems are not considered violations of this rule.

**R410** ***BUMPERS must come off.** BUMPERS must be designed for installation and removal to facilitate inspection and weighing.

> As a guideline, BUMPERS should be able to be installed or removed by 2 people in fewer than 5 minutes.

**R411** ***BUMPERS indicate your ALLIANCE.** Each ROBOT must be able to display red or blue BUMPER covers to reflect their ALLIANCE color, as assigned in the MATCH schedule distributed at the event. Contrasting markings on outward, upward, or downward facing surfaces of the BUMPER cover, other than the following, are prohibited:

A. those required per R412,
B. hook-and-loop tape, snap fasteners, or functional equivalents backed by the hard parts of the BUMPER,
C. solid white FIRST logos between 4.75in (12.06cm) and 5.25in (13.33cm) wide (i.e. comparable to those available in the Virtual Kit), and
D. narrow areas of underlying cloth exposed at seams, corners, or folds.

**R412** ***Team number on BUMPERS.** Team numbers must be displayed and positioned on the BUMPER cover in at least 3 locations approximately 90 degrees apart such that an observer walking around the perimeter of the ROBOT can unambiguously tell the team's number from most points of view, from as far as approximately 720.0in (18.29m). Team numbers must meet the following additional criteria:

A. consist of only white Arabic numerals at least 3.75in (9.53cm) high, at least 0.5in (1.27cm) in stroke width,

B. must not wrap around sharp corners (less than 135.0°) of the ROBOT PERIMETER,

C. must not split individual digits such that the team number is ambiguous, and

D. may not substitute logos or icons for numerals.

> The 0.5in (1.27cm) stroke width requirement applies to the majority of the stroke. Font elements less than 0.5in (1.27cm) such as serifs, rounded edges, small hairlines or gaps, etc. are permitted as long as the majority of the stroke meets the sizing requirement and the numbers are unambiguous. As a guideline, spacing between digits or groups of digits which exceeds approximately 4.0in (10.16cm) may be ambiguous.
>
> There is no prohibition against splitting team numbers onto different sections of BUMPER. The intent is that the team's number is clearly visible and unambiguous so that Judges, REFEREES, Announcers, and other teams can easily identify competing ROBOTS. This marking is intended to display the team number only, not to intentionally change the surface characteristics of the BUMPER. Excessive material usage as part of any team number marking will invite close scrutiny.

## 7.5 Motors & Actuators

**R501** ***Allowable motors.** The only motors and actuators permitted include the following (in any quantity):

Table 7-1 *Motor allowances*

| Motor Name | Part Numbers Available |
|---|---|
| AndyMark 9015 | am-0912, AndyMark 9015 |
| AndyMark NeveRest | am-3104 |
| AndyMark PG | am-2161 (alt. PN am-2765), am-2194 (alt. PN am-2766) |
| AndyMark RedLine Motor | am-3775, am-3775a |
| AndyMark Snow Blower Motor | am-2235, am-2235a |
| Banebots | am-3830, M5 – RS550-12, M7-RS775-18, RS550VC-7527, RS775WC-8514, RS550 |
| CIM | FR801-001, PM25R-45F-1004, M4-R0062-12, PM25R-45F-1003, AM802-001A, PMR25R-45F-1003, 217-2000, PMR25R-44F-1005, PM25R-44F-1005, am-0255 |
| CTR Electronics Minion | 24-777378, WCP-1691 |

| Motor Name | Part Numbers Available |
|---|---|
| CTR Electronics/VEX Robotics Falcon 500 | 217-6515, 19-708850, am-6515, am-6515_Short |
| Current/former KOP automotive motors | Denso AE235100-0160, Denso 262100-3040, Denso 5-163800-RC1, Bosch 6 004 RA3 194-06, Denso 262100-3030, Johnson Electric JE-PLG-149, Johnson Electric JE-PLG-410 |
| Playing with Fusion Venom | BDC-10001 |
| REV Robotics HD Hex | REV-41-1291 |
| REV Robotics NEO Brushless | REV-21-1650 (v1.0 or v1.1), am-4258, REV-21-1653, am-4258a |
| REV Robotics NEO 550 | REV-21-1651, am-4259 |
| REV Robotics NEO Vortex | REV-21-1652, am-5275 |
| Thrifty Bot Pulsar 775 | TTB-0350 |
| VEX BAG | 217-3351 |
| VEX Mini-CIM | 217-3371 |
| West Coast Products Kraken X44 | WCP-0941 |
| West Coast Products Kraken X60 | WCP-0940, am-5274 |
| West Coast Products RS775 Pro | 217-4347 |
| Fans, no greater than 120mm (nominal) size and rated electrical input power no greater than 10 watts (W) continuous duty at 12 volts (VDC) | |
| Hard drive motors part of a legal COTS computing device | |
| Factory installed vibration and autofocus motors resident in COTS computing devices (e.g. rumble motor in a smartphone) | |
| PWM COTS rotational servos with stall current ≤ 4A and mechanical output power ≤ 8W at 6V. PWM COTS linear servos with max stall current ≤ 1A at 6V. | |
| Motors integral to a COTS sensor (e.g. LIDAR, scanning sonar, etc.), provided the device is not modified except to facilitate mounting | |
| 1 compressor compliant with R806 and used to compress air for the ROBOT'S pneumatic system | |
| COTS brushed motors, linear actuators, electrical solenoid actuators, or electromagnets rated for 12V and wired to an approved power regulating device downstream of a breaker 20A or less. Electrical solenoid actuators or electromagnets used at 24V must be rated for 24V. | |

> For servos, note that the roboRIO is limited to a max current output of 2.2A on the 6V rail (12.4W of electrical input power). Teams should make sure that their total servo power usage remains below this limit at all times. Servo mechanical output power is approximated by: Mechanical Output Power (in W) = 0.25 x (Stall Torque in N-m) x (No Load Speed in rad/s), using 6V data reported by the manufacturer.
>
> Given the extensive amount of motors allowed on the ROBOT, teams are encouraged to

consider the total power available from the ROBOT battery during the design and build of the ROBOT. Drawing large amounts of current from many motors at the same time could lead to drops in ROBOT battery voltage that may result in tripping the main breaker or trigger the brownout protection of the roboRIO.

AndyMark PG Gearmotors are sold with labeling based on the entire assembly. Assemblies labeled am-3651 through am-3656 contain legal motors specified in Table 7-1. These motors may be used with or without the provided gearbox.

**R502** ***Only 4 propulsion motors.** A ROBOT may not have more than 4 propulsion motors. A propulsion motor is a motor that enables the ROBOT to move around the FIELD surface (i.e., carpet). Motors that generate small amounts of thrust as a secondary or incidental feature are not considered propulsion motors.

Examples that are not considered propulsion motors include:

A. motors that primarily alter the alignment of a wheel in contact with the FIELD surface (such as a swerve steering motor),
B. motors that run MECHANISM wheels (e.g. for SCORING ELEMENT manipulation) that occasionally happen to contact the carpet, but without enough force to generate significant thrust,
C. motors that change the speed of the drive wheels using a shifting MECHANISM without significantly contributing to propulsion, and
D. motors that enable the ROBOT to move via contact with non-carpeted surfaces of FIELD elements (e.g. the CHECKOUT ZONE tables).

**R503** ***Don't modify motors (mostly).** The integral mechanical and electrical system of any motor must not be modified. Motors, servos, and electric solenoids used on the ROBOT shall not be modified in any way, except as follows:

A. The mounting brackets and/or output shaft/interface may be modified to facilitate the physical connection of the motor to the ROBOT and actuated part.
B. The electrical leads may be trimmed to length as necessary and connectors or splices to additional wiring may be added.
C. The locking pins on the window motors (P/N 262100-3030 and 262100-3040) may be removed.
D. The connector housings on KOP automotive motors listed in Table 7-1 may be modified to facilitate lead connections.
E. Servos may be modified as specified by the manufacturer (e.g. re-programming or modification for continuous rotation).
F. Minimal labeling may be applied to indicate device purpose, connectivity, functional performance, etc.
G. Any number of #10-32 plug screws may be removed from the Falcon 500 and the Kraken X60.
H. Insulation may be applied to electrical terminals.
I. Repairs, provided the original performance and specifications are unchanged.
J. Maintenance recommended by the manufacturer.

> The intent of this rule is to allow teams to modify mounting tabs and the like, not to gain a weight reduction by potentially compromising the structural integrity of any motor.

**R504** ***Power (most) actuators off of approved devices.** With the exception of servos, fans, or motors integral to sensors of COTS computing devices permitted in R501, each actuator must be controlled by a power regulating device. The only power regulating devices for actuators permitted on the ROBOT include:

A. motor controllers:
   a. Koors40 Motor Controller (P/N am-5600),
   b. Spark Flex Motor Controller (P/N REV-11-2159, am-5276),
   c. Spark Motor Controller (P/N REV-11-1200, am-4260),
   d. Spark MAX Motor Controller (P/N REV-11-2158, am-4261),
   e. Talon FX Motor Controller (P/N 217-6515, 19-708850, am-6515, am-6515_Short, WCP-0940, WCP-0941) for controlling integral Falcon 500, Kraken X60, Kraken X44 only,
   f. Talon FXS Motor Controller (P/N 24-708883, WCP-1692),
   g. Talon Motor Controller (P/N CTRE_Talon, CTRE_Talon_SR, and am-2195),
   h. Talon SRX Motor Controller (P/N 217-8080, am-2854, 14-838288),
   i. Thrifty Nova (P/N TTB-0100),
   j. Venom Motor with Controller (P/N BDC-10001) for controlling integral motor only,
   k. Victor SP Motor Controller (P/N 217-9090, am-2855, 14-868380), and
   l. Victor SPX Motor Controller (P/N 217-9191, 17-868388, am-3748).

B. relay modules:
   a. Spike H-Bridge Relay (P/N 217-0220 and SPIKE-RELAY-H),
   b. Automation Direct Relay (P/N AD-SSR6M12-DC-200D, AD-SSRM6M25-DC-200D, AD-SSR6M40-DC-200D), and
   c. Power Distribution Hub (PDH) switched channel (P/N REV-11-1850) for controlling non-actuator CUSTOM CIRCUITS only.

C. pneumatics controllers:
   a. Pneumatics Control Module (P/N am-2858, 217-4243) and
   b. Pneumatic Hub (P/N REV-11-1852).

D. servo controllers:
   a. Servo Hub (P/N REV-11-1855).

> Note: The Automation Direct Relays are single directional. Per this rule they may not be wired together in an attempt to provide bi-directional control.
>
> Motor controllers allowed in 2022 but no longer listed (e.g. Jaguar, DMC 60, SD540, Victor 884/888, Nidec Dynamo) are not legal for FROZEN BLITZ.

**R505** ***Don't overload controllers.** Each power regulating device may control electrical loads per Table 7-2. Unless otherwise noted, each power regulating device shall control 1 and only 1 electrical load.

Table 7-2 *Power regulating device allotments*

| Electrical Load | Motor Controller | Relay Module | Pneumatics Controller |
|---|---|---|---|
| AndyMark RedLine Motor, Banebots, CIM, CTR Electronics Minion, REV Robotics NEO Brushless, REV Robotics NEO 550, REV Robotics NEO Vortex, ThriftyBot Pulsar 775, VEX Mini-CIM, WCP RS775 Pro | Yes | No | No |
| AndyMark 9015, VEXpro BAG | Yes (up to 2 per controller) | No | No |
| AndyMark PG, KOP Automotive Motors, NeveRest, Snow Blower Motor, REV Robotics HD Hex | Yes (up to 2 per controller) | Yes | No |
| Other Brushed Motor, Linear Actuator | Yes (20A breaker max) | Yes (20A breaker max) | No |
| CTR Electronics/VEX Falcon 500, Playing With Fusion Venom, WCP Kraken X44, WCP Kraken X60 | Yes (integrated controller only) | No | No |
| Compressor | No | Yes | Yes |
| Pneumatic Solenoid Valves | No | Yes (multiple) | Yes (1 per channel) |
| Electric Solenoids | Yes (multiple) | Yes (multiple) | Yes (1 per channel) |
| CUSTOM CIRCUITS | Yes (multiple) | Yes (multiple) | Yes (multiple) |

**R506** ***Control servos safely.** Servos must be connected to, and only to, 1 of the following:

A. PWM ports on the roboRIO,
B. PWM ports on a WCP Spartan Sensor Board (P/N WCP-0045),
C. REV Robotics Servo Power Module (P/N REV-11-1144), or
D. REV Robotics Servo Hub (P/N REV-11-1855).

## 7.6 Power Distribution

In order to maintain safety, the rules in this section apply at all times while at the event, not just while the ROBOT is on the FIELD for MATCHES.

**R601** ***Battery limit – everyone has the same power.** The only legal source of electrical energy for the ROBOT during the competition, the ROBOT battery, must be 1 and only 1 non-spillable sealed lead acid (SLA) battery with the following specifications:

A. Nominal voltage: 12V
B. Nominal capacity at 20-hour discharge rate: minimum 17Ah, maximum 18.2Ah

C. Shape: Rectangular

D. Nominal Dimensions: 7.1in x 3.0in x 6.6in, +/- 0.1in for each dimension (18.03cm x 7.62cm x 16.76cm, +/-0.25cm for each dimension)

E. Nominal weight: 11.0lb to 14.5lb (4.99kg to 6.57kg)

F. Terminals: Nut and bolt style

G. Battery vents must not be obstructed during charging.

> "Nut and bolt style" refers to any style battery terminal where the connector is secured to the battery using a threaded fastener.
>
> Examples of batteries which meet these criteria include: Enersys (P/N NP18-12, NP18-12B, NP18-12BFR), MK Battery (P/N ES17-12), Battery Mart (P/N SLA-12V18), Sigma (P/N SP12-18), Universal Battery (P/N UB12180), Power Patrol (P/N SLA1116), Werker Battery (P/N WKA12-18NB), Power Sonic (P/N PS-12180NB), Yuasa (P/N NP18-12B), Panasonic (P/N LC-RD-1217), Interstate Batteries (P/N BSL1116), and Duracell Ultra Battery (P/N DURA12-18NB).
>
> Teams should be aware that they may be asked to provide documentation of the specifications of any battery not listed above. Batteries should be charged in accordance with manufacturer's specification. While teams should generally attempt to leave the battery vent as unobstructed as possible during charging, as long as the short edges of the vent rectangle are not sealed or directly covered the vent will be considered not obstructed.

**R602** ***Other batteries for cameras or computers only.** COTS USB battery packs with a capacity of 100Wh or less (27000mAh at 3.7V) with 5V/5A max output or 12V/5A max output using USB-PD per port, batteries integral to and part of a COTS computing device or self-contained camera (e.g. laptop batteries, GoPro style camera, etc.), or batteries used to power CMOS/RTC features may be used to power COTS computing devices and any peripheral COTS input or output devices connected to the COTS computing device provided they are:

A. securely fastened to the ROBOT,

B. connected only using unmodified COTS cables, and

C. charged according to manufacturer recommendations.

> A COTS computing device is a non-roboRIO device used to process or collect sensor information (e.g. a "smart flashlight" is not a COTS computing device).

**R603** ***Charge batteries with safe connectors.** Any battery charger used to charge a ROBOT battery must have the corresponding Anderson SB connector installed.

**R604** ***Charge batteries at a safe rate.** Any battery charger used to charge a ROBOT battery may not be used such that it exceeds 6-Amp average charge current.

**R605** ***Batteries are not ballast.** No batteries other than those allowed per R601 and R602 are allowed on the ROBOT, whether or not they are being used to supply power.

> For example, teams may not use additional batteries as extra weight on their ROBOTS.

**R606** ***Secure the battery.** The ROBOT battery must be secured such that it will not dislodge during vigorous ROBOT interaction including if the ROBOT is turned over or placed in any arbitrary orientation.

**R607** ***Insulate battery connections.** Each electrical terminal on the ROBOT battery, main breaker, and their connections (lugs, stripped wire ends, etc.) to the wire must be fully insulated at all times.

**R608** ***Limit non-battery energy.** Non-electrical sources of energy used by the ROBOT (i.e., stored at the start of a MATCH) shall come only from the following sources:

A. compressed air stored in the pneumatic system that has been charged in compliance with R806 and R807,
B. a change in the altitude of the ROBOT center of gravity,
C. storage achieved by deformation of ROBOT parts,
D. closed-loop COTS pneumatic (gas) shocks, or
E. air-filled (pneumatic) wheels.

**R609** ***Connect main power safely.** The following devices shall be connected with 6 AWG (7 SWG or 16 mm²) copper wire or larger as shown in Figure 7-10:

A. 1 ROBOT battery
B. a single pair of Anderson Power Products (or APP) 2-pole SB type connectors
C. a single main 120-Amp (120A) surface mount circuit breaker (Cooper Bussman P/N CB185-120, CB185F-120, CB285-120, CB285F-120, CB285120F or Optifuse P/N 153120, 253120)
D. a single main power distribution device (PD): CTR Electronics Power Distribution Panel (PDP) P/N am-2856, 217-4244, 14-806880; CTR Electronics Power Distribution Panel 2.0 (PDP 2.0) P/N 24-806880, WCP-1690; REV Robotics Power Distribution Hub (PDH) P/N REV-11-1850; or AndyMark Power Distribution (AMPD) P/N am-5754

No additional devices or modifications shall be permitted with the following exceptions:

A. Monitoring circuitry permitted by R625
B. SB-50 to SB-120 adapters, provided they have been presented for Inspection
C. Termination of wires with appropriate cable lugs

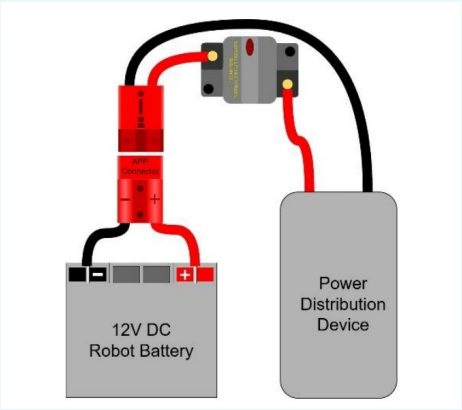



*Figure 7-10 Electrical connection diagram*

"SB type" refers to SB type only (e.g. SB-50, SB-120, etc.), not SBS or any other part type beginning with SB. While pure copper wire is recommended, copper-clad aluminum wire is

considered copper wire.

**R610** ***1 breaker/fuse per circuit.** All circuits, with the exceptions of those listed in R615 and R617, must connect to, and have power sourced solely by, a single protected connector pair of the PD. Circuits must not connect to the main power input of the PD.

**R611** ***The ROBOT frame is not a wire.** All wiring and electrical devices shall be electrically isolated from the ROBOT frame. The ROBOT frame must not be used to carry electrical current.

Compliance with this rule is checked by observing a >120 ohm resistance between either the (+) or (-) post within the APP connector that is attached to the PD and any point on the ROBOT.

All legal motor controllers with metal cases are electrically isolated. They may be mounted directly to ROBOT frame COMPONENTS.

Note that some cameras, decorative lights, and sensors (e.g. some encoders, some IR sensors, etc.) have grounded enclosures or are manufactured with conductive plastics. These devices must be electrically isolated from the ROBOT frame to ensure compliance with this rule.

**R612** ***Must be able to turn ROBOT on and off safely.** The 120A circuit breaker must be quickly and safely accessible from the exterior of the ROBOT. This is the only 120A circuit breaker allowed on the ROBOT.

Examples considered not “quickly and safely accessible” include breakers covered by an access panel or door, or mounted on, underneath or immediately adjacent to moving COMPONENTS.

It is strongly recommended that the 120A circuit breaker location be clearly and obviously labeled so it can be easily found by FIELD STAFF if needed. While the main breaker must be accessible, consider positioning or shielding it such that it’s protected from accidental actuation (e.g. it’s unlikely to be hit by a SCOOP during game play).

**R613** ***Electrical system must be inspectable.** The PD, associated wiring, and all circuit breakers must be visible for inspection.

“Visible for inspection” does not require that the items be visible when the ROBOT is in STARTING CONFIGURATION, provided the team can make the items viewable during the inspection process.

**R614** ***No high voltage allowed.** Any active electrical item that is not an actuator (specified in R501) or core control system item (specified in R710) is considered a CUSTOM CIRCUIT. CUSTOM CIRCUITS shall not produce measurable voltages exceeding 24V with the exception of COTS Power-over-Ethernet (PoE) injector devices used with COTS Ethernet cables and COTS receiving devices (i.e. not a breakout or adapter).

**R615** ***Power roboRIO as specified.** The roboRIO power input must be connected directly to a non-switched pair of protected output terminals of a PD with a 10A fuse or circuit breaker installed. No other electrical load can be connected to the breaker or fuse supplying this circuit.

**R616** ***Power radio as specified – Part 1.** The wireless bridge (radio) power for VH-109 radios must be sourced from one or both of:

A. injected into the "RIO" port of the radio using a passive injector or modified Ethernet cable connected directly to a PD, and/or
B. wired directly to the 12V power input of the radio from a PD.

> Please reference "How to Wire an FRC Robot" in the FIRST Robotics Competition Control System documentation for wireless bridge wiring information.

**R617** ***Power radio as specified – Part 2.** The device supplying power to the wireless bridge per R616 must be connected directly to a non-switched pair of protected output terminals of a PD with a 10A fuse or circuit breaker installed with the exception of the PDP 1.0 shared VRM/PCM pairs which may be protected with a 20A fuse or circuit breaker. No other electrical load can be connected to the breaker or fuse supplying this circuit with the exception of a single PCM if using the shared VRM/PCM ports of a PDP 1.0.

**R618** ***Use PD terminals as designed.** Only 1 wire shall be connected to each terminal on the PD.

> If multi-point distribution of circuit power is needed (e.g. to provide power to multiple PCMs and/or VRMs from a single 20A circuit), then all incoming wires may be appropriately spliced into the main lead (e.g. using an insulated terminal block, crimped splice or soldered wire splice), and the single main lead inserted into the terminal to power the circuit.

**R619** ***Only use specified circuit breakers in a PD.** The only circuit breakers permitted for use in the PD are:

A. Snap Action VB3-A Series or AT2-A, terminal style F57, 40A rating or lower,
B. Snap Action MX5-A or MX5-L Series, 40A rating or lower,
C. REV Robotics ATO auto-resetting breakers 40A rating or lower,
D. CTR Electronics ATO auto-resetting breakers 40A rating or lower, and
E. any ATM circuit breaker with value less than or equal to the fuses permitted per R620.

**R620** ***Only use specified fuses in PD.** The only fuses permitted for use in the PD are automotive blade fuses with the following values:

A. for the PDP, ATM style fuses with values matching the value printed on the device's corresponding fuse holder,
B. for all PDs, ATC/ATO style fuses with values 10A or lower, and

C. for the PDH, ATM style fuses with values 15A or lower with the exception of a single 20A fuse for powering a PCM or PH.

> Note that these fuses must be pressed very firmly to seat properly. Improper seating can cause a device to reboot upon impact.

**R621** ***Protect circuits with appropriate circuit breakers.** Each branch circuit must be protected by 1 and only 1 circuit breaker or fuse on the PD per Table 7-3. No other electrical load can be connected to the breaker or fuse supplying this circuit with the exception of devices downstream of a permitted motor power adapter board placed between the PD and a motor controller (WCP-1380, WCP-1903, WCP-1904, RF-4003, RF-4004, RF-4005).

Table 7-3 *Branch circuit protection requirements*

| Branch Circuit | Circuit Breaker/Fuse Value | Quantity Allowed Per Breaker |
|---|---|---|
| Motor Controller | Up to 40A | 1 |
| CUSTOM CIRCUIT | Up to 40A | No limit |
| Automation Direct Relay 40A (*6M40*) | Up to 40A | 1 |
| Fans permitted per R501 and not already part of COTS computing devices | Up to 20A | No limit |
| Spike Relay Module | Up to 20A | 1 |
| Automation Direct Relay 25A (*6M25*) | Up to 20A | 1 |
| PCM/PH – with compressor | Up to 20A | 1 |
| Servo Power Module/Servo Hub | Up to 20A | 1 |
| Additional VRM (non-radio)/Additional PCM/PH (non-compressor) | Up to 20A | 3 total |
| Automation Direct Relay 12A (*6M12*) | Up to 10A | 1 |

> This rule does not prohibit the use of smaller value breakers in the PD or any fuses or breakers within CUSTOM CIRCUITS for additional protection. Fans permitted per R501 may also be included within CUSTOM CIRCUITS.

**R622** ***Use appropriately sized wire.** All circuits shall be wired with appropriately sized insulated copper wire (SIGNAL LEVEL cables don't have to be copper):

Table 7-4 *Breaker and wire sizing*

| Application | Minimum Wire Size |
|---|---|
| 31 – 40A breaker protected circuit | 12 AWG (13 SWG or 4 mm²) |
| 21 – 30A breaker protected circuit | 14 AWG (16 SWG or 2.5 mm²) |

| Application | Minimum Wire Size |
|---|---|
| 6 – 20A breaker protected circuit; 11-20A fuse protected circuit; between the PDP dedicated terminals and the VRM/RPM or PCM/PH; compressor outputs from the PCM/PH | 18 AWG (19 SWG or 1 mm²) |
| Motor power adapter board protected circuit; ≤5A breaker protected circuit; ≤10A fuse protected circuit | 22 AWG (22 SWG or 0.5 mm²) |
| VRM 2A circuits; ≤2A fuse protected circuit | 24 AWG (24 SWG or 0.25 mm²) |
| VH-109 Passthrough per R626 | Cat5e/6/7/8 cable, 2 pairs total (1 pair V+, 1 pair ground) |
| roboRIO PWM port outputs; ≤1A fuse protected circuit | 26 AWG (27 SWG or 0.14 mm²) or Cat5e/6/7/8 cable, single pair total |
| SIGNAL LEVEL circuits (i.e. circuits which draw ≤1A continuous and have a source incapable of delivering >1A, including but not limited to roboRIO non-PWM outputs, CAN signals, PCM/PH Solenoid outputs, VRM 500mA outputs, RPM outputs, and Arduino outputs) | 28 AWG (29 SWG or 0.08 mm²) |

Wires that are recommended by the device manufacturer or originally attached to legal devices are considered part of the device and by default legal. Such wires are exempt from this rule, provided they are powered by the smallest value fuse or breaker which permits proper device operation.

> In order to show compliance with these rules, teams should use wire with clearly labeled sizes if possible. If unlabeled wiring is used, teams should be prepared to demonstrate that the wire used meets the requirements of this rule (e.g. wire samples and evidence that they are the required size).

**R623** ***Use only appropriate connectors.** Branch circuits may include intermediate elements such as COTS connectors, splices, COTS flexible/rolling/sliding contacts, and COTS slip rings, as long as the entire electrical pathway is via appropriately gauged/rated elements.

> Slip rings containing mercury are prohibited per R203.

**R624** ***Use specified wire colors (mostly).** All non-SIGNAL LEVEL wiring with a constant polarity (i.e., except for outputs of relay modules, motor controllers, or sensors) shall be color-coded along their entire length from the manufacturer as follows:

A. red, yellow, white, brown, or black-with-stripe on the positive (e.g. +24VDC, +12VDC, +5VDC, etc.) connections,

B. black or blue for the common or negative side (-) of the connections.

Exceptions to this rule include:

A. wires that are originally attached to legal devices and any extensions to these wires using the same color as the manufacturer,

B. Ethernet cable used in PoE cables.

**R625** ***Don't modify critical power paths.** CUSTOM CIRCUITS shall not directly alter the power pathways between the ROBOT battery, PD, motor controllers, relays (per R504-B), motors and actuators (per R501), pneumatic solenoid valves, or other elements of the ROBOT control system (items explicitly mentioned in R710). Custom high impedance voltage monitoring or low impedance current monitoring circuitry connected to the ROBOT'S electrical system is acceptable, if the effect on the ROBOT outputs is inconsequential.

> A noise filter may be wired across motor leads or PWM leads. Such filters will not be considered CUSTOM CIRCUITS and violate neither this rule nor R712. Acceptable signal filters must be fully insulated and must be 1 of the following:
>
> – 1 microfarad (1 µF) or less, non-polarized, capacitor may be applied across the power leads of any motor on your ROBOT (as close to the actual motor leads as reasonably possible) or
> – a resistor may be used as a shunt load for the PWM control signal feeding a servo.

**R626** ***VH-109 PoE passthrough.** The VH-109 PoE output may be used only under the following conditions:

A. The device being powered is a COTS device or COTS adapter connected to a single COTS device with current draw ≤ 2A at 12V.
B. The connection is made using standard Cat5e/6/7/8 cable.
C. The VH-109 is powered using the 12V input terminals with 18AWG wire or larger (it may additionally be powered using the PoE input if desired).

## 7.7 Control, Command & Signals System

**R701** ***Control the ROBOT with a roboRIO.** ROBOTS must be controlled via 1 programmable NI roboRIO or roboRIO 2.0 (P/N am3000 or am3000a, both versions referred to throughout this manual as "roboRIO"), with image version 2026_v1.2 or later.

> There are no rules that prohibit co-processors, provided commands originate from the roboRIO to enable and disable all power regulating devices. This includes motor controllers legally wired to the CAN bus.

**R702** ***Communicate with the ROBOT with the specified radio.** 1 Vivid Hosting wireless bridge (P/N: VH-109, WCP-1538), that has been configured with the appropriate encryption key for your team number at each event, is the only permitted device for communicating to and from the ROBOT during the MATCH.

> The OpenMesh OM5P radios used in 2022 are no longer legal for FROZEN BLITZ.

**R703** ***Use specific Ethernet port for roboRIO.** The roboRIO Ethernet port must be connected as specified:

A. For VH-109 v1.5 radios: to the port labeled "RIO" either directly, via a Cat5 Ethernet pigtail, or via a passive PoE injector cable or adapter.

B. For VH-109 v1.0 radios (either directly or via Cat5 Ethernet pigtail): to the radio port labeled "RIO" via a passive PoE injector cable or adapter (whether or not it is used to power the radio), or an Ethernet cable with the appropriate wires removed on the roboRIO end, with all wires or adapters used fully insulated; or to the radio port labeled "AUX 1" or "AUX 2" with the corresponding DIP switch in the off (default) position.

> Note: Placing a switch between the roboRIO and radio may impede the ability for FIELD STAFF to troubleshoot roboRIO connection issues on the FIELD. Teams may be asked to connect directly between the radio and the roboRIO as part of troubleshooting efforts.
>
> If not using the "RIO" port of the VH-109 1.0, it is strongly recommended to cover the port to prevent accidental damage to devices such as laptops which may occur if attached to this port.

**R704** ***Only use allowed ports and bandwidth to communicate with the ROBOT.** Communication between the ROBOT and the OPERATOR CONSOLE may not exceed 7.0Mbits/second and is restricted to network ports listed in Table 7-5.

Table 7-5 *Open FMS ports*

| Port | Designation | Bi-directional? |
|---|---|---|
| UDP/TCP 1180-1190 | Camera data from the roboRIO to dashboard software when the camera is connected the roboRIO via USB | Yes |
| TCP 1735 | SmartDashboard | Yes |
| UDP 1130 | Dashboard-to-ROBOT control data | Yes |
| UDP 1140 | ROBOT-to-Dashboard status data | Yes |
| HTTP 80 | Camera connected via switch on the ROBOT | Yes |
| HTTP 443 | Camera connected via switch on the ROBOT | Yes |
| UDP/TCP 554 | Real-Time Streaming Protocol for h.264 camera streaming | Yes |
| UDP/TCP 1250 | CTRE Diagnostics Server | Yes |
| UDP/TCP 5800-5810 | Team use | Yes |

Teams may use these ports as they wish if they do not employ them as outlined above (i.e. TCP 1180 can be used to pass data back and forth between the ROBOT and the Driver Station Software if the team chooses not to use the camera on USB).

**R705** ***Configure devices for your team number.** The roboRIO, Driver Station Software, and wireless bridge must be configured to correspond to the correct team number, per the procedures defined in the FIRST Robotics Competition Control System documentation.

**R706** ***Don't bypass the ARENA network.** All signals must originate from the OPERATOR CONSOLE and be transmitted to the ROBOT via the ARENA Ethernet network.

**R707** ***Limited wireless allowed.** The only forms of wireless communication that may be used to communicate to, from, or within the ROBOT, include:

A. those required per R702 and R706,

B. tags used for location detection systems if provided by the event, and

C. RFID or NFC systems used exclusively within the ROBOT.

> Devices that employ signals in the visual spectrum (e.g. cameras) and non-RF sensors that don't receive human-originated commands (e.g. "beam break" sensors or IR sensors on the ROBOT used to detect FIELD elements) are not wireless communication devices and thus this rule doesn't apply.

**R708** ***Wireless bridge must be visible.** The wireless bridge must be mounted on the ROBOT such that the diagnostic lights are visible to FIELD STAFF.

> Teams are encouraged to mount the wireless bridge away from noise generating devices such as motors, PCM(s)/PH(s), and VRM(s)/RPM(s).

**R709** ***ROBOTS must have a signal light.** ROBOTS must use at least 1, but no more than 2, diagnostic ROBOT Signal Lights (RSL) (P/N 855PB-B12ME522 and/or am-3583). Any RSL must be:

A. mounted on the ROBOT such that it is easily visible while standing 36.0in (91.44cm) away from at least one side of the ROBOT,

B. connected to the "RSL" supply terminals on the roboRIO, and

C. if using the 855PB-B12ME522, wired for solid light operation, by placing a jumper between the "La" and "Lb" terminals on the light per Figure 7-11.

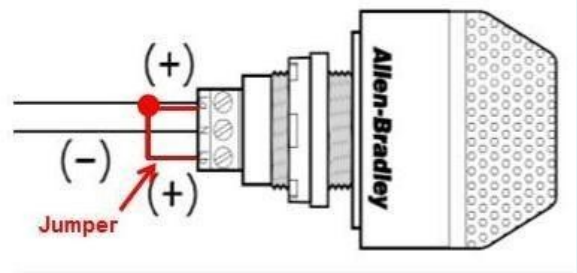



*Figure 7-11 855PB-B12ME522 jumper wiring*

**R710** ***Only specified modifications to control system devices permitted.** The Driver Station Software, roboRIO, PD, PCM(s)/PH(s), VRM(s)/RPM(s), RSL, 120A breaker, motor controllers, MXP devices used to control actuators per R713-C, relay modules (per R504-B), wireless bridge, PD breakers and fuses, Servo Power Module, Servo Hub, motor power adapter boards, and batteries shall not be tampered with, modified, or adjusted in any way (tampering includes drilling, cutting, machining, rewiring, disassembling, painting, etc.), with the following exceptions:

A. User programmable code in the roboRIO may be customized.
B. Motor controllers may be calibrated as described in owner's manuals.
C. Fans may be attached to motor controllers and may be powered from the power input terminals.
D. If powering the compressor, the fuse on a Spike H-Bridge Relay may be replaced with a VB3A-20A Snap-Action circuit breaker.
E. Wires, cables, and signal lines may be connected via the standard connection points provided on the devices.
F. Fasteners (including adhesives) may be used to attach the device to the OPERATOR CONSOLE or ROBOT or to secure cables to the device.
G. Thermal interface material may be used to improve heat conduction.
H. Labeling may be applied to indicate device purpose, connectivity, functional performance, etc.
I. Jumpers may be changed from their default location.
J. Limit switch jumpers may be removed from a Jaguar motor controller and a custom limit switch circuit may be substituted.
K. Device firmware may be updated with manufacturer supplied firmware.
L. Integral wires on motor controllers may be cut, stripped, and/or connectorized.
M. Devices may be repaired, provided the performance and specifications of the device after the repair are identical to those before the repair.
N. The cover may be removed from the Talon SRX or Talon FX data port.
O. Electrical tape may be applied to the aluminum plate inside the wireless bridge.
P. The input terminal cover from the PDP may be omitted (no other element may be installed using the threaded holes to install something in place of the PDP terminal cover).
Q. The roboRIO 2.0 SD card may be replaced with an SD card of any capacity.
R. Insulating material may be added to exposed conductors.
S. Control system power terminal blocks (e.g. RSL power connector) may be replaced with functional equivalents.
T. Tape may be applied for debris protection.
U. VH-109 v1.0 may be upgraded to VH-109 v1.5 using manufacturer materials and instructions.

> Please note that the Driver Station Software is a separate application from the Dashboard. The Driver Station Software may not be modified, while teams are expected to customize their Dashboard code.
>
> While repairs are permitted, the allowance is independent of any manufacturer's warranty. Teams make repairs at their own risk and should assume that any warranty or return options are forfeited.

**R711** ***Don't connect motor outputs to roboRIO.** Neither 12VDC power nor relay module or motor controller outputs shall be directly connected to the roboRIO, with the exception of the designated 12VDC input.

**R712** ***Control PWM controllers from the roboRIO.** Every relay module (per R504-B), servo, Servo Power Module, and PWM motor controller shall be connected to a corresponding port (relays to Relay ports, servos, and PWM controllers to PWM ports) on the roboRIO (either directly or through a WCP Spartan Sensor Board) or via a legal MXP connection (per R713). They shall not be controlled by signals from any other source.

**R713** ***Only approved MXP devices can control actuators.** If a motor is controlled via the MXP, its power regulating device must be connected by 1 of the following methods:

A. directly to any PWM pins,

B. via a network of PASSIVE CONDUCTORS used to extend the PWM pins, or

C. via 1 approved ACTIVE DEVICE: Kauai Labs navX MXP, Kauai Labs navX2 MXP, RCAL MXP Daughterboard, REV Robotics RIOduino, REV Robotics Digit Board, West Coast Products Spartan Sensor Board, or Huskie Robotics HUSKIE 2.0 Board.

> A PASSIVE CONDUCTOR is any device or circuit whose capability is limited to the conduction and/or static regulation of the electrical energy applied to it (e.g. wire, splices, connectors, printed wiring board, etc.).
>
> An ACTIVE DEVICE is any device capable of dynamically controlling and/or converting a source of electrical energy by the application of external electrical stimulus.
>
> The "network of PASSIVE CONDUCTORS" only applies to the pins being used for PWM output to motors or servos. This means that connecting an ACTIVE DEVICE, such as a sensor to 1 MXP pin does not prevent other MXP pins from being used in accordance with B.

**R714** ***Control CAN motor controllers from the roboRIO.** Each CAN motor controller must be controlled with enable/disable inputs sourced from the roboRIO and passed via either a PWM (wired per R713) or CAN bus (either directly or daisy-chained via another CAN bus device) signal, but both shall not be wired simultaneously on the same device.

> As long as the CAN bus is wired legally so that the heartbeat from the roboRIO is maintained, all closed loop control features of the CAN motor controller may be used. (That is, commands originating from the roboRIO to configure, enable, and specify an operating point for all CAN motor controller closed loop modes fit the intent of R701).
>
> "Wired directly" includes via any series of PASSIVE CONDUCTORS (i.e. star or hub configurations using only PASSIVE CONDUCTORS are permitted.)

**R715** ***Control PCM, PH, and Servo Hub from roboRIO.** Each Pneumatic Control Module (PCM), Pneumatic Hub (PH), and Servo Hub must be controlled with signal inputs sourced from the roboRIO and passed via a CAN bus connection from the built-in CAN on the roboRIO (either directly or daisy-chained via another CAN bus device).

**R716** ***Don't alter the CAN bus.** No device that interferes with, alters, or blocks communications among the roboRIO and the PCMs/PHs, and/or CAN motor controllers on the bus will be permitted.

> Only 1 wire should be inserted into each Weidmuller CAN connector terminal. For documentation on how to wire the CAN bus connections see "How to Wire an FRC Robot."

**R717** ***USB to CAN adapter permitted.** Additional CAN bus connections may be added to the roboRIO using the CTR Electronics CANivore™ (P/N 21-678682, WCP-1522) USB-to-CAN adapter.

> Any additional CAN bus added in this manner satisfies the requirements of R714 (i.e. you may connect motor controllers to this additional bus).

## 7.8 Pneumatic System

In order to maintain safety, the rules in this section apply at all times while at the event, not just while the ROBOT is on the FIELD for MATCHES.

**R801** ***Only use explicitly permitted pneumatic parts.** To satisfy multiple constraints associated with safety, consistency, inspection, and constructive innovation, no pneumatic parts other than those explicitly permitted in this section shall be used on the ROBOT.

**R802** ***No custom pneumatics and meet minimum pressure ratings.** All pneumatic items must be COTS pneumatic devices and either:

A. rated by their manufacturers for pressure of at least 125.0psi (861.8kPa, 8.618Bar), or

B. installed downstream of the primary relieving regulator (see R809), and rated for pressure of at least 70.0psi (482.6kPa, 4.826Bar).

> Any pressure specification such as "working," "operating," "maximum," etc. may be used to satisfy the requirements of this rule. It is recommended that all pneumatic items be rated by their manufacturers for a working pressure of at least 60.0psi (413.7kPa, 4.137Bar).

**R803** ***Don't modify pneumatics.** All pneumatic COMPONENTS must be used in their original, unaltered condition. Exceptions are as follows:

A. tubing may be cut,

B. wiring for pneumatic devices may be modified to interface with the control system,

C. assembling and connecting pneumatic COMPONENTS using the pre-existing threads, mounting brackets, quick-connect fittings, etc.,

D. removing the mounting pin from a pneumatic cylinder, provided the cylinder itself is not modified, and

E. labeling applied to indicate device purpose, connectivity, functional performance, etc.

> Do not, for example, paint, file, machine, or abrasively remove any part of a pneumatic COMPONENT – this would cause the part to become a prohibited item.

**R804** ***Only use specific pneumatic devices.** The only pneumatic system items permitted on ROBOTS include the following items:

A. pneumatic pressure vent plug valves functionally equivalent to those provided in the KOP (e.g. Parker PV609-2 or MV709-2),

B. pressure relief valves functionally equivalent to those provided in the KOP (e.g. Norgren 16-004-011, 16-004-003 or McMaster-Carr 48435K714); to be considered functionally equivalent the valve must be preset or adjustable to 125.0psi (861.8kPa, 8.618Bar) and capable of relieving at least 1.0scfm (471.9cm³/s),

C. solenoid valves with a maximum 1/8in (nominal, 0.31cm) NPT, BSPP, or BSPT port diameter or integrated quick connect 1/4in (nominal, 0.64cm) outside diameter tubing connection,

D. additional pneumatic tubing, with a maximum 1/4in (nominal, 0.64cm) outside diameter,

E. pressure transducers, pressure gauges, passive flow control valves (specifically "needle valve"), manifolds, and connecting fittings (including COTS pneumatic U-tubes),

F. check and quick exhaust valves, provided that the requirements of R813 are still met,

G. shutoff valves which relieve downstream pressure to atmosphere when closed (may also be known as 3-way or 3-way exhausting valves),

H. pressure regulators with the maximum outlet pressure adjusted to no more than 60.0psi (413.7kPa, 4.137Bar),

I. pneumatic cylinders, pneumatic linear actuators, and rotary actuators,

J. pneumatic storage tanks (with the exception of white Clippard tanks P/N AVT-PP-41),

K. 1 compressor that is compliant with R806,

L. debris or coalescing (water) filters, and

M. Venturi valves (note: the high-pressure side of a Venturi valve is considered a pneumatic device and must follow all pneumatic rules. The vacuum side of a Venturi valve is exempt from the pneumatic rules per "a" in the blue box below).

> The following devices are not considered pneumatic devices and are not subject to pneumatic rules (though they must satisfy all other rules):
>
> a. a device that creates a vacuum,
>
> b. closed-loop COTS pneumatic (gas) shocks,
>
> c. air-filled (pneumatic) wheels, and
>
> d. pneumatic devices not used as part of a pneumatic system (i.e. used in a way that does not allow them to contain pressurized air).

**R805** ***If using pneumatics, these parts are required.** If pneumatic COMPONENTS are used, the following items are required as part of the pneumatic circuit and must be used in accordance with this section, as illustrated in Figure 7-12.

A. 1 FIRST Robotics Competition legal compressor (per R806),

B. a pressure relief valve (per R804-B) connected and calibrated (per R811),

C. a Nason pressure switch (P/N SM-2B-115R/443) and/or REV Robotics Analog Pressure Sensor (P/N REV-11-1107) connected and wired per R812,

D. at least 1 pressure vent plug plumbed (per R813),

E. stored pressure gauge and working pressure gauge (per R810), and

F. 1 primary working pressure regulator (per R808).

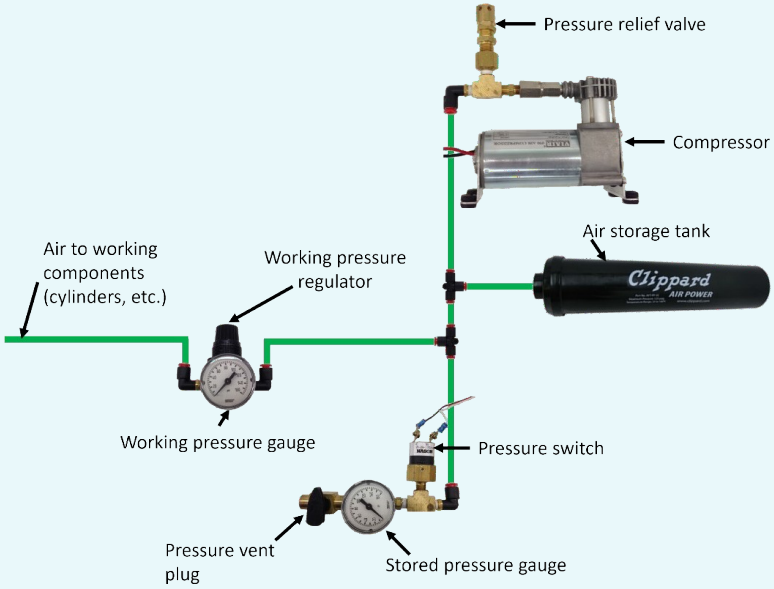



*Figure 7-12 Pneumatic circuitry*

**R806** ***Compressed air from ROBOT compressor only.** Throughout an event, compressed air on the ROBOT must be provided by its 1 onboard compressor only. Compressor specifications must not exceed nominal 1.1cfm (519.1cm³/s) flow rate @ 12VDC at any pressure.

A ROBOT'S compressor may be substituted by another compressor, but a ROBOT may only have 1 designated compressor at a time, and all compressed air on the ROBOT must be sourced from a single compressor.

Note: Viair C-series compressors, which have a max working pressure of 120.0psi, are rated for intermittent pressures greater than 125.0psi and therefore meet the requirements of this rule.

**R807** ***Air storage pressure limit.** Stored air pressure on the ROBOT must be no greater than 120.0psi (~827kPa, 8.2Bar). No stored air pressure intended for the ROBOT may be located off-board the ROBOT.

**R808** ***Working air pressure limit.** Working air pressure (air pressure used to actuate devices) on the ROBOT must be no greater than 60.0psi (413.7kPa, 4.137Bar) and must be provided through a single primary adjustable, relieving, pressure regulator. Additional regulators may be located downstream of the single primary regulator.

Examples of acceptable valves include Norgren regulator P/N R07-100-RNEA and Monnier P/N

> 101-3002-1.

**R809** ***Limited devices at high pressure.** Only the compressor, relief valve, pressure switch, pressure vent plug, pressure gauge, storage tanks, tubing, pressure transducers, filters, and connecting fittings may be in the high-pressure pneumatic circuit upstream from the regulator.

> It is recommended that all COMPONENTS in the high-pressure pneumatic circuit upstream from the regulator be rated for at least 115.0psi (792.9kPa, 7.929Bar) working pressure.

**R810** ***Pressure gauges must be visible.** Pressure gauges must be placed in easily visible locations upstream and downstream of the regulator to display the stored and working pressures, respectively. Pressure gauges must show pressure in psi, kPa, or Bar.

**R811** ***Relief valve requirements.** The relief valve must be attached directly to the compressor or attached by legal hard fittings (e.g. brass, nylon, etc.) connected to the compressor output port. Teams are required to check and/or adjust the relief valve to release air at 125.0psi (861.8kPa, 8.618Bar). The valve may or may not have been calibrated prior to being supplied to teams.

**R812** ***Pressure switch requirements.** The pressure switch must be connected to the high-pressure side of the pneumatic circuit (i.e. prior to the pressure regulator) to sense the stored pressure of the circuit. It must be either:

A. Nason P/N SM-2B-115R/443: the 2 wires from the pressure switch must be connected directly to the pressure switch input of the PCM/PH controlling the compressor or, if controlled using the roboRIO and a relay, to the roboRIO. If connected to the roboRIO, the roboRIO must be programmed to sense the state of the switch and operate the relay module that powers the compressor to prevent over-pressuring the system, and/or

B. REV Robotics P/N REV-11-1107: the analog output of the sensor must be connected directly to analog input 0 of the PH (with firmware version 22.0.2 or newer) controlling the compressor.

> The REV Robotics Analog Pressure Sensor may only be used with PH compressor control and may not be used with PCM compressor control.

**R813** ***Vent plug requirements.** Any pressure vent plug must be:

A. connected to the pneumatic circuit such that, when manually operated, it will vent to the atmosphere to relieve all stored pressure in a reasonable amount of time and

B. placed on the ROBOT so that it is visible and easily accessible.

**R814** ***Don't connect solenoid outputs together.** The output air from multiple solenoid valves must not be combined.

> Manifolds, shuttle valves, and other devices which do not combine output airflow, even though it may be plumbed into the same device, are not violations of this rule.

## 7.9 OPERATOR CONSOLE

**R901** ***Use the specified Driver Station Software.** The Driver Station Software provided by National Instruments is the only application permitted to specify and communicate the operating mode (i.e. AUTO/TELEOP) and operating state (Enable/Disable) to the ROBOT. The Driver Station Software must be version 26.0 or newer.

> Teams are permitted to use a portable computing device of their choice (laptop computer, tablet, etc.) to host the Driver Station Software while participating in MATCHES.

**R902** ***The OPERATOR CONSOLE must have a visible display.** The OPERATOR CONSOLE, the set of COMPONENTS and MECHANISMS used by the DRIVERS and/or HUMAN PLAYERS to relay commands to the ROBOT, must include a graphic display to present the Driver Station Software diagnostic information. It must be positioned within the OPERATOR CONSOLE so that the screen display can be clearly seen during inspection and in a MATCH.

**R903** ***Connect FMS Ethernet directly to the OPERATOR CONSOLE.** Devices hosting the Driver Station Software must only interface with the FMS via the Ethernet cable provided at the DRIVER STATION (e.g. not through a switch). Teams may connect the FMS Ethernet cable to the device running the Driver Station Software directly via an Ethernet pigtail, or with a single-port Ethernet converter (e.g. docking station, USB-Ethernet converter, Thunderbolt-Ethernet converter, etc.). The Ethernet port on the OPERATOR CONSOLE must be easily and quickly accessible.

> Teams are strongly encouraged to use pigtails on the Ethernet port used to connect to the FMS. Such pigtails will reduce wear and tear on the device's port and, with proper strain relief employed, will protect the port from accidental damage.

**R904** ***OPERATOR CONSOLE physical requirements.** The OPERATOR CONSOLE must not

A. be longer than 60.0in (1.524m),
B. be deeper than 16.0in (40.64cm) (excluding any items that are held or worn by the DRIVERS during the MATCH),
C. extend more than 78.0in (1.981m) above the floor, or
D. attach to the ARENA, except via the DRIVER STATION hook-and-loop tape or clamping to the DRIVER STATION support shelf (as long as the shelf is not damaged).

> Please note that while there is no hard weight limit, OPERATOR CONSOLES that weigh more than 30lb (13.61kg) will invite extra scrutiny as they are likely to present unsafe circumstances.

**R905** ***FIELD wireless only.** Other than the system provided by the FIELD, no other form of wireless communications shall be used to communicate to, from, or within the OPERATOR CONSOLE.

> Examples of prohibited wireless systems include, but are not limited to, active wireless network cards and Bluetooth devices. For the case of the FIRST Robotics Competition, a motion sensing input device (e.g. Microsoft Kinect) is not considered wireless communication and is

> allowed.

**R906** ***No unsafe OPERATOR CONSOLES.** OPERATOR CONSOLES shall not be made using hazardous materials, be unsafe, cause an unsafe condition, or interfere with other DRIVE TEAMS or the operation of other ROBOTS.

**R907** ***No AC inverters.** OPERATOR CONSOLES must not contain AC inverters.

# 08. INSPECTION & ELIGIBILITY (I)

This section describes the rules governing MATCH participation. A team has participated in a MATCH if any member of their DRIVE TEAM is in the ALLIANCE AREA, with or without the ROBOT on the FIELD, at the start of the MATCH.

At each event, the Lead ROBOT INSPECTOR (LRI) has final authority on the legality of any COMPONENT, MECHANISM, or ROBOT. The LRI may consult with additional sources, including the TideCrew Game Design Team, before making their final decision. INSPECTORS may re-inspect ROBOTS at any time to ensure compliance with the rules. Teams are encouraged to consult with INSPECTORS or the LRI if they have any questions regarding the legality of a ROBOT or about how to make a ROBOT legal.

The inspection process may progress in blocks, i.e. it may pause for a team's Practice MATCH, lunch break, etc. The process may employ various INSPECTORS throughout the process based on availability. At the team's discretion, they may request a different INSPECTOR or invite the Lead ROBOT INSPECTOR to participate in their ROBOT'S inspection.

ROBOTS may be modified after completing initial inspection; however, note that, per I104, any modifications not specifically exempt must be re-inspected. Prior to the beginning of Playoff MATCHES, all ROBOTS wishing to participate will be re-weighed to help identify any modifications that should be re-inspected per I104.

ROBOTS are permitted to participate in scheduled Practice MATCHES prior to passing inspection. However, the FTA, LRI, or Head REFEREE may determine at any time that the ROBOT is unsafe and may prohibit further participation in Practice MATCHES until the condition is corrected and/or the ROBOT passes inspection. A safe ROBOT has functional BUMPERS that approximately meet the required coverage (R401), height (R405), and construction (R402) requirements.

Prior to the start of a MATCH, any ROBOT which is unable or ineligible to participate in that MATCH, as determined by the FTA, LRI, or Head REFEREE, is declared to be BYPASSED and is DISABLED. A team whose ROBOT is BYPASSED remains eligible to receive Qualification Ranking Points or Playoff MATCH points provided that its ROBOT has passed inspection, per I102.

> An Inspection Checklist is available on the FIRST Robotics Competition website to help teams self-inspect their ROBOT before their event. Teams are strongly encouraged to self-inspect prior to the TideChallenge.

## 8.1 Rules

**I101** ***It's your team's ROBOT.** The ROBOT and its MAJOR MECHANISMS must be built by the FIRST Robotics Competition team.

A MAJOR MECHANISM is a group of COMPONENTS and/or MECHANISMS assembled together to address at least 1 game challenge: ROBOT movement, SCORING ELEMENT manipulation, FIELD element manipulation, or performance of a scorable task without the assistance of another ROBOT.

> This rule requires that the ROBOT and its MAJOR MECHANISMS were built by its team, but isn't intended to prohibit or discourage assistance from other teams (e.g. fabricating elements, supporting construction, writing software, developing game strategy, contributing COMPONENTS and/or MECHANISMS, etc.)
>
> Examples of MAJOR MECHANISMS include, but are not limited to, assemblies used to:
>
> A. manipulate SCOOPS, CONES, or ICE CREAM,
> B. position a ROBOT to engage with the CHECKOUT ZONE,
> C. manipulate a FIELD element, and
> D. move the ROBOT around the FIELD.
>
> Examples that would generally not be considered MAJOR MECHANISMS, and thus probably aren't subject to this rule include, but are not limited to: a gearbox assembly, a COMPONENT or MECHANISM that's part of a MAJOR MECHANISM, and COTS items.
>
> Neither this rule nor the language in this blue box define specific thresholds for how much of a MAJOR MECHANISM must be the result of the team's effort. This rule expects and requires the team's honest assessment of whether they built the MAJOR MECHANISMS of their ROBOT.
>
> Attempts to exploit loopholes in the definition of MAJOR MECHANISM in order to bypass this requirement are in the spirit of neither this rule nor the FIRST Robotics Competition. Examples of exploitation include: assembling pieces of a MAJOR MECHANISM provided by another team, except COTS kits, and receiving a mostly complete MAJOR MECHANISM from another team and providing a small piece.

**I102** ***Get inspected before playing a Qualification MATCH.** A team is only permitted to participate in a Qualification or Playoff MATCH and receive Ranking or MATCH Points if their ROBOT has passed an initial, complete inspection.

***Violation:** If prior to the start of the MATCH, the team is DISQUALIFIED and not eligible to participate in the MATCH. If after the MATCH, the team receives a RED CARD and the MATCH may be replayed at the discretion of the Head REFEREE.*

> Please take note of this rule. It is important that teams ensure their ALLIANCE partners have passed inspection. Allowing a partner that has not passed inspection to play puts the ALLIANCE at risk of RED CARDS. Teams should check with their ALLIANCE partners early and help them pass inspection before competing.

**I103** ***Bring it all to inspection.** At the time of inspection, the OPERATOR CONSOLE and the ROBOT must be presented with all MECHANISMS (including all COMPONENTS of each MECHANISM), configurations, and decorations that will be used on the ROBOT in MATCHES without re-inspection (per I104) and may not exceed 150.0lb (68.04kg) (note that while up to 150.0lb (68.04kg) of ROBOT MECHANISMS may be inspected together, the ROBOT configuration used in a MATCH may not violate R103). The OPERATOR CONSOLE and exceptions listed in R103 are not included in this weight.

**I104** ***Unless the change is listed below, any change to a ROBOT must get re-inspected.** A ROBOT may play MATCHES with a subset of the MECHANISMS that were present during inspection provided the reconfigured ROBOT still complies with all ROBOT Construction Rules. Only MECHANISMS that were present during the inspection may be added, removed, or reconfigured between MATCHES without re-inspection per this rule. If a ROBOT is modified after its most recent passed inspection, it must be re-inspected before it is eligible to participate in a MATCH. Exceptions are listed in A through F (unless they result in a significant change to the ROBOT'S size, weight, legality, or safety).

A. addition, relocation, or removal of fasteners (e.g. cable ties, tape, and rivets),
B. addition, relocation, or removal of labeling or marking,
C. revision of ROBOT code,
D. replacement of a COTS COMPONENT with an identical COTS COMPONENT,
E. replacement of a MECHANISM with an identical MECHANISM (size, weight, material), and
F. additions, removals, or reconfiguration of ROBOT with a subset of MECHANISMS already inspected per I103.

**I105** ***Don't exploit re-inspection.** Teams may not use the re-inspection process in I104 to circumvent the weight limit in I103.

> This restriction is not intended to prevent a team from returning to a previous configuration (e.g. due to an unsuccessful upgrade or failure of a new COMPONENT). If a team is believed to be violating this rule, the LRI will discuss the situation with the team to understand the changes and, if appropriate, the LRI in conjunction with the team will select a single configuration with which the team will compete for the duration of the event.

**I106** ***ROBOTS are off for inspection (mostly).** For the safety of all those involved, ROBOTS must be presented for inspection with the ROBOT powered off, pneumatics unpressurized, and springs or other stored energy devices in their lowest potential energy states (e.g. battery removed). Power and air pressure should only be enabled on the ROBOT during those portions of the inspection process where it is absolutely required to validate certain system functionality and compliance with specific rules (firmware check, etc.). INSPECTORS may allow the ROBOT to be powered beyond the parameters above if both criteria below are met:

A. the ROBOT design requires power or a charged stored energy device in order to confirm that the ROBOT meets volume requirements and
B. the team has included safety interlocks that mitigate unexpected release of such stored energy.

> The team may be asked to demonstrate these interlocks during the inspection process.

**I107** ***No STUDENT, no inspection (mentors' inner children don't count).** At least 1 STUDENT team member must accompany the ROBOT for any inspection efforts.

> Exceptions may be made for major conflicts, e.g. religious holidays, major testing, transportation issues, etc.

SECTION 09 · A VERY FRIENDLY MINI-TOURNAMENT

# 09. TOURNAMENT

The TideChallenge ends with a short, very friendly tournament. It follows the structure of a FIRST Robotics Competition event (Practice, Qualification MATCHES, Alliance Selection, and Playoffs) so that teams get the full event experience, but everything in it is deliberately loose. The goal is for every team to play as much FROZEN BLITZ as possible, learn something, and go home with a story.

> **This is a mini-tournament, and a very friendly one.** Event staff may adjust the schedule, the number of MATCHES, the number of ALLIANCES, and the bracket on the day to fit attendance and time. When in doubt, we pick the option that lets more teams play more ice cream.
>
> **When and where:** Saturday, December 5, 2026, at Prepa Tec Campus Santa Anita. The detailed schedule is shared in a Team Update.

## 9.1 Event Flow

The event runs in the following order. Times are approximate and announced on the day.

Table 9-1 *TideChallenge event flow*

| Phase | What happens | Counts toward |
|---|---|---|
| Load-in & inspection | Teams set up their pits and complete ROBOT inspection (Section 8). | — |
| Practice | Up to 3 Practice MATCHES per team, by sign-up. | nothing (just practice) |
| Qualifications | 3 to 6 Qualification MATCHES per team, randomly scheduled. | ranking |
| Alliance Selection | The top-ranked teams form 4 to 6 ALLIANCES. | — |
| Playoffs | Reduced double-elimination bracket and best-of-3 Finals. | Winner & Finalist awards |
| Awards & closing | Awards, photos, and ice cream (the real kind, if we can). | — |

## 9.2 Practice MATCHES

Each team may play up to 3 Practice MATCHES. Practice MATCHES are not pre-scheduled: teams sign up at the queuing table, and FIELD STAFF fill each MATCH with the next 4 teams in line, first come, first served. If the line is short, a team may be asked to play an extra Practice MATCH to fill a slot.

- Practice MATCHES do not count toward ranking.
- ROBOTS may play Practice MATCHES before passing inspection, as described in Section 8.
- All rules apply, and REFEREES call fouls as feedback so teams know what to fix before Qualifications.

## 9.3 Qualification MATCHES

Once attendance is confirmed, event staff publish the Qualification schedule. Each team plays between 3 and 6 Qualification MATCHES; the exact number depends on how many teams attend and how much time is available, and is announced with the schedule. ALLIANCE partners and opponents are assigned at random, with 2 ROBOTS per ALLIANCE in every MATCH.

If the number of MATCH slots does not divide evenly among teams, some teams are scheduled for 1 extra MATCH as a surrogate. A surrogate plays normally, but that MATCH does not count toward the surrogate team's ranking. The schedule marks surrogate MATCHES.

### 9.3.1 Ranking

Teams are ranked by Ranking Score: the average number of Ranking Points (Table 5-3) earned per Qualification MATCH, not counting surrogate MATCHES. A team that is DISQUALIFIED from a MATCH, or does not show up for it, earns 0 Ranking Points for that MATCH. Ties in Ranking Score are broken in the order listed in Table 9-2.

Table 9-2 *Qualification ranking tiebreakers*

| Order | Criteria |
|---|---|
| 1st | average TOPPER points |
| 2nd | average AUTO points |
| 3rd | average MATCH points, not including points from opponent fouls |
| 4th | random draw |

## 9.4 Alliance Selection

After Qualifications, event staff announce the number of Playoff ALLIANCES (Table 9-3). The top-ranked teams become ALLIANCE CAPTAINS, in ranking order, and the ALLIANCE CAPTAIN of ALLIANCE 1 makes the first pick.

Table 9-3 *Number of Playoff ALLIANCES*

| Teams attending | Playoff ALLIANCES |
|---|---|
| 8 to 11 | 4 |
| 12 to 14 | 5 |
| 15 or more | 6 |

A. In round 1, each ALLIANCE CAPTAIN, in order from ALLIANCE 1, invites 1 team to join their ALLIANCE.

B. If enough teams remain, a round 2 runs in reverse order (from the last ALLIANCE back to ALLIANCE 1) so that each ALLIANCE adds a third team. Only 2 ROBOTS per ALLIANCE play each MATCH; the ALLIANCE chooses which 2 before each MATCH.

C. An ALLIANCE CAPTAIN may invite any team not yet on an ALLIANCE, including a lower-seeded ALLIANCE CAPTAIN. If that captain accepts, the remaining ALLIANCE CAPTAINS move up to fill the gap.

D. An invited team may accept or decline. A team that declines cannot be picked later, but may still become an ALLIANCE CAPTAIN if the ranking reaches it.

E. Teams not on an ALLIANCE are invited to stay as backup ROBOTS. If an ALLIANCE cannot field 2 ROBOTS for a MATCH, it may call in the highest-ranked available backup team.

> With fewer than 8 teams, event staff will set up something simpler (e.g. a round-robin of random ALLIANCES). With more teams than expected, more MATCHES. Nobody gets left without ice cream.

## 9.5 Playoffs

The Playoffs are a reduced double-elimination bracket: an ALLIANCE is eliminated after losing 2 MATCHES, except where noted. In every Playoff MATCH, the higher-seeded ALLIANCE plays as the red ALLIANCE. There are no Ranking Points in the Playoffs.

Table 9-4 *Playoff bracket (4 ALLIANCES)*

| MATCH | Bracket | Red vs. Blue | Loser |
|---|---|---|---|
| M1 | upper | ALLIANCE 1 vs. ALLIANCE 4 | goes to M3 |
| M2 | upper | ALLIANCE 2 vs. ALLIANCE 3 | goes to M3 |
| M3 | lower | loser M1 vs. loser M2 | eliminated |
| M4 | upper final | winner M1 vs. winner M2 | goes to M5 |
| M5 | lower final | loser M4 vs. winner M3 | eliminated (3rd place) |
| Finals | best of 3 | winner M4 vs. winner M5 | Finalist |

With 5 or 6 ALLIANCES, a single-elimination play-in round runs first, and its winners fill seeds 3 and 4 of the bracket above:

- **5 ALLIANCES:** P1, ALLIANCE 4 vs. ALLIANCE 5. The winner is seed 4.
- **6 ALLIANCES:** P1, ALLIANCE 3 vs. ALLIANCE 6, and P2, ALLIANCE 4 vs. ALLIANCE 5. The winner of P1 is seed 3 and the winner of P2 is seed 4.
- Play-in losers are eliminated, with a round of applause.

The Finals are won by the first ALLIANCE to win 2 MATCHES. If a Playoff MATCH ends in a tie, it is broken in the order listed in Table 9-5.

Table 9-5 *Playoff MATCH tiebreakers*

| Order | Criteria |
|---|---|
| 1st | fewer foul points awarded to the opponent |

| Order | Criteria |
|---|---|
| 2nd | more TOPPER points |
| 3rd | more AUTO points |
| 4th | replay the MATCH |

Each ALLIANCE may call 1 timeout of up to 6 minutes during the Playoffs. Event staff also give extra time between MATCHES whenever a team needs it to repair a ROBOT safely; nobody gets rushed onto the FIELD.

## 9.6 Awards

The following awards are presented at the TideChallenge. Event staff may add others on the day.

- **Winner:** the ALLIANCE that wins the Finals.
- **Finalist:** the ALLIANCE that loses the Finals.
- **Single Top Score:** a special award for the highest single-MATCH score with a single ROBOT, open to every team, including teams that cannot attend in person (see section 9.6.1).

### 9.6.1 Single Top Score

FROZEN BLITZ is a fun game to build for, even if a team can't make it on the day. Any team that builds a ROBOT for the TideChallenge may submit a video of its best full MATCH. Teams that compete in person may also submit one.

A. **One full MATCH, one take.** The video shows 15 seconds of AUTO followed immediately by 2 minutes and 15 seconds of TELEOP in a single, unedited take, with the MATCH timer visible or audible (e.g. a match timer app or Driver Station practice mode).

B. **Show the setup first.** The video begins with a short walk-around of the ROBOT (with its team number visible) and the staged FIELD.

C. **One ROBOT, no opponents.** Only the team's own ROBOT is on the FIELD. The team supplies its own 2 HUMAN PLAYERS, who follow the HUMAN PLAYER rules in Section 6.

D. **A real FIELD.** The FIELD elements used during the run must follow the official drawings. Elements the run does not use may be omitted, and construction materials may differ as long as dimensions match. SCORING ELEMENTS and HOOPS are staged for 1 ALLIANCE as described in section 5.3.4.

E. **A legal ROBOT.** The ROBOT must comply with Section 7. The TideCrew Game Design Team may ask for photos or a weigh-in video.

Submissions are scored per Section 5 for 1 ALLIANCE, with the following adjustments:

- Ranking Points, the REACH BONUS, and the CHECKOUT BONUS do not apply.
- Fouls the video shows the team committing are subtracted from its score (4 points per MINOR FOUL and 8 points per MAJOR FOUL).
- Ties are broken by TOPPER points, then AUTO points, then the earliest submission.

Teams may submit more than once; their highest valid score counts. Submissions are reviewed and scored by the TideCrew Game Design Team, and its decision is final. Submissions are due by the day of the in-person event (December 5, 2026); the submission link is shared in a Team Update.

This is the only place in the TideChallenge where video counts. On the FIELD, the Head REFEREE'S call stands (section 5.7).

SECTION 10 · WORDS WE USE

# 10. GLOSSARY

| Term | Definition |
|---|---|
| **ACTIVE DEVICE** | any device capable of dynamically controlling and/or converting a source of electrical energy by the application of external electrical stimulus |
| **ALLIANCE** | a cooperative of up to 4 FIRST Robotics Competition teams |
| **ALLIANCE AREA** | a 330.0in (~8.38m) wide by 106.0in (~2.69m) deep, infinitely tall volume formed by, and including, the ALLIANCE STATIONS and ALLIANCE colored tape; it includes neither the FREEZER nor the STORAGE |
| **ALLIANCE CAPTAIN** | the designated STUDENT representative of each ALLIANCE'S lead team |
| **APRILTAG** | a 36h11 visual marker mounted on the FIELD to help ROBOTS find their position; there are 8 on the FIELD, IDs 1 to 8 (see section 4.9) |
| **ARENA** | includes all elements of the game infrastructure that are required to play FROZEN BLITZ: the FIELD, SCORING ELEMENTS, and all equipment needed for FIELD control, ROBOT control, and scorekeeping |
| **ARENA FAULT** | an error in ARENA operation |
| **AUTO** | the first 15 seconds of the MATCH, in which the FMS blocks any DRIVER control, so ROBOTS operate with only their pre-programmed instructions |
| **BUMPER** | a required assembly which attaches to the ROBOT frame. BUMPERS protect ROBOTS from damaging/being damaged by other ROBOTS and FIELD elements |
| **BUMPER ZONE** | a space between 2.5in (6.35cm) and 5.75in (14.61cm) from the floor |
| **BYPASSED** | any ROBOT which is unable or ineligible to participate in that MATCH, as determined by the FTA, LRI, or Head REFEREE |
| **CENTER LINE** | a white line at the center of the FIELD, parallel to the ALLIANCE WALL, that bisects the length of the FIELD |
| **CHECKOUT** | MATCH points earned by a ROBOT completely within, or completely above and fully supported by, its ALLIANCE'S CHECKOUT ZONE at the end of the MATCH |
| **CHECKOUT ZONE** | 1 of 2 (1 per ALLIANCE) 112.0in (~2.84m) wide by 51.0in (~1.30m) deep, infinitely tall volumes bounded by and including the ALLIANCE colored tape and divided by the CENTER LINE, containing the CHECKOUT ZONE tables |
| **CHUTE** | the opening in the base of the DELIVERY ZONE structure through which the DELIVERY ZONE HUMAN PLAYER returns recycled SCOOPS to the FIELD (see section 4.4) |

| Term | Definition |
|---|---|
| **COMPONENT** | any part in its most basic configuration, which cannot be disassembled without damaging or destroying the part or altering its fundamental function |
| **CONE** | a 9.0in (~22.9cm) tall orange plastic sport cone; a SCORING ELEMENT |
| **CONTINUOUS** | describes durations that are more than approximately 10 seconds |
| **CONTROL** | an action by a ROBOT in which a SCORING ELEMENT is fully supported by, travels with, is held or shielded by, or is prevented from leaving the DELIVERY ZONE by the ROBOT (see G407) |
| **COTS** | a standard (i.e. not custom order) part commonly available from a VENDOR for all teams for purchase |
| **CUSTOM CIRCUIT** | any active electrical item that is not an actuator (specified in R501) or core control system item (specified in R710) |
| **DELIVERY TRAY** | 1 of 2 trays in each DELIVERY ZONE structure on which ICE CREAM is delivered; holds 1 ICE CREAM and is lowered by the HUMAN PLAYER to clear it after delivery (see section 4.4) |
| **DELIVERY ZONE** | 1 of 2 (1 per ALLIANCE) 200.0in (~5.08m) wide by 67.25in (~1.71m) deep, infinitely tall volumes bounded by and including the ALLIANCE colored tape, containing 2 DELIVERY TRAYS, 1 CHUTE, 1 PURPLE PLANE, and a SAFE ZONE |
| **DISABLED** | the state in which a ROBOT is commanded to deactivate all outputs, rendering the ROBOT inoperable for the remainder of the MATCH |
| **DISQUALIFIED** | the state of a team in which they receive 0 MATCH points and 0 Ranking Points in a Qualification MATCH or causes their ALLIANCE to receive 0 MATCH points in a Playoff MATCH |
| **DRIVE COACH** | a guide or advisor |
| **DRIVE TEAM** | a set of up to 5 people from the same FIRST Robotics Competition team responsible for team performance for a specific MATCH |
| **DRIVER** | an operator and controller of the ROBOT |
| **DRIVER STATION** | 1 of the assemblies within an ALLIANCE WALL behind which a DRIVE TEAM operates their ROBOT |
| **END GAME** | the final 30 seconds of TELEOP (timer 0:30 – 0:00) |
| **FABRICATED ITEM** | any COMPONENT or MECHANISM that has been altered, built, cast, constructed, concocted, created, cut, heat treated, machined, manufactured, modified, painted, produced, surface coated, or conjured partially or completely into the final form in which it will be used on the ROBOT |
| **FIELD** | an approximately 360.0in (~9.14m) by 552.0in (~14.02m) carpeted area bounded by and including the guardrails and ALLIANCE WALLS |
| **FIELD STAFF** | the collective group of people working on or near the FIELD responsible for making sure the MATCHES are cycled through efficiently, fairly, safely, and with a spirit of cooperation, Gracious Professionalism, and generosity of spirit |
| **FMS** | the FIELD Management System |

| Term | Definition |
|---|---|
| **FREEZER** | 1 of 2 (1 per ALLIANCE) 122.0in (~3.10m) wide, 53.5in (~1.36m) deep, infinitely tall volumes defined by the ALLIANCE STATION, guardrail, and ALLIANCE colored boundaries, where SCOOPS are staged and REFILLED |
| **FTA** | FIRST Technical Advisor |
| **HANDED** | a CONE delivered to the SAFE ZONE of its ALLIANCE'S SCOOPING STATION after having contacted the floor; once per trip out of STORAGE (see 5.5.1) |
| **HOOP** | a white, 3D-printed 4.00in (~10.2cm) torus ring placed by a HUMAN PLAYER on top of a SCOOP so another SCOOP can be stacked on it; not a SCORING ELEMENT |
| **HUMAN PLAYER** | a SCORING ELEMENT manager |
| **ICE CREAM** | a stack (a CONE with 1 to 5 SCOOPS) whose CONE tip has completely left the SCOOPING STATION, treated as a single SCORING ELEMENT |
| **INSPECTOR** | a person determined to accurately and efficiently assess the legality of a given part of a ROBOT |
| **KOP** | the Kit of Parts |
| **LRI** | the Lead ROBOT INSPECTOR |
| **MAJOR FOUL** | a credit of 8 points towards the opponent's MATCH point total |
| **MAJOR MECHANISM** | a group of COMPONENTS and/or MECHANISMS assembled together to address at least 1 game challenge: ROBOT movement, SCORING ELEMENT manipulation, FIELD element manipulation, or performance of a scorable task without the assistance of another ROBOT |
| **MATCH** | the 2-minute and 30-second period in which a ROBOT is enabled to play FROZEN BLITZ |
| **MECHANISM** | an assembly of COMPONENTS that provide specific functionality on the ROBOT. A MECHANISM can be disassembled (and then reassembled) into individual COMPONENTS without damage to the parts |
| **MINOR FOUL** | a credit of 4 points towards the opponent's MATCH point total |
| **MOMENTARY** | describes durations that are fewer than approximately 3 seconds |
| **MXP** | myRIO Expansion port, the expansion port on the roboRIO |
| **OPERATOR CONSOLE** | the set of COMPONENTS and MECHANISMS used by the DRIVERS and/or HUMAN PLAYERS to relay commands to the ROBOT |
| **PASSIVE CONDUCTOR** | any device or circuit whose capability is limited to the conduction and/or static regulation of the electrical energy applied to it (e.g. wire, splices, connectors, printed wiring board, etc.) |
| **PCM** | Pneumatic Control Module |
| **PD** | a main power distribution device per R609 (PDP, PDP 2.0, PDH, or AMPD) |
| **PDH** | Power Distribution Hub |

| Term | Definition |
|---|---|
| **PDP** | Power Distribution Panel |
| **PH** | Pneumatic Hub |
| **PIN** | an action by a ROBOT that is preventing the movement of an opponent ROBOT by contact, either direct or transitive (such as against a FIELD element) |
| **PURPLE PLANE** | the vertical plane marked in purple on each DELIVERY ZONE structure beyond which DRIVE TEAM members may not reach (see Figure 4-9 and G430) |
| **REACH** | a ROBOT whose BUMPERS have partially entered its ALLIANCE'S FREEZER at any point during AUTO |
| **RED CARD** | issued by the Head REFEREE for egregious ROBOT or team member behavior or rule violations which results in a team being DISQUALIFIED for the MATCH |
| **REFEREE** | an official who enforces the rules of FROZEN BLITZ |
| **REFILL** | a neutral or ALLIANCE color SCOOP inside its ALLIANCE'S FREEZER at the end of the MATCH, capped by the number of SCOOPS delivered (see 5.5.1) |
| **REPEATED** | describes actions that happen more than once within a MATCH |
| **ROBOT** | an electromechanical assembly built by the FIRST Robotics Competition team to play the current season's game and includes all the basic systems required to be an active participant in the game – power, communications, control, BUMPERS, and movement about the FIELD |
| **ROBOT PERIMETER** | the part of a ROBOT contained within the BUMPER ZONE and established while in the ROBOT'S STARTING CONFIGURATION, that is comprised of fixed, non-articulated structural elements of the ROBOT (formerly "FRAME PERIMETER") |
| **RP** | Ranking Points |
| **RPM** | Radio Power Module |
| **RSL** | ROBOT Signal Light |
| **SAFE ZONE** | 1 per SCOOPING STATION and 1 per DELIVERY ZONE structure: the infinitely tall volume between the structure and the ALLIANCE colored tape located 30.0in (~76.2cm) from it, bounded by and including that tape. A ROBOT at least half within its own SAFE ZONE is protected (G425), and opponents may not enter it (G426) |
| **SCOOP** | an 8.0in (~20.3cm) diameter inflatable, felt-covered tennis ball: neutral (green), red ALLIANCE (pink or red), or blue ALLIANCE; a SCORING ELEMENT |
| **SCOOPING STATION** | the ALLIANCE structure inside each SCOOPING ZONE where the HUMAN PLAYER places HOOPS and ROBOTS build stacks; includes a SAFE ZONE |
| **SCOOPING ZONE** | 1 of 2 (1 per ALLIANCE) 200.0in (~5.08m) wide by 67.25in (~1.71m) deep, infinitely tall volumes bounded by and including the ALLIANCE colored tape, containing the SCOOPING STATION, where CONES are placed |
| **SCORING ELEMENT** | a SCOOP, a CONE, or an ICE CREAM |

| Term | Definition |
|---|---|
| **SIGNAL LEVEL** | a term used to characterize circuits which draw ≤1A continuous and have a source incapable of delivering >1A, including but not limited to roboRIO non-PWM outputs, CAN signals, PCM/PH Solenoid outputs, VRM 500mA outputs, RPM outputs, and Arduino outputs |
| **STACK** | a CONE with 1 or more SCOOPS stacked on it while the tip of its CONE is still inside the SCOOPING STATION; each CONE and SCOOP in it is an individual SCORING ELEMENT |
| **STARTING CONFIGURATION** | the physical configuration in which a ROBOT starts a MATCH |
| **STARTING LINE** | a white line spanning the width of the carpet, located 28.0in (~71.1cm) from the back of the DRIVER STATION diamond plate panel to the near edge of the tape |
| **STORAGE** | 1 of 2 (1 per ALLIANCE) 122.0in (~3.10m) wide, 52.5in (~1.33m) deep, infinitely tall volumes defined by the ALLIANCE STATION, guardrail, and ALLIANCE colored boundaries, where CONES are staged |
| **STUDENT** | a person who has not completed high-school, secondary school, or the comparable level as of September 1 prior to Kickoff |
| **SURROGATE** | a team randomly assigned by the FMS to play an extra Qualification MATCH |
| **TECHNICIAN** | a resource for ROBOT troubleshooting, setup, and removal from the FIELD |
| **TELEOP** | the Teleoperated Period, the final 2 minutes and 15 seconds of the MATCH |
| **TIER** | the number of SCOOPS in an ICE CREAM (1 to 5) |
| **TOPPER** | the top SCOOP of a delivered ICE CREAM of TIER 2 or higher when it is the delivering ALLIANCE'S color; its points add to the ICE CREAM points |
| **VENDOR** | a legitimate business source for COTS items that satisfies criteria defined in section 7 ROBOT Construction Rules |
| **VERBAL WARNING** | a warning issued to a team by event staff or the Head REFEREE |
| **VRM** | Voltage Regulator Module |
| **YELLOW CARD** | issued by the Head REFEREE for egregious ROBOT or team member behavior or rule violations |